Loki III: Reverse Engineering Structural Netlists without a Language Model
How far do provenance, a Bayesian L1 and a graph L2 get on their own?

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***Abstract*—A synthesis tool turns a circuit description into a netlist and throws information away on the way. Loki III asks how much of the design can be recovered *without* a language model, from provenance (the link from each netlist object back to the source text), a Bayesian hypothesis flow, graph analysis and, newest, a search that grows blocks from anchors and joins them.**

The answer key works: it covers all 54,306 gate nodes of 150 designs, and the pipeline is naming-neutral over 105,646 renamed identifiers. Structures that synthesis already reverse engineered are recovered (hard adders 67 of 67), and calibration cuts the calibration error from 0.650 to 0.217. The newest result is block recovery. Blocks grown from anchors to the end of their path and then joined by a conductance rule are the right block for 0.715 of 1141 held-out constructs, against 0.385 for bounded cones, and a kind guess on those blocks reaches a macro F1 of 0.392. Structures that synthesis erased recover weakly, clustering does not find the designer's hierarchy, and a split step, a word-first detector and Loki's hints to a language model did not help; a model with the netlist does far better than without it.

Every claim is a hypothesis with one experiment and one verdict. Of 104 register rows, 14 are USEFUL, 12 POSSIBLY GOOD, 19 DISMISSED, 34 INCONCLUSIVE and 25 are measurements. Round-by-round records are in the appendix. No area, speed or power is measured.

***Index Terms*—reverse engineering, netlist, provenance, Bayesian inference, graph neural networks, FPGA CAD**

I. Introduction

A designer writes a circuit in a hardware description language (HDL) such as Verilog. At the register-transfer level (RTL) the HDL text says what the registers hold and how they change. A synthesis tool turns that text into gates, the basic logic elements such as AND, OR and NOT, and a netlist, the list of those gates and the wires that join them. Synthesis throws information away. It merges operators, flattens the module hierarchy and leaves no source names on the gates. Reverse engineering a netlist means getting back as much of the RTL structure as possible. Gate-Breaker [1] is an earlier study in which a language model reads a netlist with hidden wire names and writes the RTL; it ends by naming the graph tools that its model needed: clustering to name buses, a classification of signals as datapath or control, and partitioning of the netlist.

Loki III (https://github.com/drpaj12/loki) asks how much of the RTL can be recovered *without* a language model. It stands on provenance: the link from each netlist object back to the HDL that produced it. Odin III, the synthesis front end of the Yggdrasil III suite, keeps that link through elaboration, optimisation and mapping. The link yields an answer key (also called ground truth): what the tool recorded about which netlist objects form which HDL construct. The answer key is what makes recovery measurable, because a recovered structure can be compared with a structure the tool knew.

The short version. The paper asks five questions. Each answer is stated here with its headline number; Section V gives the evidence and Appendix A the record of each experiment.

1) *Can we build netlists where the right answer is known and the names carry nothing?* Yes. The answer key covers all 54,306 gate nodes of 150 designs, and scrambling every name changes no answer (476 byte-identical run pairs).
2) *How much does a Bayesian flow recover?* What synthesis already reverse engineered, almost perfectly (hard adders 67 of 67). What synthesis erased, weakly (a score of 0.354), and on a gate netlist not at all (0 of 144). The stated probabilities become far closer to the observed rates (error 0.650 to 0.217).
3) *Does graph analysis find the designer's hierarchy?* No. The best clustering scores a normalised mutual information of 0.3220 against the module path. A graph network labels gate roles ahead of a baseline that sees only the neighbourhood of each gate (0.4241 against 0.2717), but on one split of 13 designs only.
4) *Can blocks of the netlist be found?* Yes, and this is the newest and strongest result (L3). Blocks are grown from anchors, the places where a known pattern is easy to see, to the end of their path, and touching blocks that are one construct are joined. A block is right for a construct when it covers at least 0.8 of it with a Jaccard overlap of at least 0.5. On 1141 held-out constructs the joined

blocks are right for 0.715, against 0.385 for bounded cones (per design and cell, geomean 1.318, 1.276 to 1.364, W394 / L46 / T417). A kind guess on the joined blocks reaches a macro score of 0.392. The pieces that did not help are listed in Table XI.

5) *Does a language model do better with Loki's hints?* No. On 137 problems with three draws, a model given the netlist and Loki's hints passes as often as a model given the netlist alone (mean pass@1 0.878 against 0.888). The netlist itself matters: without it the model passes 0.560.

We write every claim as a hypothesis: a claim that an experiment can confirm or fail to confirm. For each one the paper gives the experiment, the result and one verdict. USEFUL means that the experiment confirms the hypothesis under its own decision rule. POSSIBLY GOOD means that it confirms it only narrowly or fragilely, so the paper states the fragility and never uses such a result alone as evidence for an innovation. INCONCLUSIVE means that the numbers do not decide; we never invent a threshold to force a verdict. DISMISSED means that the experiment does not confirm it. Ideas whose experiments prove their value are the innovations of the main paper; dismissed ideas are one line each in Table XI and in the research log (Appendix C).

Loki has three layers: L0, labelled and anonymised netlist datasets with the answer key; L1, a Bayesian flow that proposes structure hypotheses and settles them with probabilities and equivalence checks; and L2, graph analysis. The innovations that the experiments confirm are these four, in plain English, each with the hypotheses that confirmed it. A hypothesis is named by the id of its *register row*, one line of the research log (Appendix C); an id such as L0a is only a pointer to that line.

1) *A benchmark in which the answer is known and the names carry nothing.* We build the netlists with a front end that remembers which source text made each gate, give every object a neutral name, and show that scrambling all names changes no answer (naming neutrality: 476 byte-identical run pairs). The answer key covers every gate node of the 150 designs that build (module coverage 1.000). Confirmed: L0a, L0b.
2) *A hypothesis flow that is honest about what it knows.* The tool proposes "this group of gates is an adder, a register or a mux", gives each claim a probability learned from training designs, and uses an equivalence check to settle the claims it can. Structures that synthesis already reverse engineered are recovered almost perfectly, and the tool's stated probabilities are far closer to the observed rates after fitting. Confirmed: L1h, L1c.
3) *A limit on what any recovery can do, a synthetic test bed, and a count of repeated structure.* A set of twin designs measures how much coding style synthesis destroys (H14a); a generated set of idiom designs builds at every mapping level of the flow (H12a); and the large designs of the VTR suite are mostly structure that repeats (H13a), with identical neighbourhoods inside the repeats (H13c). Confirmed: H14a, H12a, H13a, H13c.
4) *A block search that grows from anchors and joins.* Blocks grow cell by cell from anchors to the end of their path, extra anchors cover constructs that no block touches, and a join step merges touching blocks that are one construct. On 1141 held-out constructs the blocks are right for 0.715, against 0.385 for bounded cones. Confirmed: QL7a, QL7b, QL9-a, QL9-b. The stop rule adds nothing over growing to the end of the path (QL7e, INCONCLUSIVE), and the split step is DISMISSED (QL9-d).

Candidate innovations. Three more ideas have only POSSIBLY GOOD support: each experiment leans the right way but is narrow or fragile, so none is a contribution of this paper. (i) *Graph analysis with a network.* Treat the netlist as a graph and train a graph network to label each gate's role (L2b: a score of 0.4241 against 0.2717 for a baseline that sees only the neighbourhood of each gate, one run on one split of 13 designs). What would confirm it: several random splits of the designs with an interval over designs, and a second benchmark suite. (ii) *A score that sees recovery.* Comparing recovered Verilog text with the reference cannot tell whether the right adders were found; counting the operator types recovered can (R2-L7: small margins on few pairs, and the flow with fitted probabilities recovers no operator). What would confirm it: a set with many more tie pairs and a flow that recovers operators. (iii) *A loss audit.* When synthesis merges the logic of two source statements, the construct can no longer be pulled apart from its neighbours. Using the provenance the front end already keeps, we record which pass merged what (L15a, L15d: a mux is found at the gate snapshot and lost after the final mapping pass, but the two ground truths differ in size and the footprints are not like for like). What would confirm it: the same audit with equal-sized ground truths at the two snapshots and on the big designs.

The other answers to the paper's question are negative or undecided. Clustering does not find the designer's hierarchy (L2a, DISMISSED); the structures that synthesis erased recover weakly; a pattern finder that looks for multi-bit signals first is not shown to beat a published method; a full language-model run (137 problems, three draws) shows that Loki's hints add nothing measurable to the netlist; and in the block search a split step and a better growth order did not help (Section V).

Loki's measures are recovery measures (how much of the answer key is found) and not circuit measures. Area, speed and power are not measured by any experiment in this paper, and we do not claim they correlate with the recovery scores. Mímir III, the suite's circuit knowledge base, is where an answer key of this kind would be stored and queried across designs; Loki does not use it yet.

Loki III is one tool of the Yggdrasil III suite. Odin III is the trunk of the suite: its one provenance-tracked IR is what Loki III reads and what supplies its answer key. Loki III runs as a step of Wyrd III, the experiment runner, and Mímir III, the circuit knowledge base, holds the benchmarks and the records

that this paper uses. The suite is drawn as a tree, with the pieces and their links, in the hub repository at https://github.com/drpaj12/YggdrasilIII; the tree is shown in the Odin III paper.

Figure 1 shows Loki III as one flow, from the Odin III netlist and answer key to the recovered blocks that are scored against the key. The stages are L0 (datasets), L1 (the Bayesian flow), L2 (graph analysis) and L3 (blocks); the language-model harness is a side path with no Loki stage inside it, and a second path hands Loki's blocks to the model one block per prompt. Each circled mark is the subsection of the Results (Section V) or of the language-model section (Section VI) that tests the stage it sits on, and the dashed box holds the ideas that did not work.

II. Background

This section introduces, one step at a time, the terms the rest of the paper uses. We start with the cells of an FPGA and the CAD step that covers a netlist with them. Next we describe provenance and what Odin III records, then the earlier study, Gate-Breaker, whose measures and language-model protocol we reuse. We then describe the benchmark designs and the levels at which they are cut, the way Loki proposes and scores structure hypotheses, and finally the graph methods and the published baseline that Loki is compared with. The basic FPGA and CAD terms follow the thesis of Jamieson [2] (Chapter 2) and the thesis of Luu [3] (Sections 2.1.1 and 4.1.2). The Introduction has already introduced the HDL, RTL, gates and netlists that Loki reads, provenance, the answer key, hypotheses and the three layers.

A. Storage and logic cells of an FPGA

A netlist of gates is the form in which a design leaves a simple synthesis flow. A real FPGA offers two other kinds of cell. The first is storage: a one-bit element that keeps its value between clock edges is called a *flip-flop*. The second is logic: a small memory that implements any logic function of K inputs is called a *look-up table* (LUT), with $K = 6$ in this paper. In most FPGAs a LUT and a flip-flop together form one basic logic element [2], [3]. The CAD step that covers the logic of a netlist with the cells that the target offers is called *technology mapping*: gates for a gate library, LUTs and flip-flops for an FPGA [2]. Mapping is one of the steps that throws structure away, which is why the same design can be cut at several levels (Section II-D). Mapping to LUTs is a research problem with a long literature; ABC's mapper, which our flow links, is described by Mishchenko et al. [4]. In the experiments, these cells are the nodes of every netlist that Loki reads, and the levels at which a design is cut (Section II-D) differ in how much structure the mapping left.

B. The provenance problem and Odin III

Synthesis modifies the designer's functional intention: flattening removes hierarchy, optimization merges operators, and mapping covers logic with cells that carry no source names. Linking each netlist object back to the HDL that produced it is the provenance problem [5]. Odin III, the provenance-tracked front end under Loki, keeps that link: every object of its one IR records its source location, and the link survives elaboration, optimization, partial mapping and the linked ABC flow. Loki stands on that link: its L0 datasets carry the answer key the synthesis tool kept, which is what makes the recovery statistics in this paper possible at all. The problem was stated for FPGA netlists by Gharibian et al. [5], who recovered system-level structure from flat netlists; Odin II [6] is the open-source synthesis tool from which Odin III descends. The first two Results arguments check that the key is complete and independent of names, and every later argument is scored against it.

The newest form of the link is Odin III's version-2 name map, which we call the *D14 provenance map*. For each object it records the source-statement units that the object blends, and it keeps labelled join records for the places where objects merge. The map answers a question that the loss audit (Section III) asks after every pass. A *unit* is the statement-level anchor of a construct; a unit is *separable* while some object has exactly that unit set, and it is *lost* at the first later snapshot where no object has. To say whether a loss is more than chance, the audit compares it with two null models. The *random-set null* draws, at each pass, random unit sets of the real sizes. The *persistence null* instead starts from the objects of the previous snapshot and merges them at random until the real object count is reached. We introduced the map and the audit for this paper, so there is no earlier citation; the loss-audit argument of the Results uses them.

C. Gate-Breaker as the method baseline

Gate-Breaker measures recovery by functional correctness and by the similarity of the recovered and reference RTL: Success Rate (the fraction of K samples that simulate correctly) and Success@k (the fraction of designs solved within the first k samples). It also measures how alike two RTL texts are, by *AST and CFG similarity*: AST similarity is the mean subtree Jaccard overlap of the syntax trees over normalized identifiers (depth $d = 3$), and CFG similarity is a normalized shortest-path node kernel over the control-flow graphs. Its complexity score weights sequential blocks, generates, loops, instantiations, always blocks, buses, multiplexers, cases, arithmetic operators, assignments and concatenations. Ordering the designs by that score and cutting the order into four equal parts gives the *quartiles* Q1 (simple combinational) through Q4 (complex sequential).

Gate-Breaker also fixes how a language model is asked. The pilot and the full run of this paper vary the prompt in three *arms A, B and C*: arm A gives the model the gate netlist, arm B the netlist plus Loki's hints, and arm C only the interface line, which controls for memorisation. Gate-Breaker was introduced by Faruque et al. [1], who report that its model needed graph tools such as clustering and partitioning; Loki is the attempt to build those tools with provenance. Loki reports its own statistics on the same axes where they apply, and the Gate-Breaker metrics on the replication runs; the language-model

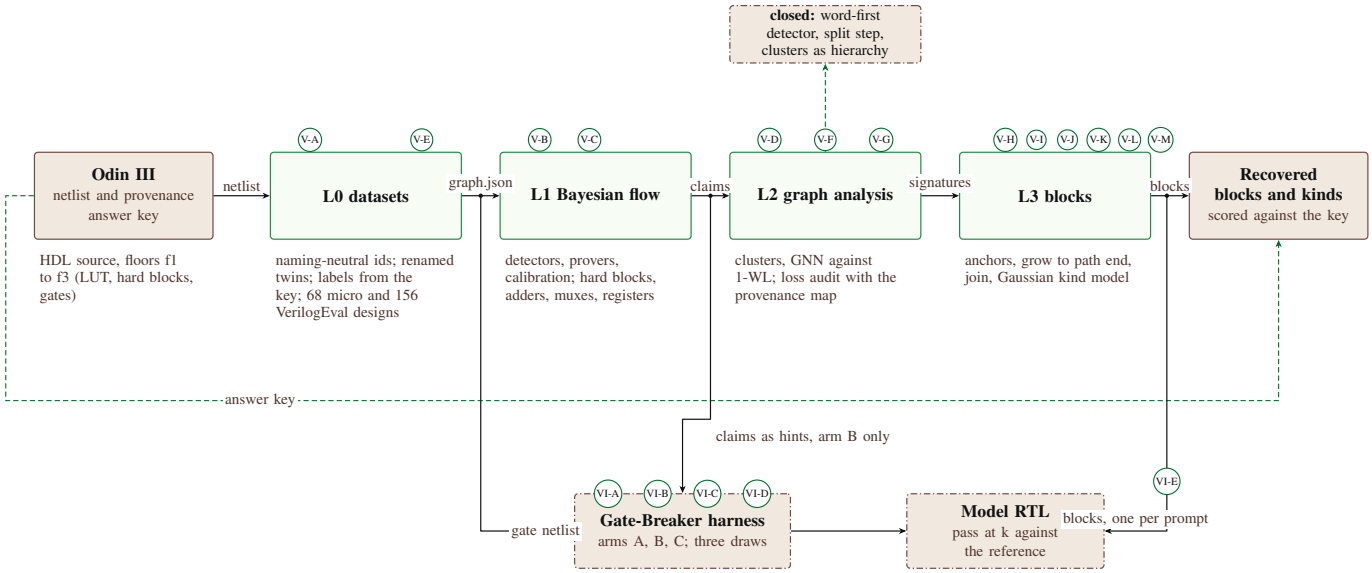


Fig. 1. Loki III as a flow. Boxes are stages, the text under a box lists the features of the stage, and the labels on the arrows are the data that move between stages. The answer key from Odin III runs under the whole pipeline to the score. Each circled number is the subsection of the Results or of the language-model section that tests the stage; the dashed box holds closed ideas, and the language-model harness is a side path that takes the gate netlist and, in one prompt, Loki's hints; a second arrow carries Loki's blocks to the model one block per prompt (Q-L10).

section of the Results reproduces the baseline and adds Loki's hints in arms B.

D. Benchmark designs and how they are cut

Loki uses two benchmarks. The first is the 68 small synthesizable RTL designs (counters, registers, arithmetic, ALUs, muxes, comparators, small state machines), which we call the *micro designs*. The second is the 156 reference RTL designs of the VerilogEval spec-to-RTL set of Liu et al. [7], the *VerilogEval references*.

A design can be written down as a netlist at several levels. We call the abstraction level and technology mapping at which a design is cut, and written as the netlist to reconstruct, a *floor* (f1-k6 and f1-earch, f2, f3; Section III describes each). A recovery method must not read the names that the designer chose. To test that, the *renamed-instance protocol* renames every identifier through one shared permutation, runs the pipeline, and maps the tokens back; equal results show that the pipeline does not use names.

Two further sets are built from these. A seeded generator makes 1200 labelled idiom designs in twelve families, with train, iid and ood splits; we call this the *synthetic corpus*. A table from a cell's cone class and WL colour to the construct it usually is, mined from the synthetic corpus, is the *signature dictionary*. The floors, the renamed-instance protocol, the corpus and the dictionary are our own constructions, so they have no earlier citation. The renamed-instance protocol is what the first Results argument runs, and the floors are what the later arguments compare.

E. Structure hypotheses and how they are scored

A plug-in that proposes structure hypotheses for a netlist and reports features of them, never a likelihood, is a *detector*. A structure hypothesis can be settled by a *prover*, an engine (cone check, truth table, transfer) that proves the hypothesis equal to the netlist; a proof makes it near certain. For the hypotheses that no engine settles, the probabilities come from *calibration*: fitting priors and likelihood ratios on a training split so that predicted probabilities match observed rates. The quality of a calibration is the *expected calibration error* (ECE), the average gap between predicted probability and observed match rate over ten bins. The ideas of calibration and ECE come from the machine-learning literature; their original sources are not in this paper's bibliography, so we only define them here. Detectors and provers are part of Loki. The Results use these definitions in the arguments on recovering kept and erased structure.

To score hypotheses against the answer key we need a rule for which are kept and which are right. A hypothesis is *counted* when it is committed or has probability at least 0.5, and a counted hypothesis *matches* an answer-key instance by an overlap rule (Section IV-A). The *precision* is the share of counted hypotheses that match, the *recall* is the share of answer-key instances matched, and the *F1* is their harmonic mean. The F1 averaged over the kinds of structure is the *macro-F1*, and the F1 pooled over all instances is the *micro-F1*.

Many structures in a netlist are several bits wide. A group of cells that acts as one multi-bit signal (a register, a datapath operator, a bus) is a *word*. The *operator type-F1* is the F1 of the recovered word-level operator types (adder, mux, comparator, ...) against the answer key's operators. A candidate that looks wrong but is in fact part of a real construct is a *fragment*. A detector that first groups flip-flops into register words by shared clock, enable and reset, and then builds the

rest around them, is a *word-first detector*. Related word-level work is WordRev [8], which derives word-level structures from bit-level gates; the word-first detector and the loss audit are our own, and the Results compare the word-first detector with a published method (below).

F. Graph methods and a published baseline

Loki's second layer treats the netlist as a graph. Community detection groups the nodes: *Louvain* is such a method, *modularity* scores how dense the clusters are, and *NMI* (normalized mutual information) scores the agreement of a clustering with the known module hierarchy. A graph neural network labels each node from its neighbours; this is a *GNN*. To judge it we compare it with the *1-WL baseline*, colour refinement, whose distinguishing power a message-passing GNN cannot exceed, a result of Xu et al. [9]. GNN-RE [10] applied graph networks to the reverse engineering of gate-level netlists. K-means [11] is one of the clusterings we compare. The graph argument of the Results asks whether any of these methods recovers the designer's hierarchy and gate roles.

HAL, the Hardware Analyzer of the Embedded Security Group [12], [13], is an open framework (MIT licence) for the reverse engineering of gate-level netlists. Its dataflow plugin implements *DANA* [14], a published method that groups flip-flops into register words from their shared control nets (clock, enable, set, reset) and from common predecessors and successors in the flip-flop dataflow graph, without reading the logic between them. DANA is therefore a word-finding method that is independent of the cell library once the flip-flops are known, which makes it the natural external baseline for the register words that Loki's word-first detector (Appendix A-I) targets. Appendix A-I runs both on the same netlists and scores both against the same Odin III answer key. We credit HAL and DANA in the README and the bibliography; no HAL code is part of this repository, only our gate-library files, a converter and the scorer.

Last, two paired tests compare methods on the same designs. The *McNemar test* is an exact test on discordant pairs, and the *Wilcoxon test* is a paired rank test. Both are standard tests, whose sources are not in this paper's bibliography. The language-model arguments of the Results use the McNemar test and the word-first argument uses the Wilcoxon test.

III. The method as it stands

The Introduction describes each innovation in plain English. This section describes them technically: the floors and the L0 datasets (including the renamed-instance protocol) and the L1 hypothesis flow, which are confirmed innovations, and L2 graph analysis, the operator-recovery score and the loss audit, which are candidate innovations (POSSIBLY GOOD, not contributions). The experiment that confirms or fails to confirm each one is summarised in Section V; the full records are in Appendix A.

A. Floors: the abstraction level of the netlist

A *floor* is the abstraction level and technology mapping at which an Odin III design is cut and written out as the L0 netlist, the object the reconstruction receives. Every floor starts from the same elaborated design, the common front `proc; fsm; opt; flatten; opt -full; wreduce` (`FRONT` in `loki/odin.py`), and then maps it with a floor-specific script (`floor_script` in the same file). Table I lists the floors. Four exist: `f1-k6` and `f1-earch` (one floor per FPGA architecture file; the directory name is `f1-` plus the architecture's lower-case prefix), `f2`, and `f3`.

What a floor's netlist contains decides what is left to reconstruct. At `f1` the flow's `hardmap` step has already recognised adders, multipliers and memories, so they appear as typed hard cells; the construct they implement is part of the netlist, and the labels still name it. At `f2` the same logic is only LUT truth tables: an adder is a carry chain of LUTs that has to be found again. At `f3` the logic is a network of two-input gates, the form Gate-Breaker's language model receives. The `f2` output does not depend on the architecture file, and `f3` reads none, so their files are byte-identical across architectures.

Loki uses several floors because what a synthesis flow erases depends on the mapping, and recovery has to be measured against each. `f1` keeps the structures the flow already reverse engineered, which is what lets the `hard` detector of L1 read them from Odin III's `unmap` facts; `f2` removes that crutch with the same source and the same answer key; `f3` makes the input comparable to Gate-Breaker's and is the floor on which the AST and CFG similarities are computed (Appendix A-B). Within one design the floors share one source of truth, the provenance spans of the same elaborated IR, so a difference between floors is a difference of the mapping.

B. L0: provenance ground-truth datasets

In plain English. The datasets are the test bed: netlists built from Verilog by a front end that remembers which source text made each gate, with every name hidden, so that the tool under test cannot cheat by reading names and its answers can be checked against the front end's record.

Technically. The first benchmark set is the 68-design micro suite (counters, registers, arithmetic, ALUs, muxes, comparators, small FSMs) that Odin III already compiles and checks against Yosys; the second, the 156 VerilogEval references, is described in Section IV-D. L0 compiles each design on every floor of Table I. Every netlist object carries its provenance (source file and line), a neutral `n`k name from one shared counter, and a label: the structure kind (`construct_type`) is the answer key in `labels.csv`, and the netlist-visible hard-block kind (`cell_type`) travels in the graph labels, so structures that are already reverse engineered stay visible to the reconstruction. `loki check` validates every dataset: 68 designs, 269 floors, a label row for every node, zero errors. The two hard-block designs the v1.0 front end refused now compile on the v1.1.0 SystemVerilog front end, so the set is complete.

Renamed instances. A separate protocol asks the naming question directly: rename every identifier of every netlist (105,646 identifiers across 269 netlists) through a shared permutation, run the pipeline on the renamed floors, and

TABLE I
THE FLOORS OF L0. ALL FLOORS RUN THE COMMON FRONT FIRST; THE MAPPING SCRIPT IS THE SECOND PART OF `FLOOR_SCRIPT`. NODE KINDS ARE THOSE OF `GRAPH.JSON` (`PI`, `PO`, `CONST`, `LUT`, `FF`, `HARD`, `GATE`).

Floor	Reads	Mapping script	Netlist contents
`f1-k6`	`read_arch k6_frac_N10_ frac_chain_mem32K_40nm`	`hardmap; lower; abc --lut 6`	LUT6, flip-flops, and hard adder, multiplier and RAM cells that keep their Odin III type
`f1-earch`	`read_arch` of the EArch file	same as `f1-k6`	same kinds, EArch hard blocks
`f2`	`read_arch` (cell types only)	`lower; abc --lut 6`	LUT6 and flip-flops only; arithmetic is soft, so no `hard` node exists
`f3`	`read_techlib lib/cmos6.o3lib`	`lower; abc --gates cmos6; dfflibmap cmos6`	library gates (BUF, INV, NAND2, NOR2) and DFF flip-flops

inverse-map every token back. The verifier reports token-for-token equality with the original, zero failures, and the base and renamed score digests are byte-identical (sha256 over all run scores, 204 of 204 pairs) – the pipeline is naming-neutral, so nothing it recovers is an artefact of a name.

C. L1: the Bayesian hypothesis flow

In plain English. The flow proposes "this group of gates is an adder, a register or a mux", gives each claim a probability learned from training designs, and settles the claims it can by an equivalence check.

Technically. L1 is a round-based hypothesis flow over one netlist, with no language model. It reads a floor netlist into a fresh Odin III design (level 0, never edited) in which every cell is a function, and keeps a working copy that grows one level of hierarchy per committing round.

Hypotheses and detectors. A hypothesis is a claim that a set of level-0 cells is a structure of some kind, for example: a flip-flop with enable or synchronous reset (`ff_en`, `ff_srst`), a `register`, a `bus` of isomorphic slices, an `adder`, `subtractor`, `comparator` or constant comparison (`eq_const`), a `mux` or `mux_tree`, a `counter`, `shift_reg`, `lfsr` or `fsm`, a datapath or control `unit`, a per-cell `role`, or a hard block (`hard_add`, `hard_mul`, `hard_mem`). Nine detector plugins (arith, bus, ff_pattern, fsm, hard, mux, role, seqfn, unit) propose them. A detector reports features as *bin labels* (a width of 5 to 8, whether an enable function is exact, the NPN class of a cell) and never a likelihood; the likelihoods come from calibration.

Bayesian inference. Each hypothesis h of kind k from detector d carries a log-odds $L(h)$, the logarithm of the odds that it is a real structure, and $p(h) = 1/(1 + e^{-L(h)})$ is its probability. L is a naive-Bayes sum: a prior plus evidence items, each the log-likelihood ratio (LLR) of an observation,

$$L(h) = \pi_{d,k} + \mathrm{clip}_{\pm 6.91}\Big(\sum_f w_{\phi(f)}\, \lambda_{k,f}(b_f) + \min\big(1.10\, n_{\mathrm{conf}},\, 3.30\big) - 2.30\, n_{\mathrm{ref}} \Big) + e_h, \quad (1)$$

where $\pi_{d,k}$ is the prior log-odds of that detector and kind; $\lambda_{k,f}(b_f)$ is the LLR of feature f in bin b_f; $w_{\phi(f)}$ weights the feature's family (structural, WL colour, NPN class or L2; the `f3` calibration uses the first three only); n_{conf} and n_{ref} are the numbers of confirmed (or committed) and refuted child hypotheses (a bus slice, a register's flip-flops), which raise and lower their parent; and e_h collects fixed records (a sampled check adds $\ln 100 = 4.61$). The clip at $\pm \ln 1000 = 6.91$ limits the double counting that a naive-Bayes sum of correlated features would otherwise allow. A proof sets $L(h) \geq \ln 9999 = 9.21$ (the odds become 9999 to 1), and a counterexample sets $L(h) = -\infty$.

All three tables are *calibrated* from the training split of L0 (`loki calibrate`). The flow runs with default tables on every training design; each hypothesis is labelled true when it matches an L0 construct (Section IV-A). The prior is the Laplace-smoothed precision of the detector, $\pi_{d,k} = \mathrm{logit}\,\frac{tp+1}{n+2}$ over n proposals of which tp are true. For example, the `f3` comparator detector made 31,036 proposals of which 2 were true, so $\pi = \ln \frac{3}{31035} = -9.24$; the committed value is -9.244259 (`experiments/f3-recovery/calib/f3.json`). A bin's LLR is $\lambda = \ln \frac{(pos+1)/(POS+B)}{(neg+1)/(NEG+B)}$, with pos and neg the true and false proposals in the bin, POS and NEG the totals of the kind, and B the number of bins. The family weights w are the grid values in $\{0, 0.1, \ldots, 1\}$ that maximise the Bernoulli log-likelihood of the labels, fitted by three sweeps of coordinate ascent on out-of-fold rows (two folds, split by design). Without calibration (`--calib none`) the priors are 0 and the LLRs are the defaults of `bayes.py`. The reliability check is the expected calibration error over ten equal-width bins of the predicted probability, $\mathrm{ECE} = \sum_b \frac{n_b}{N} |\bar{p}_b - r_b|$ with r_b the observed match rate.

Proof engines. A hypothesis with a functional claim ("these 9 LUTs compute $A + B$ at width 5") can be proved. Three engines prove claims: `cone_check` first when the claim has a template, the truth table otherwise or when the cone cannot be bound to the template, and transfer after an exact proof. (i) `cone_check`, an Odin III pass, proves that the claim's cone of level-0 cells is equivalent to a template word cell (for example `$add`, `$sub`, `$eq`, `$mux`) with its ports bound to the cone's boundary nets, under a timeout; the reference functions of the claims (`claims.py`) cover addition, subtraction, the six comparisons, constant comparison, 2:1 and N:1 multiplexers, counters, shift registers and LFSRs. (ii) The *truth-table* prover enumerates the claim's inputs and side nets when together they number at most 16, an exact proof or a counterexample, and otherwise simulates 4096 seeded

random vectors, which can refute a claim but only raises it by $\ln 100$. (iii) *Proof transfer* groups open hypotheses by operation, parameters and the 1-WL colour of their output drivers; when one member is proved exactly, the others are tried by a lockstep `cone_match` of the two cones, and one transferred member per group is proved for real as a spot check, revoking the transfers of the group if it fails. A sequential claim is proved as a next-state function (state in, next state out); an FSM gets no proof.

Deciding the structure. Hypotheses that claim the same cells compete. Two hypotheses *conflict* when both kinds are exclusive, their cell sets intersect, and neither is a strict subset of the other (a strict subset is nesting: the smaller becomes a child of the larger). For each connected component C of the conflict graph the competitive probability is

$$p_{\mathrm{comp}}(h) = \frac{e^{L(h)}}{1 + \sum_{j \in C} e^{L(j)}}, \tag{2}$$

where the "1" is the option that none of them is real; a hypothesis without conflicts has $p_{\mathrm{comp}} = p$. Two confirmed hypotheses with equal cells are two true forms of the same logic and count as one member of C; the one whose kind ranks higher (a counter above an adder, for example) wins at commit and the other is kept as an *alternative*. A proof is spent on hypotheses with $p_{\mathrm{comp}} \geq 0.2$, most probable first, at most 200 per round; a hypothesis below 0.01 is rejected, and re-enters when later evidence lifts it. A hypothesis is *selected for commit* when $p_{\mathrm{comp}} \geq 0.9$, every child that could commit is already committed, and it does not conflict with a committed piece. Selection is greedy in the order (p_{comp} descending, kind rank, cell count descending, id), and the losers of a conflict become rejected or alternative.

Commit and the final structure. A selected hypothesis is rebuilt on a copy of the working design: a hypothesis is *raised* to a single word cell only if its kind is one of adder, subtractor, comparator or multiplexer, its claim has a `cone_check` template, and it is proved by `cone_check` (directly or by transfer); the cone is re-checked by `cone_check` unconditionally before the raise. A claim proved only by truth table is wrapped, not raised, and counters, shift registers, LFSRs, FSMs, registers, constant comparisons, multiplexer trees and carry-in adders are never raised. Every selected piece except buses and roles (which only set attributes) is wrapped by `submod` in a module named for its kind, whose instance is tagged with the hypothesis id, kind, confidence p_{comp}, proof status and the recovered function. If `raise` or `submod` fails, the offending hypothesis is rejected with a record of the error and the commit is retried once without it (`reject_failed`). The structure that L1 reports is therefore exactly the set of committed pieces in `design.v`; the alternatives, rejected and open hypotheses stay in `overlay.json` with their evidence. Algorithm 1 summarises the loop. It stops when a round commits nothing, proposes nothing new and changes no log-odds by more than 10^{-6} (a fixed point), after eight rounds, or at the time budget, then flattens and lowers the final design and checks it against the input with `equiv-check`; a result of `differs` is a bug and the run exits with an error.

Algorithm 1 The L1 round loop (`loki/re/loop.py`).

```
level_0 ← read netlist as functions; work ← level_0
if floor f1 with hard cells then run unmap on work
for r = 1 to 8 do
    propose: every detector adds hypotheses over the view of work
    score: L(h) by Eq. 1; compete: p_comp by Eq. 2
    prove claims with p_comp ≥ 0.2: cone_check, truth table, transfer
    re-score and compete; reject p_comp < 0.01
    select p_comp ≥ 0.9 (children first); commit by raise and submod
    if nothing committed, nothing new, |ΔL| ≤ 10^-6 then
        break
    end if
end for
write design.v, overlay.json, report; equiv-check against the input
```

The `hard` detector reads the `unmap` pass's facts: a hard adder, multiplier or memory that the flow already reverse engineered is recovered directly, with the decline reasons the pass recorded (carry-in chains and the like) kept as open hypotheses. The `hard` detector does not run on `f2` or `f3`.

D. L2: graph analysis

L2 reads the L0 `graph.json` files and nothing else: never `key.json` and never the BLIF. It has three parts.

Structural features. Each node gets numeric features (fan-in, fan-out, logic level, LUT width, hard-port count), its kind and, on hard cells, its `cell_type`, plus three families: colours of 1-WL colour refinement (a hash of the node's label and the multiset of its neighbours' colours, per round, so that equal colours mean equal unfolded neighbourhoods), the NPN class of a LUT's function (invariant under permutation and negation of the inputs and the output), and the ORCA orbit counts of the 4-node graphlets around the node. Before graphlets are counted, the global signals (clock, reset, constants, global enables) are classified by role and their edges removed, so that a clock does not connect everything.

Clustering. Four methods partition a netlist graph into candidate blocks: k-means on the feature space ($k = \mathrm{clip}(\mathrm{round}\sqrt{N/2}, 2, 200)$), affinity propagation on a sample of at most 3000 nodes with the remaining nodes assigned to the nearest exemplar, spectral clustering (a spectral embedding of the largest connected component, then k-means), and Louvain community detection (resolution 1, no k). The first two use node features; the last two use the graph structure. A partition is scored by its *modularity* on the undirected simple projection of the graph, by its *NMI* against the known `module_path` and `construct_type` labels and by its *ARI* against `module_path` only over the LUT, flip-flop, hard and gate nodes. These two numbers are a quality report, never an input to a method: modularity says whether the clusters are dense subgraphs, NMI whether they are the designer's modules.

GNN against 1-WL. A node classifier predicts each node's `construct_type`: a two-layer GraphSAGE (GCN as the variant) of hidden width 64 with ReLU and dropout 0.2 over the graph and its reverse edges, trained with Adam (learning rate 0.01, weight decay $5 \cdot 10^{-4}$) for a fixed 100 epochs, full-batch, with class weights $1/\sqrt{\text{count}}$. The split is L0's, by design, with the class vocabulary taken from the training designs only (rarer values, and values first seen in a test design, are one class `__other__`); there is no validation set and no early stopping, so nothing is tuned on a test design. Loss and metrics cover only the LUT, flip-flop, hard and gate nodes with a known target. The question is whether message passing earns anything over what the colour refinement already knows, because a message-passing GNN is at most as discriminating as 1-WL [9]. So the 1-WL baseline predicts, for each node, the most common training class of its WL colour (backing off from round 4 to round 0 when a colour has fewer than five training nodes), and the GNN has beaten it only if its test macro-F1 is higher. Two further baselines are reported on the same nodes, the majority class and a logistic regression on the features alone. The task is that of GNN-RE [10], which maps each gate of a flat netlist to the module it belongs to; Loki's model is a baseline for the same task, not a re-implementation, and the numbers are not compared with that paper's.

E. Operator type-F1 and the loss audit

In plain English. Comparing recovered Verilog text with the reference cannot tell whether the right adders were found, so we also count the operator types that were recovered. The loss audit asks a different question: when synthesis merges the logic of two source statements, the construct can no longer be pulled apart, and the audit records which pass did the merging, using the provenance map that the front end already keeps.

Technically. The L0 key lists the word cells of the elaborated reference. A run's recovered operators are its counted hypotheses, mapped to an operator class and an operand width (add, sub, mul, eq, ne, lt, le, gt, ge, mux, register). Precision and recall are multiset overlaps, so the score does not depend on which cells carry an operator; `type` ignores the width and `tw` requires it. The loss audit reads the D14 provenance map after each pass of the Odin III flow (proc, fsm, opt, flatten, opt `-full`, wreduce, lower, abc) and tracks units: a unit is separable at a snapshot when some object holds exactly that unit set, and lost at the first later snapshot where none does (Appendix A-J gives the protocol).

F. L3: patterns, anchors, growth and join

L3 is the third level of Loki. L1 proposes single constructs from the facts of one synthesis pass. L3 asks for the blocks of a design: which cells of the netlist form a state machine, a counter, a shift register, a register file or a plain register. The motive is that the failures of the model baseline (`Prob151_review2015_fsm`, `Prob156_review2015_fancytimer`) are compositions: a sequence-detector state machine, a shift register that loads, a prescaler, a down counter and a handshake. The specification is `docs/specs/2026-10-30-l3-patterns-design.md`. A *cell* below is one synthesis tool and target pair (Odin III or Yosys; gates, four-input LUTs, or six-input LUTs with or without hard blocks).

Signatures. The Odin III `graphlets` pass counts, for every netlist cell, the graphlet orbits it takes part in (73 undirected, 129 directed). The directed counts of a block of cells form its signature. A library of 700 generated circuits (counters, shift registers, LFSRs, registers, timers, register files, state machines and glue, every width from 1 to 32) in the five cells gives the signature of each pattern at each width; signatures of wider circuits are made on demand. The earlier *bounded-cone detector*, *cones* below, cuts a cone of bounded depth around a group of flip-flops and compares its signature with that library.

Rule detectors (v0). A rule detector reads the level-one claims and the netlist through a JSON catalogue with one signature per target. For example, on the hard-block target a counter is a hard-adder claim whose sum bits loop through flip-flops back to the adder with a one-bit or constant operand. The detectors work for some patterns and not for others (Section V).

Anchor and grow. Anchors are the places where a known pattern is easy to see: a group of flip-flops that feed only each other, a shift chain, a triangular counter, a decoder-selected memory, a register, an adder chain. Three more anchor classes cover constructs that no block touches: flip-flops that share the same global control, open feedback loops of flip-flops, and the constant cells (one anchor that does not grow). Every anchor grows at once, one neighbouring cell per step. Each block is grown to the end of its path (the frontier is empty or the block has 300 cells); no tested stop rule did better than this.

Join. A join step then looks at every pair of touching blocks and merges them when at least a fifth of the link ends of the smaller block lie on links to the other (a conductance rule; the threshold 0.2 was chosen on validation designs among four rules). Merging is greedy and capped at 50 merges per design and cell. A split step (cut a block along the Fiedler vector of its cell graph when two parts explain it better than one) is part of the code but is not in the headline: no setting gained (QL9-d).

Kind guess. Each block carries a probability over five construct kinds (finite-state machine, counter, shift register, register file, register) and glue. The kind likelihood used is the *Gaussian kind model*: a full-covariance Gaussian per kind on 24 principal components of features that include orbit instances per cell, how the block touches the rest of the netlist (conductance, frontier fan-in and fan-out), the share of flip-flops, hard and adder cells, and the feedback structure of the block. It is fitted on answer-key blocks of held-out designs only.

Scoring. The answer key supplies the constructs. A block is right for a construct when it covers at least 0.8 of the construct and the Jaccard overlap is at least 0.5. The headline measure is the share of constructs for which some block is right. Per design and cell, the geometric mean of $(g+1)/(d+1)$

compares right counts g of one method against d of another, with a 90 percent bootstrap over designs, and the counts of designs where one method wins, loses or ties (W / L / T) come first, as in the other chapters of the CAD literature.

IV. Methodology

The setup is: the floors and the two benchmark sets (Section IV-D), the scoring of L1 and L2 against the answer key (Section IV-A), the metrics and the statistics (below), and the language-model baseline of Gate-Breaker. All experiments except the language-model pilot and the full run use no language model.

A. Evaluation protocol: how L1 and L2 are scored

Gate-Breaker evaluates a stochastic model: for each design the model is sampled K times, each sample's RTL is extracted and simulated against the VerilogEval testbench, and the results are aggregated per complexity quartile as the success rate (the fraction of the K samples that pass), Success@k (the fraction of designs with a passing sample among the first k), and the AST and CFG similarity of the recovered RTL to the reference. Loki's L1 and L2 are deterministic and their output is a structure, not an RTL text, so the same four questions are asked in different terms. Table II sets the two side by side.

L1, step by step.

1) *Split.* L0 assigns each design to train or test by a hash of the design name ($\mathrm{sha256}(\text{seed:design}) \bmod 100 < 80$ is train), so the split is fixed by the name and independent of the run.
2) *Calibrate* on the training designs of one floor (`loki calibrate`): the priors, bin LLRs, family weights and reliability curve of Section III-C.
3) *Run* `loki re` on every test netlist, calibrated and, as a diagnostic, with `--calib none`. The run never reads the answer key.
4) *Ground truth.* From `words.json`, the L0 word cells give ground-truth instances (type, F_g) with F_g the set of final cells of the word; counters, mux trees and FSMs are derived from the word connectivity.
5) *Count and match.* A hypothesis *counts* when it is committed or its $p_{\text{comp}} \geq 0.5$ (`--score-p`); it *matches* a ground-truth instance of an accepted type when the Jaccard of their cell sets is at least 0.5, greedily one-to-one by Jaccard. Precision is matched over counted, recall matched over ground truth, and average precision sweeps the threshold from 0.1 to 0.9.
6) *Buses, roles and hierarchy.* A bus matches a ground-truth bus (nodes sharing a module path and bus name, at least two bits) at the same Jaccard; its bit order is scored exactly and up to reversal. Roles are scored as precision and recall of the predicted control and datapath cells. The hierarchy similarity compares the clusters of the module paths to the committed pieces with the Dice score $F(g,h) = 2|g \cap h|/(|g| + |h|)$, $\mathrm{HS}_{gt} = \sum_g |g| \max_h F(g,h) / \sum_g |g|$ and its converse HS_{l1}, and reports their harmonic mean.

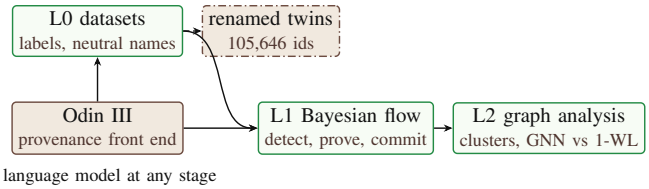


Fig. 2. The Loki III pipeline. L0 compiles the benchmarks through Odin III into provenance-tracked, labelled, anonymised datasets (and their renamed twins). L1 proposes and proves structures in a Bayesian flow that commits into the same IR. L2 reads the graphs back and scores clustering and a GNN against a 1-WL baseline. No stage has a language model.

7) *Aggregate.* `loki score` writes one `score.json` per run and `loki score-all` the floor-by-kind tables, the reliability curve of p_{comp} with its ECE, and, with `--compare`, the ablation against `--features none`.
8) *Naming neutrality.* The renamed twin of the dataset is run and scored against the original-named key; the base and renamed `score.json` must be byte-identical.

Of the Gate-Breaker metrics, the AST and CFG similarities are used on `f3` as in the paper: AST similarity is the mean, over the subtrees of the reference, of the best Jaccard similarity to a subtree of the candidate (depth 3, node kinds normalised); Gate-Breaker computes both on functionally correct samples only, Loki's are unconditional; CFG similarity is the node kernel over shortest-path histograms of the control-flow graphs. They are computed twice per design, for the gate netlist (L0-stop) and for `design.v` (L1-stop), and the L1 delta is their difference. The success rate and Success@k need samples; a deterministic flow has one attempt per design, so they are reported only for the language-model replication (Section VI-A). Formal `final_equiv` does not play the role of the testbench: it compares the flattened final design with the *input netlist*, so it shows that the rewrite preserved the netlist (it is trivial on `f3`, where L1 commits nothing), not that the design is correct against the reference RTL. Loki has no behavioural pass rate against the reference.

L2, step by step. (1) The same split applies. (2) `loki analyze cluster` writes a partition per method and design, with its modularity and NMI/ARI against `module_path` and `construct_type`; the table reports the mean over designs. (3) `loki analyze gnn` fits the vocabulary and the model on the training designs and writes `metrics.json`: accuracy and macro-F1 on the training and test nodes for the GNN, the majority class, the feature-only logistic regression and the 1-WL baseline, and `beats_wl`. (4) The neutrality check re-runs both on a permuted copy of every identifier.

B. One design through the pipeline

The steps below follow one design, the 4-bit comparator micro `bm_DL_4_bit_comparator` (three comparison outputs), from source to score. Each step names the file or code that defines it.

TABLE II
THE GATE-BREAKER EVALUATION AND THE LOKI EVALUATION, STEP BY STEP.

Step	Gate-Breaker	Loki
Input	cmos6 gate netlist of a VerilogEval reference, internal nets renamed n_⟨SHA-1 prefix⟩, ports kept	a floor netlist (`f1`, `f2`, `f3`) with names nk; for the neutrality check a renamed twin
Producer	a language model, K samples per design	the L1 flow, one deterministic run per design (fixed seed); no sampling
Correctness	simulate the extracted RTL against the testbench: success rate, Success@k	`equiv-check` of the flattened final design against the *input netlist*: `final_equiv` (formal; on `f1` hard cells are cut points and only soft logic is checked). A self-consistency check of the rewrite, not correctness against the reference RTL; Loki has no behavioural pass rate against the reference
Match to the truth	AST and CFG similarity to the reference RTL	each counted hypothesis against the L0 answer key: Jaccard ≥ 0.5, one-to-one; precision, recall, F1, average precision per kind and floor; hierarchy similarity; AST and CFG of `design.v` against the reference on `f3`, unconditionally (Gate-Breaker's are computed on functionally correct samples only)
Aggregate	per complexity quartile	micro-averaged per kind and floor over the test split; AST and CFG per quartile of `f3` node count

1) *HDL to Odin III with provenance.* `read_verilog` and the common front elaborate the design. Every IR object keeps its source spans (`Node.sources()`, the only provenance the binding exposes). Before mapping, Loki snapshots every word-level cell (type, parameters, spans) and every instance site.
2) *Map to the floor.* `lower` keeps each word cell's exact span on every gate it creates; `abc -lut 6` gives each LUT the union of the spans of its cone. A LUT that merges the three comparators therefore lists all three spans.
3) *Join and label.* A final cell is matched to a word cell when the word's span set is a subset of the cell's. The primary match gives `construct_type` (here `$eq`, `$gt`, ...); a cell with several matches is flagged (`n_construct_cells` above one) and a cell with none stays `unknown`: a label is an answer key, never a guess. The module path comes from the instance spans, bus name and bit index from net names and aliases (`AeqB[0]`), and the datapath or control role from a rule table.
4) *Anonymise and write the L0 files.* Nets, cells and port bits get names nk from one counter. Loki writes `netlist.blif` and `netlist.v`, `graph.json` (nodes with features and labels, edges), `labels.csv` (one row per node), `words.json` (the word-level connectivity the scorer needs) and `key.json`. Only `key.json` holds original names; no analysis reads it.
5) *Verify.* Odin III's `equiv-check` compares the written BLIF with the floor's own netlist and the anonymised netlist with the original; both must read `equivalent` before the floor counts as built.
6) *L1: normalise.* `loki re` reads `netlist.blif` (never `key.json`) into a fresh Odin III design, level 0, in which every logic cell is a function (a LUT by its truth table, a gate by its library function). On `f1` with hard cells the `unmap` pass runs on the working copy and its facts feed the `hard` detector.
7) *L1: hypothesis rounds.* Each round runs the detectors, scores every hypothesis (Section III-C), proves the ones worth a proof, resolves conflicts and commits the confident ones (Algorithm 1); the next round sees the committed level.
8) *L1: commit and output.* A committed hypothesis is rebuilt as a word cell (`raise`) wrapped in a module (`submod`) tagged with its kind, confidence and recovered function. At the fixed point Loki writes `design.v`, `overlay.json` (every hypothesis with its evidence) and `report.json`, and checks the final design against the input with `equiv-check`.
9) *L2.* `loki analyze` reads `graph.json` only: structural features, four clusterings and a GNN (Section III-D).
10) *Score.* `loki score` compares the overlay with the answer key (Section IV-A); `loki score-all` aggregates every run into per-kind tables.

In this micro the `f3` comparator, as re-synthesised by the minimal `.v` front, has 26 logic cells (L1 spec, Section 3); the `f2` LUTs carry the comparator spans that the join reads back.

C. Metrics and statistics

Recovery metrics. Precision, recall and F1 (micro and macro) of the counted hypotheses against the answer key; the expected calibration error (ECE); AST and CFG similarity and the operator type-F1 of recovered text against the reference; modularity and NMI of a clustering; the functional pass rate of recovered RTL; and the share of units lost at a pass. All are defined in the Glossary and in Section IV-A.

Area, speed and power. Loki's measures are recovery measures. No experiment of this paper measures area, speed (critical-path delay or near-critical paths) or power. We do not claim that the recovery scores correlate with any of them, and we cite no work that says so; they are not measured here.

Statistics. Paired comparisons over designs use the Wilcoxon signed-rank test; paired pass/fail comparisons use the exact two-sided McNemar test on discordant pairs; intervals are bootstrap intervals (clustered over designs where the designs come in families); the null models (random cell sets

of the same size, random merges) are named where they are used. A result is called a difference only when the interval or the exact test shows one; otherwise it is INCONCLUSIVE.

D. The VerilogEval benchmark and its structure

1) VerilogEval spec-to-rtl as the Gate-Breaker benchmark: Gate-Breaker evaluates on VerilogEval [7], taking the spec-to-rtl variant: 156 problems, each a natural-language prompt, a reference solution (`*_ref.sv`, module `RefModule`) and a testbench (`*_test.sv`). The Gate-Breaker flow synthesizes the reference to a cmos6 gate netlist, hands the netlist to the model, and checks the model's RTL against the testbench. Loki uses the same 156 references as L0 sources and reads the reference module as the design; the prompts and testbenches are not inputs to L0, L1 or L2. The references are SystemVerilog, read through Odin III's slang adapter (Appendix A-A).

2) The complexity score and its quartiles: Gate-Breaker groups the problems by a complexity score (its Eq. 2), a weighted sum of counts of HDL constructs:

$$C = 1000\,c_{\mathrm{Seq}} + 200\,(c_{\mathrm{FG}} + c_{\mathrm{F}}) + 20\,c_{\mathrm{Mo}} + 15\,(c_{@A} + c_{\mathrm{B}}) + 10\,(c_{\mathrm{M}} + c_{\mathrm{Ca}}) + 5\,c_{\mathrm{E}} + 3\,c_{\mathrm{A}} + 2\,c_{\mathrm{C}}. \quad (3)$$

The paper names the terms but does not define them, so the implementation we reuse (`complexity.py` of the second Gate-Breaker replication) fixes the reading: c_{Seq} counts clocked `always` blocks (a `posedge` or `negedge` in the sensitivity list), c_{FG} for-generate constructs, c_{F} `for` loops, c_{Mo} module instantiations, $c_{@A}$ all `always` blocks, c_{B} declarations wider than one bit, c_{M} multiplexers (ternary operators plus non-default `case` items), c_{Ca} `case` statements, c_{E} arithmetic operators (`+ - * / %`), c_{A} continuous assignments and c_{C} concatenations. A clocked block weighs 1000 on its own, so a design in which the counter recognises one scores at least 1000, which is Q3 or Q4.

The quartiles are the paper's absolute bins (its Table II), not rank quartiles of the benchmark: Q1 is 3 to 23, Q2 24 to 111, Q3 112 to 1064 and Q4 1065 to 1326. The 156 references fall 38, 39, 40 and 39 into Q1 to Q4, close to the paper's 39 per quartile. The 150 designs that build on every floor are split 38, 39, 37 and 36: the six missing ones are `Prob031`, `Prob046` and `Prob145` (Q3) and `Prob078`, `Prob151` and `Prob156` (Q4). Table III gives the counts.

Deviation: pyverilog AST against a regex fallback. The score is computed on the pyverilog AST of the first module when the file parses, and by regular-expression counts of the same terms otherwise, with the method recorded per design. Because the references are SystemVerilog, pyverilog parses only 60 of the 156; the other 96 use the regex counts. The split is skewed by quartile (Table III): 36 of Q1 and 23 of Q2 are AST-scored, but 1 of Q3 and none of Q4. The regex counts operator characters and keywords textually, so they differ from an AST count (the observed Q4 range, 1069 to 3259, exceeds the paper's bin top of 1326 for this reason or because the problems differ; we do not separate the two). The regex also never counts `always_comb`, `always_ff` or `always_latch` blocks, which 34 references use, so `c@A` and `cSeq` are undercounted in the 96 regex-scored designs, most of Q3 and Q4; the quartile split and the feature trends below rest on approximate scores. The 156 scores equal those of the replication that computed them first, exactly, so the quartiles here and in Table XII are the same partition.

3) The feature matrix: Gate-Breaker characterises its benchmark by the HDL structures it contains. We build the same kind of matrix for the 150 built designs, from the `f3` floor of each, with rules fixed in advance (`summarize.py`): *arithmetic* counts the word cells of type `$add $sub $mul $div $mod $neg $pos $pow $alu $macc`; *select/mux* those of type `$mux $pmux $bmux $demux`; *storage* the register words (`$dff`, `$dffe`, `$adff`, `$adffe`, `$sdff`, `$sdffe`, `$sdffce`, `$dffsr`, `$aldff`, `$dlatch`, `$adlatch`); *FSM* is 1 when Odin III's `fsm` pass marked any word; *control* is 1 when any label row has role `control` or `mixed`; and *wide buses* counts the distinct bus names with a bit index of at least 1. The matrix uses one symbol per cell: ✗ for count 0, ✓ for one, ✓✓ for two or more; FSM and control are 0/1 flags, so they show ✗ or ✓ only.

The bus column is `buses_wide`, not `buses`. The latter counts every named signal, including single-bit ports, so it is at least 1 for all 150 designs and at least 2 for 147, and would put a check mark in every row. The wide count is the quantity a "multi-bit bus" row means; both columns are committed in `features.csv`.

Table IV shows 12 designs, three per quartile at fixed ranks by size, so the sample is not hand-picked; Table V counts, per quartile, the designs that show each feature. The storage trend is partly tautological: a clocked block adds 1000 to the score (Eq. 3) and storage counts register words, so storage rises with the quartile by construction; the other trends are conditioned on the score as well. Storage is absent from Q1 and Q2 (0 of 77 designs) and present in 31 of 37 Q3 and all 36 Q4 designs. The other features grow with the quartile: select/mux 8% in Q1 to 56% in Q4, control 3% to 83%, FSMs 0 of 77 below Q3 and 15 of 36 in Q4. Arithmetic is rare throughout (21 of 150, 14%), so the benchmark leans on control, selection and storage more than on arithmetic datapaths. Wide buses are common from Q2 up (90%, 59%, 100%) but nearly absent in Q1 (8%), whose members mostly have only single-bit signals. The FSM flag is Odin III's detection, not a reading of the source: `Prob107_fsm1s` in Table IV has storage and control cells but no FSM mark.

4) Benchmark size per technology mapping: Table VI gives the size of the 150 designs on each floor. The averages hide a heavy tail. Over all designs the mean node count is 163.5 on `f1-k6`, 111.4 on `f2` and 362.0 on `f3`, but the medians are 14.0, 13.5 and 25.0: most references are small (125 of 150 have at most 100 nodes at `f3`) and a few are large (9 exceed 1000; the largest, `Prob144_conwaylife`, has 15,487). The gate mapping multiplies the node count of the LUT mapping by about 3 on the mean and 1.9 at the median (362.0 against 111.4 on average, 25.0 against 13.5 at the median), as expected when a LUT of up to six inputs becomes

TABLE III
THE VERILOGEVAL SPEC-TO-RTL REFERENCES BY GATE-BREAKER COMPLEXITY QUARTILE (ABSOLUTE BINS OF TABLE II; SCORE RANGE OBSERVED IN THE 156 REFS; PYVERILOG AST SCORES AGAINST REGEX-FALLBACK SCORES; BUILT = DESIGNS WITH ALL THREE FLOORS).

Quartile	Bin	Observed	Refs	AST/regex	Built
Q1	3–23	3–23	38	36/2	38
Q2	24–111	24–90	39	23/16	39
Q3	112–1064	120–1060	40	1/39	37
Q4	1065–1326	1069–3259	39	0/39	36
All			156	60/96	150

TABLE IV
BENCHMARK FEATURE MATRIX, A SAMPLE OF 12 OF THE 150 DESIGNS (THE DESIGNS AT RANKS $\lfloor n/6 \rfloor$, $\lfloor n/2 \rfloor$ AND $\lfloor 5n/6 \rfloor$ BY F3 NODE COUNT INSIDE EACH QUARTILE). ✗: FEATURE ABSENT; ✓: ONE; ✓✓: MORE THAN ONE; FSM AND CONTROL ARE 0/1 FLAGS. FEATURES ARE READ FROM THE F3 FLOOR (SECTION IV-D3).

Design	Q	f3 nodes	Arith	Sel	Stor	FSM	Ctl	Bus
Prob028_m2014_q4a	Q1	3	✗	✗	✗	✗	✗	✗
Prob036_ringer	Q1	8	✗	✗	✗	✗	✗	✗
Prob091_2012_q2b	Q1	17	✗	✗	✗	✗	✗	✓
Prob125_kmap3	Q2	15	✗	✗	✗	✗	✗	✗
Prob015_vector1	Q2	32	✗	✗	✗	✗	✗	✓✓
Prob123_bugs_addsubz	Q2	166	✓✓	✓	✗	✗	✗	✓✓
Prob107_fsm1s	Q3	11	✗	✓	✓	✗	✓	✗
Prob117_circuit9	Q3	28	✓	✓	✓✓	✗	✓	✓
Prob045_edgedetect2	Q3	89	✗	✗	✓✓	✗	✗	✓✓
Prob128_fsm_ps2	Q4	30	✗	✗	✓	✓	✓	✓✓
Prob140_fsm_hdlc	Q4	52	✗	✗	✓	✓	✓	✓
Prob115_shift18	Q4	1121	✗	✓✓	✓✓	✗	✓	✓✓

a network of cells from a library of one- and two-input gates (BUF, INV, NAND2, NOR2). `f1-k6` sits above `f2` on average and in Q3 and Q4 (109.4 against 77.4; 453.9 against 269.8) while the two coincide in Q1 and Q2 (6.3 and 6.1; 100.2 and 100.0): the `f1-k6` floor keeps hard cells (21 designs hold hard adders) where `f2` spreads arithmetic over LUTs, which is consistent with the excess but is not isolated here. By quartile on `f3`, the mean grows from 9.9 nodes in Q1 to 972.1 in Q4, but the Q4 median is only 51.5: the complexity score tracks the number and kind of HDL constructs, not the netlist size. For scale, the micro suite's `f3` floors hold 50,047 nodes over 65 designs, a mean of about 770.

V. RESULTS: WHAT THE EVIDENCE SAYS NOW

Each result in this section is one argument with seven parts, always in the same order: the **Hypothesis** (what we claim), its **Implication** (why it matters if true), the **Experiment** (what is run and against what), the **Methodology** (the measure, the split and the statistics), the **Measurements** (the numbers), the **Interpretation** (what the numbers mean) and the **Conclusion** with one verdict word (Section I defines the four words). The arguments are ordered so that each builds on the ones before it. How we came to each experiment, and the fix rounds on the way, are in the appendix records (Appendix A) and not here. The Implication parts are our own reasoning about use, not measured results.

Each argument ends with the *register rows* it rests on, in square brackets. A register row is one line of the research log (Appendix C) with the hypothesis, data, measure and verdict of one experiment; its short id (for example L0a) is a pointer and not a concept. Table VII lists the results in the order of this section.

A. Names carry nothing, and the answer key covers every gate

Hypothesis. The pipeline uses no names, so scrambling every identifier changes no answer; and the provenance answer key labels every gate node of the benchmark designs.

Implication. If both hold, every later recovery score measures structure and not name guessing, and the key is a complete target. If either failed, no number in this paper could be trusted.

Experiment. We run the pipeline on the netlists of the 150 designs that build, once with the original names and once with every identifier renamed through one permutation (the renamed-instance protocol), and compare the two runs. We separately count the gate nodes that carry a label in the answer key.

Methodology. The measure for names is byte identity of the L1 scores and of the L2 clusterings between the two runs; for the key it is the share of gate nodes with a label, a cell type, an HDL span and a module. There is no split: both are checks of the dataset, not of a trained model.

Measurements. The key covers all 54,306 gate nodes of the 150 designs (module coverage 1.000). Renaming 105,646 identifiers gives 476 byte-identical run pairs; the L2 clusters are identical on 271 of 272 partitions, the exception being a nondeterminism of spectral clustering that we report. The protocol also exposed one defect of our own rename script, which removed every hard-block proposal on renamed floors and is fixed.

Interpretation. No answer depended on a name, so a score is a score of structure. The one spectral exception is a nondeterminism of one method, which we report, and not a use of names.

Conclusion. The benchmark has a complete key and neutral names. **USEFUL** [L0a, L0b; Appendix A-A].

B. Recovering what synthesis kept, and honest probabilities

Hypothesis. The Bayesian flow recovers almost perfectly the structures that synthesis already reverse engineered (hard adders and multipliers), and fitting priors on training designs makes its stated probabilities honest.

Implication. A tool that reports a probability is only useful if the probability can be believed; recovering the structures synthesis kept is the base on which anything harder stands.

Experiment. We run the flow on the micro designs at the floor that keeps hard blocks, scoring its hard-adder and hard-multiplier claims against the key, and we run it with and without the fitted priors and likelihood ratios.

Methodology. Precision and recall of the claims, and the expected calibration error over ten bins, with the fit done on

TABLE V

DESIGNS SHOWING EACH FEATURE, PER COMPLEXITY QUARTILE: COUNT (PERCENT OF THE QUARTILE). THE LAST COLUMN COUNTS DESIGNS WITH TWO OR MORE INSTANCES OF THE FEATURE (ALWAYS 0 FOR THE 0/1 FLAGS). WIDE BUSES ARE DISTINCT MULTI-BIT BUSES; A DESIGN WITH ONLY SINGLE-BIT PORTS HAS NONE.

Feature	Q1 (n=38)	Q2 (n=39)	Q3 (n=37)	Q4 (n=36)	All (n=150)	≥ 2
Arithmetic	2 (5%)	4 (10%)	7 (19%)	8 (22%)	21 (14%)	9
Select/mux	3 (8%)	10 (26%)	12 (32%)	20 (56%)	45 (30%)	12
Storage	0 (0%)	0 (0%)	31 (84%)	36 (100%)	67 (45%)	12
FSM	0 (0%)	0 (0%)	1 (3%)	15 (42%)	16 (11%)	0
Control	1 (3%)	4 (10%)	15 (41%)	30 (83%)	50 (33%)	0
Wide buses	3 (8%)	35 (90%)	22 (59%)	36 (100%)	96 (64%)	55

TABLE VI

BENCHMARK SIZE PER TECHNOLOGY-MAPPING FLOOR, 150 DESIGNS: MEAN / MEDIAN NODE COUNT (TOP), AND PER COMPLEXITY QUARTILE (BOTTOM; NODES). NODES ARE ALL GRAPH NODES OF THE FLOOR NETLIST: PRIMARY INPUTS AND OUTPUTS, CONSTANTS, CELLS.

All 150 designs	f1-k6	f2	f3
Nodes	163.5 / 14.0	111.4 / 13.5	362.0 / 25.0
Cells	114.6 / 5.0	62.5 / 4.0	313.2 / 13.5
Nets	218.3 / 12.0	93.5 / 10.0	344.2 / 21.0
Nodes by quartile	**f1-k6**	**f2**	**f3**
Q1 (n=38)	6.3 / 5.0	6.1 / 5.0	9.9 / 8.0
Q2 (n=39)	100.2 / 26.0	100.0 / 26.0	238.2 / 32.0
Q3 (n=37)	109.4 / 18.0	77.4 / 13.0	260.6 / 28.0
Q4 (n=36)	453.9 / 26.5	269.8 / 26.5	972.1 / 51.5

a training split of designs and the scores on held-out designs (Section IV-A).

Measurements. Hard adders 67 of 67 recovered (precision 0.985) and hard multipliers 30 of 30. The calibration error is 0.650 before fitting and 0.217 after.

Interpretation. Where synthesis has already typed a block, the flow finds it essentially every time. An error of 0.217 means that a stated probability is on average off by about one fifth from the observed rate, much closer than the 0.650 of the unfitted flow, though not exact.

Conclusion. Kept structures are recovered, and fitting makes the probabilities usable. **USEFUL** for both [L1h, L1c; Appendix A-B].

C. Recovering what synthesis erased

Hypothesis. The same flow recovers the soft structures that synthesis erased, on LUT netlists and on the gate netlist GateBreaker uses.

Implication. This is the actual reverse-engineering task. A positive answer would mean that a designer could recover registers, muxes and adders from a flat netlist without a language model.

Experiment. We score the flow's soft-structure claims against the key at the two LUT floors, and on the 15 test designs of the gate floor (144 answer-key instances).

Methodology. Macro-F1 over the structure kinds, where a hypothesis is counted when it is committed or has probability at least 0.5 and matches by the overlap rule (Section IV-A). No threshold for success was set beforehand for the LUT floors.

Measurements. At the first floor the macro-F1 is 0.354 (271 counted, 107 matched, 688 in the key); at the second floor it is 0.108. On the gate netlist 0 of 144 instances are counted and none is committed.

Interpretation. With LUTs the flow matches 107 of the 688 key instances (about one in six), and the gate netlist gives it nothing to hold on to: its single-level claims do not see through the mapping.

Conclusion. Weak on LUT netlists, **INCONCLUSIVE** because no bar was set; on the gate netlist **DISMISSED** [L1s, F3; Appendix A-B].

D. Graph analysis: clusters and a graph network

Hypothesis. Clustering the netlist graph finds the designer's module hierarchy, and a graph network that sees node, graphlet, colour-refinement and NPN features labels gate roles better than a baseline that sees only the neighbourhood of each gate (the 1-WL baseline).

Implication. If clusters matched modules, the hierarchy that synthesis flattens could be restored from the graph alone. If a network beat 1-WL, learned graph features would add information beyond structure that colour refinement already sees.

Experiment. Four clusterings (k-means, affinity propagation, spectral, Louvain) are scored against the module path of 68 designs. A graph network is trained to predict each node's construct type and compared with 1-WL on 13 held-out designs (3,932 nodes).

Methodology. NMI against the module path for the clusterings; macro-F1 and accuracy for the network, one run, one split, no interval.

Measurements. NMI is 0.3220 for k-means, 0.2708 for affinity propagation, 0.1378 for spectral clustering and 0.1360 for Louvain. The network scores macro-F1 0.4241 against 0.2717 for 1-WL (accuracy 0.8863 against 0.8492); a network on node features alone loses (0.2413 against 0.2711).

Interpretation. An NMI of at most 0.32 means that the clusters share under a third of the information of the real hierarchy, and the community methods are the worst. The

TABLE VII
THE CURRENT RESULTS, WITH THE VERDICT OF THE REGISTER ROW (APPENDIX C).

Question	Result	Verdict	Rows
Answer key covers the gates	54,306 of 54,306 gate nodes of 150 designs labelled	USEFUL	L0b
Pipeline ignores names	476 byte-identical run pairs, 105,646 renamed identifiers	USEFUL	L0a
Recover what synthesis kept	hard adders 67 of 67 (precision 0.985)	USEFUL	L1h
Calibration	error 0.650 uncalibrated, 0.217 calibrated	USEFUL	L1c
Recover what synthesis erased	F1 0.354 (271 counted, 107 matched)	INCONCLUSIVE	L1s
Graph network for roles	0.4241 against 0.2717 (1-WL), one split of 13 designs	POSSIBLY GOOD	L2b
Style survives synthesis	ceiling 0.50 on the gate floor	USEFUL	H14a
Blocks grown to path end	right for 0.528, cones 0.385, ceiling 0.556	USEFUL	QL7a, QL7b
Join touching blocks	right for 0.715, was 0.550; 157 of 251 split constructs repaired	USEFUL	QL9-a, QL9-b
Netlist helps a model	B minus C $+0.319$ (0.241 to 0.394)	USEFUL	LLM4

network's lead is large in the point estimate but rests on one run and 13 designs.

Conclusion. Clustering does not find the hierarchy, **DISMISSED**. The network is **POSSIBLY GOOD**: it needs several splits with an interval and a second benchmark [L2a, L2b; Appendix A-C].

E. What synthesis loses: a limit on style, and repeated structure

Hypothesis. The flow destroys information about coding style, so even a perfect recovery cannot restore it; and large designs are mostly repeated structure.

Implication. The first sets an upper bound for every method here. The second says that a method that finds one instance of a pattern finds many.

Experiment. Twin designs that compute the same function in different styles are put through the flow, and the best possible recovery from the netlist is computed. The six large VTR designs are cut into units and the units grouped by signature.

Methodology. The ceiling is the best score a recovery that sees only the netlist can reach, with the pre-stated bound of at most 0.60. The repeated share is compared with a threshold of 0.50 and the identical-neighbourhood share with 0.70.

Measurements. The ceiling is 0.50 on the gate floor (0.4823 and 0.4697 at the two LUT floors). The share repeated is 0.9283 (6,653 of 7,167 units), and 0.8606 of the repeat groups have identical neighbourhoods.

Interpretation. Half of the twin pairs cannot be told apart from the netlist. In the big designs more than nine units in ten are copies of another unit, so a block detector pays off once per pattern and not once per instance.

Conclusion. Both hold. **USEFUL** for the style limit and for repetition [H14a, H13a, H13c; Appendix A-F].

F. The loss audit

Hypothesis. Each kind of construct is first lost at one dominant Odin III pass, and a pass can be shown to fool a detector.

Implication. If the pass that loses a construct is known, a flow could preserve the construct by changing that pass.

Experiment. The D14 provenance map (Section II) is read after each pass for 1,436 source units; we count the units first lost at each pass and compare with the random-set and persistence null models. Detector recall is measured before and after each pass.

Methodology. Share of first losses at the dominant pass, with the null models as the chance level; recall change per kind with at least 10 answer-key instances.

Measurements. There are 580 first losses (419 not subsumed by an earlier loss). The first `opt` pass is dominant for registers, comparisons, muxes and arithmetic under the random-set null, but the persistence null gives the same placement by chance. Only the mux word is demonstrably fooled, by `abc`: its recall falls from 0.55 to 0.07, although the two answer keys differ in size (174 and 166 instances).

Interpretation. Most apparent loss at the first `opt` is what random merging of the same objects also gives. The mux result is a real fall in recall, but the two sets are not like for like.

Conclusion. Not a firm finding, **POSSIBLY GOOD** for both [L15a, L15d; Appendix A-J].

G. A word-first detector against a published method

Hypothesis. A detector that groups flip-flops into register words first tiles the words of real designs better than the earlier detectors, and recovers words better than DANA run through HAL.

Implication. Words are the unit a designer names. A method that finds them would give the first handle for the bus and datapath naming that Gate-Breaker asks for.

Experiment. The detector and the baseline detectors tile the words of held-out real designs at the floor with only LUTs and flip-flops (f2). On 100 designs, the pooled word overlap of the detector is compared with that of DANA.

Methodology. Tiling quality with a Wilcoxon test, and pooled word overlap F_J with a 95 percent interval of the difference. Both comparisons were fixed before the runs, and the scorer was fixed before any DANA score was computed.

Measurements. Tiling is 0.023 against 0.022 for the baseline (null model 0.203; Wilcoxon $p = 1.0$). The pooled F_J is 0.169 for Loki against 0.261 for DANA, a difference of -0.092 with

TABLE VIII
RULE DETECTORS ON HELD-OUT DESIGNS (ROWS L3A, L3C). THE BAR FOR A WORKING DETECTOR IS 0.80.

Pattern	Result	Works
State machine	precision 0.85 to 0.95, recall 0.95	yes
Counter	no false alarm; recall 0.86 to 0.90 (gates, LUTs), 21 of 21 (hard blocks)	yes
Glue logic	precision 0.98 to 1.0	yes
Shift register	precision 0.14 to 0.17	no
Timer	recall at most 0.17	no
Register file	recall 0 to 0.31	no
Composites	parts found 0.41 to 0.50, wires 0.12 to 0.18 (bar 0.70)	no

interval $[-0.317, 0.043]$ over 41 designs with a ground-truth word.

Interpretation. Tiling is no better than the baseline and the detector does not beat DANA; the interval contains both a clear loss and a small gain, so the two are not separated.

Conclusion. Tiling **DISMISSED**; against DANA **INCONCLUSIVE** [H16a, RQ-H1; Appendix A-I].

H. Level three: single-pattern detectors

Hypothesis. Rule detectors on the level-one claims find state machines, counters, shift registers, timers, register files and glue logic in whole designs at a precision and recall of at least 0.80, and find composites of them with the right wiring.

Implication. Patterns are the next level above single constructs. If rules sufficed, block recovery would need no further search.

Experiment. The detectors of Section III-F are scored on 252 held-out single designs and on 82 held-out composite benchmarks wired from 27 VerilogEval references with a known wiring. Two follow-ups try to raise the weak detectors: a state-machine claim from the turning-on relation of state bits, and graphlet signatures with a library of every width and learned classifiers.

Methodology. Precision and recall per pattern against the key, with 0.80 as the bar for a working detector and 0.70 for composites; the rules are stated before the scoring.

Measurements. Table VIII gives the results. A state-machine claim read from how the state bits turn each other on recovers 98 of 100 one-hot machines and 99 of 100 binary ones, but its pooled precision is 0.17, and a rule on the next-state function lifts it to 0.68, short of 0.80. Graphlet signature matching with a library of every width reaches 0.80 for no pattern on whole designs, and with learned classifiers for four cells only.

Interpretation. Three patterns work (state machines, counters and glue logic) and four do not. Most misses are a missing level-one claim, not a wrong composition: the rule sees the pattern's parts but not the whole. Rules alone do not close the gap.

Conclusion. Detectors for state machine, counter and glue work; shift register, timer, register file and composites do not. The next-state rule is **DISMISSED** (precision), with **POSSIBLY GOOD** for its recall; all graphlet follow-ups are **INCONCLUSIVE** [L3A, L3C, FSM2a, FSM2b, FSM3, GL1 to GL9; Appendices A-O to A-S].

I. Level three: why the cone detectors miss

Hypothesis. Most failed detections of the bounded-cone detector (cones) are near misses, whose gap is the partition, a missing width in the library or the threshold, and not missing structure.

Implication. If the misses were near misses, tuning the cone detector would be enough. If the partition is the gap, the blocks must be found another way.

Experiment. We take the 695 single-design constructs and 1,044 composite parts that cones miss, and for each ask what the fixed rule calls the miss, and how close the true block (cut out by the provenance) is to the pattern.

Methodology. The share of misses classed as near misses by a rule and the share whose true block is close to its pattern, on the test designs, with the rule and the thresholds committed before the run.

Measurements. The rule calls 80 percent of the misses near misses, but the true block is close to its pattern for only 39 percent of them.

Interpretation. For most misses the cone is the wrong group of cells, not a slightly wrong signature: the largest gap is the partition.

Conclusion. Cones need a better partition. The near-miss reading is **POSSIBLY GOOD**; the claim that the true block is close is **INCONCLUSIVE** [QL4a, QL4b; Appendix A-T].

J. Level three: growing blocks from anchors

Hypothesis. Blocks grown cell by cell from anchors, the places where a known pattern is easy to see, are the right block for more constructs than cones are.

Implication. Growing finds blocks of any shape and size, so detection no longer needs a library entry for each width.

Experiment. Starting from the anchors, every block grows one neighbouring cell per step. We compare four ways to decide when a block stops (a Bayes-factor stop, three other stop rules, and never stopping before the end of the path), and a growth order that respects structure, against cones. All are scored on the same 1,141 test constructs of 857 design and cell units.

Methodology. A block is right for a construct when it covers at least 0.8 of it with a Jaccard overlap of at least 0.5. The measure is the share of constructs for which some block is right, and per design and cell the geomean of $(g+1)/(d+1)$ with a 90 percent bootstrap interval over designs and win / loss / tie counts (Section III-F). The plan, the split and the rules were committed before the test runs. The ceiling is the share for which some prefix of the grown path is right.

Measurements. Table IX gives the share at each step. The Bayes-factor stop gives 0.053 against 0.385 for cones, because

TABLE IX
RIGHT-BLOCK SHARE ON 1141 TEST CONSTRUCTS OF 857 UNITS, ONE ROW PER STEP.

Step	Pull request	Right-block share
Bounded cones		0.385
Grow, Bayes-factor stop	#73	0.053
Grow to the end of the path (the best stop rule equals it)	#74	0.528
Attachment growth order (not adopted)	#76	0.526
Extra anchors	#77	0.550
Join touching blocks	#78	0.715
Ceiling: some prefix of the #74 paths is right	#74	0.556

the blocks stop early (a median of 11 cells against 35). The growth paths contain the right block for 0.556 of the constructs. Growing every block to the end of its path gives 0.528 (geomean 1.146 per unit against cones, 1.105 to 1.190), which closes 94 percent of the gap between the Bayes-factor stop and the ceiling (0.927 to 0.961). No stop rule beats never stopping; they are 1 to 9 percent worse. A growth order that respects structure gives 0.526, recovering 12 of 449 constructs with no right prefix.

Interpretation. About 5 of 10 test constructs are now recovered as one block, against fewer than 4 of 10 for cones, and almost all of what the growth paths can reach. The gain comes from not stopping blocks early, not from a cleverer stop or order.

Conclusion. Growing to the end of the path is **USEFUL**; the Bayes-factor stop is **DISMISSED**; a stop rule better than never stopping and the structural order are **INCONCLUSIVE** [QL6a, QL7a, QL7b, QL7e, QL8a; Appendices A-U to A-X].

K. Level three: anchors for untouched constructs

Hypothesis. Three more anchor classes (flip-flop banks that share global control, open feedback loops of flip-flops, and constant cells) touch and recover constructs that no block touched, at no cost to the constructs recovered before.

Implication. Without them a fixed share of the constructs can never be found, whatever the later steps do.

Experiment. The 58 constructs that no block touched are re-run with the extra anchors, and the whole test set is re-scored against growing to the end of the path.

Methodology. The share of the 58 touched and right, and the right-block share and per-unit geomean on all 1,141 constructs, as above.

Measurements. All 58 are touched and 47 are right. The overall share rises from 0.528 to 0.550, a tie per unit (geomean 1.001, 0.980 to 1.024), because some constructs found before are lost.

Interpretation. The anchors reach what was out of reach, and the price is some constructs lost elsewhere, so the net gain is within noise.

TABLE X
RIGHT-BLOCK SHARE BY SCOPE BEFORE AND AFTER THE JOIN (TEST, 1141 CONSTRUCTS).

Scope	Constructs	Cones	Before join	After join
All	1141	0.385	0.550	0.715
Gate netlists	351	0.242	0.538	0.610
Four-input LUT	460	0.439	0.602	0.754
k6 and hard-map	330	0.461	0.491	0.773
Single designs	635	0.356	0.539	0.742
Composite parts	506	0.421	0.565	0.682

Conclusion. The coverage is **POSSIBLY GOOD**; the claim of no cost is **INCONCLUSIVE** [QL8b-a, QL8b-b; Appendix A-Y].

L. Level three: joining touching blocks

Hypothesis. Joining touching blocks that are one construct raises the right-block share above 0.550 without losing single designs, repairs many of the 251 constructs that growth cut across several blocks, recovers many of the 95 constructs that one block swallowed, and a split step that cuts a block in two helps.

Implication. A construct that growth cuts in several pieces counts as missed. A join step is the cheapest way to repair it, and if a split step helped, blocks that are too large could be repaired as well.

Experiment. A join rule is applied to every pair of touching blocks after growth. We compare four join rules (a conductance rule, two Bayes-factor rules, and shared control nets) and a split along the Fiedler vector of the block's cell graph, on the same 1,141 test constructs.

Methodology. As in the growth argument. The rule is merged greedily, capped at 50 merges per design and cell; the conductance threshold 0.2 was chosen among four rules on validation designs, and the test designs were run once.

Measurements. The conductance join raises the share from 0.550 to 0.715, against 0.385 for cones. Per unit the geomean is 1.149 (1.123 to 1.177), W194 / L9 / T654, against the blocks before the join, and 1.318 (1.276 to 1.364), W394 / L46 / T417, against cones. It repairs 157 of the 251 constructs cut across blocks. Of the 95 swallowed constructs, 42 are right after the step and were right before it, and it loses 29 constructs that were right, all in composite parts. The Bayes-factor joins give 0.680 and 0.677, shared control nets 0.617. Every split threshold loses more constructs than it repairs. Table X gives the share by scope.

Interpretation. About 7 of 10 held-out constructs are recovered as one block, where cones recover fewer than 4 of 10. The join repairs the pieces that growth cut and does not touch the swallowed constructs; a split never pays for itself. The limits are that designs above 400 cells and blocks above 300 cells are not scored, that composite parts built with Yosys have no provenance key and are not scored, that the join is greedy, and that this one split family did not help.

Conclusion. The join and the repair of split constructs are **USEFUL**; recovering swallowed constructs is **POSSIBLY GOOD**; the split step is **DISMISSED** [QL9-a, QL9-b, QL9-c, QL9-d; Appendix A-Z].

M. Level three: guessing the kind of a block

Hypothesis. A stronger likelihood for the kind of a block (a Gaussian on richer directed graphlet features, the *Gaussian kind model*) raises the macro-F1 over the five kinds by 0.10 over the earlier likelihood.

Implication. A recovered block is useful to a reader only with a name: counter, shift register, state machine, register file or register.

Experiment. The Gaussian kind model replaces the earlier likelihood on the same grown blocks, and is then applied to the joined blocks.

Methodology. Macro-F1 over five kinds (precision over all claims, recall over constructs), with 90 percent bootstrap interval of the gain; the gain of 0.10 was fixed before the test run as "well above".

Measurements. The macro-F1 rises from 0.211 to 0.262 (gain 0.051, 0.019 to 0.081; 232 constructs recovered with their kind against 213). Given the answer-key block it is only 0.487. On the joined blocks it is 0.392, against 0.237 before the join; per kind, finite-state machine 0.282 to 0.476, counter 0.132 to 0.509, shift register 0.287 to 0.317, register file 0.296 to 0.393 and register 0.188 to 0.266.

Interpretation. The kind guess lags the block itself: it improves, by about half of the gain we asked for, and even a perfect block gives a macro-F1 below 0.5.

Conclusion. The gain is positive but short of the stated bar, **DISMISSED** for the 0.10 hypothesis and **POSSIBLY GOOD** for the direction. The per-kind scores on joined blocks are a measurement with no verdict [QL8c-a, QL8c-b, QL9-e; Appendices A-W and A-Z].

N. What did not work

Table XI lists the hypotheses that were DISMISSED, and the main INCONCLUSIVE ones, one line each. Their records are in the appendix.

VI. The language-model replication and Loki's hints

Gate-Breaker's own method gives a language model the netlist with hidden wire names and asks for the RTL. This section reproduces that baseline and then adds Loki's hints to the prompt. The arguments use the same seven parts as Section V. The model prompts come in the arms of Section II: arm A (the netlist), arm B (the netlist and Loki's hints) and arm C (the interface line only). The earlier pilot (40 problems, one draw) is Appendix record A-K.

A. The Gate-Breaker baseline, replicated

Hypothesis. None. This is a measurement of the baseline that the language-model arguments below compare against.

Implication. Loki recovers structure and not a new RTL text, so a direct comparison with Gate-Breaker's metrics is possible only by running its harness ourselves. The numbers tell how far a model alone gets from the gate netlist.

Experiment. We run the Gate-Breaker harness itself, with the only language model available on the host. The first run uses the simple benchmarks of the fallback micro set. The second uses the VerilogEval designs of its quartiles Q1 and Q2: one run, $K = 10$, four workers, 180 s per call, 740 calls.

Methodology. The metrics are the Gate-Breaker ones (Success Rate, Success@k, AST and CFG similarity; Section II). The AST and CFG kernels are recomputed with common-width feature vectors; the first run padded per graph and truncated in the zip, so those columns are not comparable to the first run. Seventy-four of the 78 designs of Q1 and Q2 are usable: three are excluded because their synthesized netlists fail the official testbench self-check, one because its 241 KB prompt exceeds the OS argument limit.

Measurements. On the fallback micro set the success rate is 0.933 on Q1 and 0.829 on Q2, Success@10 is 1.000 in both, AST similarity 0.576 and 0.444, CFG similarity 1.000 and 0.736, with 169 of 174 extracted candidates functionally correct. On VerilogEval the results are in Table XII. Against the paper's O3-Mini the deltas are: Q1 success rate -0.092, AST -0.030, CFG -0.011; Q2 success rate -0.039, Success@5 $+0.125$, Success@10 $+0.111$, AST -0.164, CFG $+0.083$.

Interpretation. Every scored design is solved by attempt 5, so a model that sees the gates recovers function almost always, and names nothing reliably. The Q1 gap to the paper is harness no-output, not wrong RTL; the Q2 AST gap is truth-table and case references against the model's boolean expressions. Two caveats: the host's only model path routes through a gateway that aliases a DeepSeek backend as Claude Opus, so the model column is not verifiably the paper's comparison class, and the CLI exposes none of the paper's sampling flags (T=0.85, p=0.95). The two lines of work are complementary: the model recovers behaviour, Loki recovers the structures the flow preserved and proves what it claims. The Gate-Breaker future work (bus naming, datapath and control classification, partitioning) is exactly the work of level two.

Conclusion. Both runs are **MEASUREMENT**: they carry no verdict and are not counted [GB1, GB2].

B. Do Loki's hints help a model that has the netlist? The full run

Hypothesis. A model given Loki's hints recovers RTL better than the same model given the netlist alone (arm B above arm A); and the netlist helps beyond what the model knows from the interface line alone (arm B above arm C).

Implication. If the hints helped, Loki's structure would be a usable aid to language-model reverse engineering. If the netlist

TABLE XI
WHAT DID NOT WORK. NUMBERS ARE FROM THE REGISTER ROWS NAMED IN THE FIRST COLUMN.

Row	Idea	Result	Verdict	Record
F3	L1 recovers structure from the gate netlist	0 of 144 counted, 0 committed	DISMISSED	A-B
L2a	Clustering finds the designer's modules	NMI at most 0.3220 (k-means); community methods lowest	DISMISSED	A-C
H9b	A word's fragments tile the word	fragments cover 0.065 of the words	DISMISSED	A-E
H12c	GNN transfer to real designs improves with synthetic corpus size	slope per doubling -0.0026, interval $[-0.0097, 0.0094]$	DISMISSED	A-F
H16a	The word-first detector tiles words	0.023 against 0.022 for the baseline	DISMISSED	A-I
RQ-H1	The word-first detector beats DANA	F_J 0.169 against 0.261, difference -0.092 $[-0.317, 0.043]$	INCONCLUSIVE	A-I
FSM2a	State-machine claims are selective	pooled precision 0.17 against 0.70 to 0.92	DISMISSED	A-O
FSM3	A next-state rule makes the claims selective	precision 0.676 against the bar of 0.80; recall 1.000	DISMISSED	A-O
GL1 to GL9	Graphlet signatures find the patterns the rules miss	no pattern reaches 0.80 on whole designs	INCONCLUSIVE	A-R, A-S
QL6a	Blocks grown with a Bayes-factor stop beat the cones	0.049 against 0.371 (1255 constructs); blocks stop early (median 11 cells against 35)	DISMISSED	A-U
QL7e	A stop rule beats never stopping	none does; they are 1 to 9 percent worse	INCONCLUSIVE	A-V
QL8a	A structural growth order adds recovery	0.526 against 0.528; 12 of 449 recovered	INCONCLUSIVE	A-X
QL8c-b	A richer kind likelihood lifts macro F1 by 0.10	+0.051 (0.262 against 0.211)	DISMISSED	A-W
QL9-d	A split step helps	every threshold loses more constructs than it repairs	DISMISSED	A-Z
LLM3	Loki's hints help a model that has the netlist	pass@1 0.878 with hints, 0.888 without	DISMISSED	VI-B
LLM6	Wrong hints hurt the model	W 0.828 against A2 0.802; W is not below A2	DISMISSED	VI-C

TABLE XII
GATE-BREAKER METRICS ON THE VERILOGEVAL QUARTILES (ONE RUN, K=10, 4 WORKERS, 180 S PER CALL, 7,440 S WALL CLOCK, 740 CALLS; DELTAS VS THE PAPER'S O3-MINI IN THE TEXT).

Quartile (n)	SR	S@1	S@5	S@10	AST	CFG
Q1 (36)	0.903	0.944	1.000	1.000	0.894	0.942
Q2 (37)	0.778	0.811	1.000	1.000	0.761	0.890

TABLE XIII
LLM FULL RUN: PASS RATES PER ARM, 137 PROBLEMS, 3 DRAWS, STRICT SCORER. MAJ: PASSES IN AT LEAST 2 OF 3 DRAWS.

Arm	Calls	pass@1	pass@3	MAJ
A netlist	411	0.888	0.942	123/137
B netlist + hints	411	0.878	0.942	124/137
C interface only	411	0.560	0.686	77/137

TABLE XIV
LLM FULL RUN: PRIMARY TESTS (MAJ), EXACT MCNEMAR PER QUARTILE AND STRATIFIED; DISCORDANT PROBLEMS (FIRST ONLY, SECOND ONLY), HOLM p OVER TEN TESTS.

Comparison	Stratum	Discordant	p	Holm p
A vs B	Q1	0, 2	0.5000	1.0000
A vs B	Q2	1, 0	1.0000	1.0000
A vs B	Q3	0, 0	1.0000	1.0000
A vs B	Q4	2, 2	1.0000	1.0000
A vs B	all	3, 4	1.0000	1.0000
B vs C	Q1	19, 0	< 0.0001	< 0.0001
B vs C	Q2	12, 1	0.0034	0.0273
B vs C	Q3	9, 1	0.0215	0.1504
B vs C	Q4	12, 3	0.0352	0.2109
B vs C	all	52, 5	< 0.0001	< 0.0001

did not help beyond the interface line, the benchmark would be testing memorisation of the reference designs.

Experiment. The earlier pilot could not separate the arms, so the same question is asked on every eligible problem of the four quartile files: 137 problems (144 in the files, 7 excluded: four prompts over the 40000-byte cap, one reference that fails its own testbench, two whose hints cannot be built). Each arm has three independent draws per problem: 1,233 calls to fresh `claude-sonnet-5-5` subagents, with the pilot's prompts, scorer and answer format.

Methodology. The primary outcome is that a problem passes in at least 2 of its 3 draws (*MAJ*). Per quartile we use the exact two-sided McNemar test, across quartiles the exact stratified form (discordant problems pooled), and Holm adjustment over the ten tests; a claim needs an adjusted $p < 0.05$. The plan (arms, scorer, primary outcome, tests and stop rules) was written and committed before the runs.

Measurements. All 1,233 calls were answered (0 API refusals, 0 no-file calls, 0 timeouts). Table XIII gives the pass rates and Table XIV the tests. Arms A and B are level: mean pass@1 0.888 against 0.878, A minus B $+0.010$ with 95% interval $[-0.017, +0.036]$, and 3 problems that only A passes against 4 that only B passes. Arm C, with no netlist, still passes 0.560. B minus C is $+0.319$ ($[+0.241, +0.394]$), with 52 problems that only B passes against 5 that only C passes (Holm $p < 0.0001$). By quartile the B-against-C difference passes the rule in Q1 and Q2; Q3 and Q4 have the same sign but Holm $p = 0.1504$ and 0.2109.

Interpretation. The interval rules out a gain from the hints larger than about 0.04 of the pass rate. Over half of the problems pass with only the interface line, so the benchmark carries some memorisation, and the netlist adds the other third of the pass rate. The threats are these. (i) A Sonnet dispatcher

subagent spawned the call subagents, so the controller session did not. (ii) The call subagents were not tool-less: a probe lists Agent, Artifact, Bash, Edit, Read, Skill, ToolSearch, Write and the Docs tools, with the workspace instructions, a memory index and a terse-style hook text in context, and the reference files are on the same disk. The transcript audit of all 1,233 calls found 0 calls touching any path other than their own call file and reply file; 37 used more than two tool calls (paged reads, double writes, and in 3 calls a shell search of their own call file). All are scored as made. This is evidence from behaviour, not a technical block. (iii) The 7 excluded problems and the prompt-size cap under-represent the largest designs. (iv) Arm A is near the ceiling in Q1 and Q2, which limits how much any hint could show. (v) The system prompt mentions Loki hints in every arm, and sampling is not controlled.

Conclusion. The hints add nothing measurable, **DISMISSED**. The netlist helps, **USEFUL**; per quartile Q1 and Q2 pass the rule and Q3 and Q4 are **INCONCLUSIVE** [LLM3, LLM4; Appendix A-M].

C. Better hints, and wrong hints

Hypothesis. A model given Loki's linked hints of the second version recovers RTL better than the netlist alone, and the model uses the hints, so hints with wrong content hurt.

Implication. The full run found no gain from the first version of the hints. Two defects were fixed: the claims were weak (no counter kind, empty state-machine claims, duplicates), and the model was never told what the hints are. If the second version helps, the first result was a defect of the hints and not of the idea. If wrong hints hurt, the model reads the hints at all.

Experiment. We take the 64 problems of Q3 (34) and Q4 (30), where arm A of the full run was not at the ceiling. Four arms are run: *A2* (netlist only, new prompt), *B1'* (the first-version hints with the new prompt), *B2* (second-version hints) and *W* (the second-version claims moved to wires they do not describe; 5 problems whose wrong claims stayed too close to the true ones are left out of the W tests). The second-version hints write one plain-English line per claim with a probability, link it into the netlist as a comment on the line of each cell the claim covers, and the prompt explains the hints and asks the model to name the claims it used.

Methodology. Three draws per problem and arm: 768 calls to fresh `claude-sonnet-5-5` subagents, with the scorer and tests of the full run (at least 2 of 3 draws; exact stratified McNemar; Holm over nine tests). The plan was committed before the calls, and the stop was set at 97 percent of the weekly meter before the calls; the meter never passed 89.

Measurements. 762 calls produced a reply file and 6 did not (the model answered in chat); there were 0 API refusals and 0 timeouts. Outcomes: pass 645, runtime 109, compile 8. Table XV gives the pass rates. In Q4 pass@1 is 0.933 for B2 against 0.744 for A2, 0.800 for B1' and 0.789 for W; in Q3 all arms sit between 0.85 and 0.87. Over all 64 problems B2 minus A2 is $+0.099$ ($[+0.052, +0.151]$) and B2 minus B1' is $+0.073$ ($[+0.031, +0.115]$). Pass@3 is 0.906 for A2, B1' and B2 (W 0.875). The primary test does not reach the rule: 4 problems that only B2 passes against 0 that only A2 passes, Holm $p = 1.0000$. The model names a claim in 107 of 192 B2 calls; state-machine claims were used in 57 B2 calls and all 57 passed, counter claims in 24 and all passed. In W, state-machine claims were used in 33 calls with 27 passes, against 24 calls where one was offered and not used with 19 passes (0.82 against 0.79). W minus A2 is $+0.026$ ($[-0.021, +0.073]$), with 2 problems that only W passes and 2 that only A2 passes.

TABLE XV
LLM HINTS V2: PASS RATES PER ARM, 64 PROBLEMS, 3 DRAWS, STRICT SCORER. MAJ: PASSES IN AT LEAST 2 OF 3 DRAWS.

Stratum	Arm	Problems	pass@1	pass@3	MAJ
Q3	A2	34	0.853	0.882	30/34
Q3	B1'	34	0.853	0.882	30/34
Q3	B2	34	0.873	0.882	30/34
Q3	W	34	0.863	0.882	30/34
Q4	A2	30	0.744	0.933	24/30
Q4	B1'	30	0.800	0.933	26/30
Q4	B2	30	0.933	0.933	28/30
Q4	W	30	0.789	0.867	25/30
All	A2	64	0.802	0.906	54/64
All	B1'	64	0.828	0.906	56/64
All	B2	64	0.901	0.906	58/64
All	W	64	0.828	0.875	55/64

Interpretation. The second-version hints raise the pass rate of a single draw, mostly in Q4, but not the best-of-three ceiling, and the pre-stated test cannot show it: A2 already passes 54 of 64 problems by that measure, so few problems can differ, and 4 against 0 cannot pass after Holm. The point estimate and the claim-use table suggest that the model reads and uses state-machine and counter claims; this is a pointer for a better-powered test, not a verdict. A used wrong state-machine hint passed less often than a used right one (27 of 33 against 57 of 57), but on different problems, so this is not a paired test. The threats: (i) the test is underpowered, and the pass@1 interval and the claim-use table are not the pre-stated rule; (ii) blinding is as in the full run, and the audit of all 768 calls found 0 calls touching any path other than their own call file and reply file, and 0 using other tools; 26 used more than two tool calls (paged reads, double writes, and 6 calls that edited their own reply file with Edit instead of Write); (iii) A2 has neither the hint explanation nor the claim-use line, so B2 against A2 differs by both, while B2 against B1' holds the prompt fixed and isolates the hint content; (iv) the probability of a sequential claim is a per-kind constant checked on the netlist, not a measure of designer intent, and it was not used in any analysis; (v) wrong claims stay partly close to the true ones in small designs; (vi) memorisation and sampling are not controlled in this run.

Conclusion. Better hints: **INCONCLUSIVE** (stratified 4 against 0, Holm $p = 1.0000$). Wrong hints hurt: **DISMISSED** (2 against 2, Holm $p = 1.0000$) [LLM5, LLM6; Appendix A-N].

D. Do the hints help in some quartiles, and what does Loki alone recover?

Hypothesis. The hints help in some Gate-Breaker complexity quartiles (Q1 to Q4) and not in others. A second question, a measurement, asks what Loki alone recovers in each quartile.

Implication. A gain confined to the hard quartiles would show where structure helps a model; the Loki-only table shows where the structure itself is recovered.

Experiment. The pilot of the Appendix is split by quartile and tested per quartile; Loki-only results that are defined on the VerilogEval designs are tabulated in the four quartiles. No new run or model call was made. The partition is the Gate-Breaker quartile files (41, 36, 36 and 31 problems); 138 built designs appear in them, and 12 built designs that appear in no file are left out.

Methodology. The exact two-sided McNemar test per quartile, with the stratified exact test equal to the overall test. A test needs at least 6 discordant pairs, all in one direction, for $p < 0.05$; the largest number in any one quartile here is 3.

Measurements. Arm B is ahead of arm A by one problem in Q2 and in Q3, level in Q1 and behind by two in Q4. Overall the paired difference is 2 problems for B and 0 for A ($p = 0.50$). Arm C is behind arm B in the point estimate in every quartile except Q4, and the overall test of B against C is $p = 0.070$ (paired view) and $p = 0.18$ (all calls); no quartile reaches $p < 0.05$. The Loki-only tables, the word recovery against HAL per quartile and the side-by-side with the Gate-Breaker replication are in the appendix record A-L.

Interpretation. The pilot has too few problems per quartile for any test to decide, so it neither shows nor excludes a gain in any quartile. The full run above is the test with power.

Conclusion. The hints help in some quartiles: **INCONCLUSIVE**. The per-quartile Loki-only results are a **MEASUREMENT** [Q1, Q2].

E. Loki cuts the netlist and the model decodes one block at a time

Hypothesis. If Loki cuts the gate netlist into its joined blocks and a language model decodes one block per prompt, and a script glues the decoded blocks into one design, then (a) on designs that fit Gate-Breaker's prompt the result is as correct as the whole-netlist prompt of arm A, (b) on designs too large for one prompt it still gives correct RTL, and (c) adding Loki's kind guess for a block to its prompt does not improve correctness, as the hints did not in the whole-netlist setting.

Implication. If (a) holds, Loki, and not the model, decides how the netlist is split, and every decoded piece answers to one block with a provenance key; the model's context no longer limits the size of the design, which is (b). Statement (c) tells whether the weakest part of Loki, the kind guess, matters for a model that reads the gates of the block itself.

Experiment. The blocks are the joined blocks of Section V. In *arm D* the model sees one block at a time, only its gates and the nets that cross its boundary, with no examples and no hint. In *arm D+h* the prompt is the same plus one sentence with Loki's kind guess, for the blocks of four or more cells that have a guess; for the other blocks the two arms share one reply. A script writes the replies into one module that instantiates the decoded blocks and copies the netlist's `assign` statements. The baseline is arm A of the full run (Section VI-B) on the same problems and draws. The *multi-block designs* are the 27 of the 137 problems that Loki cuts into two or more blocks. Fifteen problems have no gate cell and are scored without a model call. The 94 single-block problems were not run: there the block is the whole netlist, so arm D would be arm A in another wrapper. Two sets that arm A cannot run are added. The *over-cap designs* are four VerilogEval problems whose prompt exceeds the 40 000-byte cap of the earlier runs (mux256to1, gates100, rotate100, shift18), scored by their testbench. The *processor modules* are five modules of picorv32 and tinyriscv (66 to 195 blocks each, one draw), scored by sequential equivalence of the assembled RTL with the netlist, because no testbench exists. One fresh Sonnet subagent answers each prompt, and 1,414 replies were collected (none empty).

Methodology. A design passes when its testbench reports no mismatch (strict scorer of Section VI-B), or for a processor module when the equivalence check says equal. Pass@1 is the share of passing draws, averaged over designs, with a 95 percent bootstrap interval over designs (2 000 resamples, seed fixed). The rules were fixed before any call: (a) holds when the lower bound of the pass@1 difference D minus A is above -0.05, and is contradicted when the upper bound is below 0; (b) holds when one design passes and at least 0.8 of the blocks pass the *block check*, which simulates a decoded block against its own gates on 400 random cycles; (c) is DISMISSED when the interval of D+h minus D lies within ± 0.05 and D+h is not better by an adjusted exact McNemar $p < 0.05$, USEFUL when it is, and else INCONCLUSIVE. The second measure is Gate-Breaker's AST and CFG similarity (Section II) to the reference RTL. Both read one module, so the assembled design is flattened with Odin III into one module; arm A is passed through the same flatten, and the *same-writer pair* (D flattened against A flattened) is the headline, since the writer alone lowers A's own score. This measure had no rule and gives no verdict. Twenty-two of the 1,339 calls broke the call template (a subagent used a shell or an editor on its own reply file); the results are repeated without the 27 rows they touch.

Measurements. Table XVI gives the pass rates. On the multi-block designs D passes 72 of 81 (design, draw) pairs and A 73 of 81: the difference is -0.012 (-0.074 to $+0.037$), W3 / L4 / T74 over draws; both reach pass@3 on 26 of 27 designs. Without the 27 template-deviation rows the difference is the same. The block check passes for 954 of 969 block replies of the over-cap and processor designs (0.985); three of the four over-cap designs pass in at least one draw (shift18 in none), and one of the five processor modules is equivalent. Replacing the blocks that failed the check by their own gates makes

TABLE XVI
BLOCK-BY-BLOCK DECODE AGAINST THE WHOLE-NETLIST PROMPT. PASS@1 OVER DRAWS, MEAN OVER DESIGNS. THE REFERENCE ARM IS A (FOR THE LAST ROW, D) AND THE TESTED ARM IS D (FOR THE LAST ROW, D+H); THE INTERVAL IS THE BOOTSTRAP 95 PERCENT INTERVAL OF TESTED MINUS REFERENCE.

Set	N	Ref.	Tested	Difference (95% interval)
Multi-block, D vs A	27	0.901	0.889	-0.012 (-0.074, $+0.037$)
Gate-free, no call	15	0.822	0.867	$+0.044$ (0.000, $+0.133$)
Over-cap, D only	4	–	0.667	–
Processor, D only	5	–	0.200	–
Hint, D+h vs D	8	0.750	0.708	-0.042 (-0.125, 0.000)

all ten checked processor assemblies equivalent. The hint arm passes 0.708 against 0.750 for D on the 8 designs with a hint that were run (6 multi-block, 2 over-cap): -0.042 (-0.125 to 0.000), W1 / L2 / T21 over draws, $p = 1.0$; of 102 hinted blocks 95 pass the check in each arm. For the second measure, on the multi-block designs the same-writer pair gives an AST difference of -0.037 (-0.063 to -0.014, 23 designs, W6 / L17) and a CFG difference of -0.042 (-0.067 to -0.021, 27 designs, W6 / L21); against A as the model wrote it the differences are -0.132 and -0.065. The hint changes neither similarity. Answer key: for 8 multi-block designs with a key, a block holds the construct as a right block for 12 of 36 key constructs (three draws) and the decoded block is correct for 10; for the two over-cap designs with a key, 6 of 6. Estimated cost (not measured): arm D is about 14 500 tokens per multi-block design over three draws against about 10 300 for A.

Interpretation. Where Loki's cut changes the prompt, the blocked decode passes about as often as the whole-netlist prompt (72 against 73 of 81), too close to tell with 27 designs: the interval allows a loss of 7 points. Its AST and CFG similarity to the reference is lower by about 0.04, in about three designs of four, even through the same writer. So the cut costs a little in how alike the text is and nothing detectable in correctness. For the over-cap designs the model decodes nearly every block correctly, but a design passes only if every block is right: with a per-block rate of 0.985 the chance that 100 blocks are all right is about 0.2, which is why the processor modules mostly fail even though each failing assembly is made equivalent by fixing the one to three blocks the check flags. The hint was never found better; the interval of eight designs is too wide to call it harmless. The limits are the single-block designs that were not run, the answer key that covers registers only (34 of the 42 designs with a key are single-block), the random-input block check, and the processor equivalence, which is a check and not a proof.

Conclusion. (a) Blocked decode is as correct as the whole-netlist prompt: **INCONCLUSIVE** (-0.012, the lower bound -0.074 is below the margin -0.05). (b) The model decodes the blocks of designs too large for one prompt: **USEFUL** (the rule is met: a design passes and 0.985 of the blocks pass the check), with whole-design correctness low for the largest designs. (c) The kind-guess hint does not improve correctness: **INCONCLUSIVE** (-0.042, W1 / L2 / T21, interval not inside ± 0.05). AST and CFG similarity, the same-writer pair: a **MEASUREMENT** of about -0.04 for D [QL10-a to QL10-e; Appendix A-AA].

VII. DISCUSSION AND LIMITS

Five observations follow from the verdicts, and then the candidate innovations.

First, *the answer depends on what the flow erased*. The structures that survived synthesis recover almost perfectly without any intelligence beyond reading the pass's own facts (L1h); the structures synthesis erased recover weakly (L1s) and, on the gate netlist, not at all (F3). Nothing in the experiments shows that the Bayesian machinery, as built, changes that: the evidence families add nothing over the per-detector priors (R2-L4), and a LUT6 re-map reaches the second floor's level only in-sample.

Second, *the graph stage does not find the hierarchy, and the network result is possibly good*. No clustering of the netlist graph finds the designer's modules well: NMI against the module path is at most 0.3220 (k-means), and the community methods score lowest (L2a, DISMISSED). The GNN beats 1-WL only when node, graphlet, WL and NPN features combine (L2b), and that is one run on one split.

Third, *the provenance answer key is the part that works*. It covers every gate node (L0b), the renamed-instance protocol shows the pipeline is naming-neutral (L0a), and the same protocol exposed two defects (a rename-script defect that removed every hard proposal on renamed floors, which was fixed, and a spectral nondeterminism, which is reported). The loss audit (Appendix A-J) uses the same map: the mux word is the one kind that a pass (`abc`) demonstrably fools (L15d), and the placement of other losses at the first `opt` is what chance merging also gives (L15a, POSSIBLY GOOD).

Fourth, *readable and verified are different properties*. The language model produces RTL whose similarity to the reference is high and whose operators it names, but nothing proves the text equals the netlist; Loki's output is formally equal to its input in all runs and recovers few operators. The pilot (LLM1, LLM2) could not separate the arms; the full run (LLM3, LLM4) shows that the hints add nothing measurable to the netlist (LLM3 DISMISSED) while the netlist itself adds a lot (LLM4 USEFUL). A second run with better, linked hints (LLM5, LLM6) gave a Q4 pass@1 of 0.933 against 0.744 but did not meet the pre-stated test (LLM5 INCONCLUSIVE, LLM6 DISMISSED); it is a pointer, not a verdict.

Fifth, *blocks are found by growing them, and the weak part is the kind guess*. Growing every block to the end of its path recovers as much as the best stop rule (0.528), a better growth order does not add (0.526), extra anchors add a little (0.550), and joining touching blocks adds the most (0.715). What the join leaves undone is visible in the numbers: it does not recover constructs that a block swallows, it loses 29 constructs that were right (a joined block takes in a neighbour), and the kind guess on the joined blocks reaches a macro F1 of only 0.392. Every threshold was fitted on training and validation

designs and the test designs were run once, but all of it is one deterministic run on small designs.

Candidate innovations. Three ideas have only POSSIBLY GOOD support and stay out of the contributions. The graph network (L2b) needs several random splits with an interval over designs and a second benchmark suite. The operator-recovery F1 (R2-L7) needs a set with many more tie pairs and a flow that recovers operators, since the calibrated flow recovers none. The loss audit (L15a, L15d) needs equal-sized ground truths at the two netlist snapshots, so that the recalls are on the same instances, and a run on the big designs.

Limits. (i) The recovery stages use no language model; the pilot and the full run are separate experiments. (ii) Recovery is structural, not functional: the output is exactly as correct as its input netlist, which itself passes the official testbench in only some designs (inferred latches and asynchronous resets are the failures). (iii) Twelve register rows are POSSIBLY GOOD and not USEFUL because they are narrow or fragile (among them L2b, R2-L7, H9a, H13b, L15a, L15d); round-1 claims (L0a to L2b) have a clear gap but no interval. (iv) The word-first detector, the signature dictionary and the synthetic-corpus GNN did not transfer from synthetic to real designs. (v) Area, speed and power are not measured, and the benchmarks are small designs: equivalence and recovery were not run on the big VTR designs.

VIII. Future work

The next question comes first, then the untested hypotheses. Each is an idea and the experiment it needs.

Finishing the block-by-block decode. Section VI-E left three things open. First, the 94 single-block problems were not run, so (a) is tested on 27 multi-block designs only; running them would add designs where the cut is the whole netlist, and would show how much of the difference is the prompt wrapper. Second, a design passes only if every block is right, and the block check names the wrong blocks (all ten checked processor assemblies were made equivalent by replacing the one to three flagged blocks); a repair round that asks the model again only for the flagged blocks is the obvious next experiment, with the same scorer. Third, the interval for the hint (eight designs) is too wide to call it harmless or useful; more multi-block designs with a kind guess are needed.

A guard for the join. The join loses 29 constructs that were right because a joined block takes in a neighbour. A guard that refuses to join two blocks whose kind posteriors are both confident and different is the obvious repair, and a split cut along the dependency graph of the flip-flops, instead of the Fiedler vector of the cell graph, is the obvious second try. Neither has been run.

L3, a cross-suite GNN. A GNN trained on one benchmark suite may transfer to the other. The experiment trains on the micro designs and tests on the VerilogEval designs and the reverse, with an interval over designs.

L6, a motif null model. The graphlet counts may differ from what a degree-preserving random graph gives. The experiment compares the ORCA graphlet counts of each netlist with those of random rewirings of the same graph.

L8, a model given the ReverseProvenance record. A language model that sees Loki's record may recover RTL better than the model alone. The experiment is the Gate-Breaker task with a record arm and a no-record arm on the full quartile files, three draws per arm. The full run of Section VI-B did this for the hints (arm B against arm A) but not yet for the ReverseProvenance record.

Target-aware transformations. The flow that makes a netlist can recode a construct for its target: a state machine may be rewritten one-hot instead of binary, a sum may map to a carry chain, a memory to a hard block. Each changes the footprint the construct leaves in the netlist. The experiment (hub question L Q3.3) measures how each transformation changes the netlist footprint of a construct and whether Loki recovers more when it speculates which transformation was applied, starting with state-machine encoding and then carry chains and hard blocks. The hints-v2 run (LLM5) gives a first reason to try: state-machine claims are the ones the model used most.

IX. Conclusion

Loki III asks how far netlist-to-RTL reverse engineering gets without a language model. We stated each claim as a hypothesis and gave it one experiment and one verdict. Of 104 register rows, 14 are USEFUL, 12 are POSSIBLY GOOD with their fragility stated, 19 are DISMISSED, 34 are INCONCLUSIVE and 25 are measurements. The provenance answer key and the renamed-instance protocol are the part that works and that we would export to any future comparison, with or without a model. The structures that synthesis already reverse engineered are recovered (hard adders 67 of 67, precision 0.985), the stated probabilities are better calibrated (error 0.217 against 0.650), and the pipeline is naming-neutral over 105,646 renamed identifiers. The newest result is the block search: blocks grown from anchors to the end of their path and joined by a conductance rule are the right block for 0.715 of 1141 held-out constructs, against 0.385 for bounded cones, though the kind guess on them is weak (macro F1 0.392). The structures that synthesis erased recover weakly, clustering does not find the designer's hierarchy, and a word-first detector, a split step and Loki's hints to a language model did not help. A model that decodes the netlist one Loki block at a time passes about as often as Gate-Breaker's whole-netlist prompt on the 27 designs the cut changes (72 against 73 of 81, inconclusive), decodes three of four designs too large for one prompt, and its text is slightly less like the reference (Section VI-E).

Appendix A
Experiment records

This appendix holds the record of each experiment in time order, so that the main text can tell the current story. Each record has the same parts: the question or hypothesis, the experiment (data, split and measure), the result (numbers; for

TABLE XVII
INDEX OF THE RECORDS (MERGE DATES OF THE LOKI PULL REQUESTS).

Record	Merged	PR
Naming neutrality, L1, L2 (round 1)	2026-10-23	#36, #39, #41, #43
Round 2	2026-10-24	#44
Round 3 (fragments)	2026-10-25	#45
Round 4 (identifiability, corpus)	2026-10-25	#46
Round 5 (word-first, HAL)	2026-10-25, 10-27	#47, #48, #51
Round 6 (loss audit)	2026-10-27	#53
Language-model pilot	2026-10-27	#49, #50, #52
State-machine encoding	2026-10-30	#64, #65
L3 census and detector	2026-10-30, 10-31	#66, #67
Graphlet library and partitions	2026-10-31	#68, #69, #70
Fuzzy graphlets	2026-10-31	#71
Q-L4 to Q-L9	2026-11-01, 11-02	#72 to #78
Q-L10	2026-11-02	#82

the Q-L series the win / loss / tie counts and the geomean with its interval), the verdict (USEFUL, POSSIBLY GOOD, INCONCLUSIVE or DISMISSED, as in the register) and the pull request. Nothing was removed: a record that the main text no longer needs is here. The records of the full language-model run, the hints-version-2 run and the per-quartile tables stay in Section VI, because they are the answer to the model question.

A. Naming neutrality and the answer key (L0a, L0b)

Hypotheses. L0a: the pipeline uses no names, so renaming every identifier changes no result. L0b: the provenance answer key reconstructs the reference on every floor with no recovery step.

Experiment. For L0a, every identifier of every netlist (105,646 identifiers across 269 netlists) is renamed through one permutation, L1 and L2 are run on the renamed and the base floors, and the tokens are mapped back. For L0b, the answer key is counted on the 156 VerilogEval references.

Result. L0a: all 476 pairs of L1 runs are byte-identical (204, 204 and 68 sets); the L2 clusters are identical on 271 of 272 partitions, and the one difference is a spectral nondeterminism that also occurs between two runs of the base design. The protocol also exposed a rename-script defect (library `.blackbox` model names were renamed, which silently removed every hard proposal on renamed floors), which was fixed. L0b: all 54,306 gate nodes of the 150 designs that build carry a label, a cell type and an HDL span; module coverage is 1.000 and signal coverage has geometric mean 0.986; six of the 156 references do not build.

Detail of L0b. The experiment is run on the 156 VerilogEval spec-to-rtl reference designs (Section IV-D) next to the micro suite. The experiment builds the three floors `f1-k6`, `f2` and `f3` for every reference (`f1-earch` is not built; one architecture is enough because `f2` and `f3` do not depend on it).

TABLE XVIII
THE L0 CEILING ON `f3`, MICRO SUITE AND VERILOGEVAL REFERENCES. BOTH COLUMNS ARE COUNTS OVER ALL BUILT DESIGNS; THE LAST ROW IS THE PER-DESIGN COVERAGE OF DECLARED SIGNALS.

	Micro	VerilogEval
Source designs	68	156
Built on `f3`	65	150
Refused by the reader	3	2
Failed equivalence gate	0	4
Gate nodes, each with a label row	50,047	54,306
Rows with `cell_type` and HDL span	100%	100%
Module scopes / reference modules	112 / 113	150 / 150
Leaf signal names / declared signals	611 / 666	841 / 745
Per-design signal coverage: median	100%	100%
minimum	55.6%	75.0%
geometric mean	–	0.986

Coverage of the build. Of the 156 references, 150 build on all three floors, and `loki check` reports zero errors on the assembled dataset. Two are skipped by Odin III's SystemVerilog reader: `Prob151_review2015_fsm` and `Prob156_review2015_fancytimer` (both Q4), with `ODIN3_ERR_PARSE ... an explicit 4-bit cast is not supported`. Four fail Loki's own equivalence gate and are recorded as failed rather than forced in: `Prob031_dff` (the reference initialises `q` to `1'hx`; the writer-versus-Odin III BLIF check reports `differs` on `f1` and `f2`, while `f3` was equivalent) and `Prob046_dff8p`, `Prob078_dualedge` and `Prob145_circuit8`, whose negedge flip-flops make `equiv-check` return an error status for both comparisons. One design, `Prob144_conwaylife`, was killed by Wyrd's default 2 GB step memory limit (peak 4.3 GB) and built after a rerun with an 8 GB limit. Reading `.sv` sources needed two fixes in Loki (SystemVerilog is read with `--nolatches` only, since `--vtr-compat` is refused for it, and the slang adapter is registered with the process); both are in `loki/odin.py` and apply to the experiment run.

The ceiling. On `f3` the 150 designs have 54,306 gate nodes. Every node has a label row, a `cell_type` and an HDL span: the label, `cell_type` and HDL coverage have geometric mean 1.000 over designs. Label coverage is rows over nodes, so it shows that a row exists, not that its fields are informative; unmatched cells stay `unknown`. The 150 module scopes of the 150 reference modules are all present (module coverage 1.000). Signal coverage, the recovered leaf signal names over the signals the reference declares, has geometric mean 0.986 (arithmetic mean 1.029): 120 designs are at exactly 1, 28 are below it, and 2 are above it, where flattened array elements add names. The micro suite shows the same shape with a slightly lower ceiling (Table XVIII): 99.1% of module scopes and 91.7% of signal names (an aggregate ratio, where VerilogEval's 841 over 745 is inflated by those two designs; capped at each design's declared count it is 715 over 745, 96.0%). The two sets agree on the message of Appendix A-B: every node has a label row, a `cell_type` and an HDL span,

and what the mapping erased is the signal names, not the label rows. (Label coverage counts rows, not informative values: a cell that no rule matched keeps the value `unknown`.)

Caveats, reported with the numbers. (i) `--nolatches` turns inferred-latch paths into don't-care and `lower --async sync` turns asynchronous resets into synchronous clears, so a built netlist can differ from the reference's behaviour even though Loki's own checks (netlist against its anonymised twin and against Odin III's writer) pass; the build gate guarantees internal consistency, not equality with the reference. (ii) The reference module, instance and signal counts come from pyverilog where it parses the reference (130 of the 150) and from a regex fallback otherwise (20); the regex counts are approximate and the reference signal count is the weakest number in this section. (iii) The same pyverilog limit decides the complexity quartiles (Section IV-D2). (iv) A train/test split by design is fixed by the manifest rule (117 train, 33 test on `f3`); it is not used here, where no model is fitted.

Verdict. L0a **USEFUL**: naming is neutral, so nothing the pipeline recovers is an artefact of a name. L0b **USEFUL**: the answer key covers every gate node and every reference module; what the mapping erased is signal names, not label rows.

Record. Merged as loki PRs #36 (2026-10-23), #39 and #41 (2026-10-24) and #43 (2026-10-24); register rows L0a, L0b.

B. L1: what the flow already reverse engineered, and what it erased (L1h, L1c, L1s)

Hypotheses. L1h: structures that the synthesis flow already reverse engineered (hard adders and multipliers) are recovered almost perfectly. L1c: calibration makes the predicted probabilities honest. L1s: the flow also recovers the structures that synthesis erased.

Experiment. L1 runs on the renamed floors with a calibration fitted on the training designs, and is scored against the answer key (precision, recall, macro-F1, ECE).

Result. Table XIX reports L1 on the renamed floors (identical to base, as the neutrality check requires). The head-line result is the `hard` detector: hard adders recover 67 of 67 with precision 0.985, hard multipliers 30 of 30, because the synthesis flow itself already reverse engineered them and `unmap` says so. The rest is the honest difficulty: on f1 the calibrated macro-F1 is 0.354 (counted 271, matched 107, ground truth 688), with `ff_en` at 0.364, registers at 0.021 and muxes at 0.018; on f2 it is 0.108 (91/19/686), with adders at F1 0.237 (precision 1.000, recall 0.134) and most other kinds near zero. Without calibration the flow proposes far more and confirms little: f1 macro-F1 0.202 with 1,128 counted (323 of them declined-adder proposals), f2 0.075. Calibration is honest about itself: ECE 0.217 calibrated against 0.650 uncalibrated.

TABLE XIX
L1 RECOVERY ON THE RENAMED FLOORS, PER KIND. PRECISION/RECALL/F1 AGAINST THE L0 GROUND TRUTH; CALIBRATED (MICRO-FIT) COLUMNS. THE HARD KINDS RECOVER FROM `UNMAP` FACTS; THE REST MUST BE RE-FOUND FROM THE GATES.

Kind (f1)	Precision	Recall	F1
hard_add	0.985	1.000	0.993
hard_mul	1.000	1.000	1.000
ff_en	–	–	0.364
register	–	–	0.021
mux	–	–	0.018
mux_bit	–	–	0.081
macro-F1 (all kinds)			0.354
Kind (f2)			
adder	1.000	0.134	0.237
subtractor	1.000	0.074	–
mux_bit	–	–	0.100
adder_cin	–	–	0.029
macro-F1 (all kinds)			0.108

The gate netlist (f3), hypothesis F3. Gate-Breaker's own input is a cmos6 gate netlist, so we built the f3 floor (65 of 68 micro designs; the gate reader refuses three hard-block sources) and asked whether L1 recovers structure from it. On the 15 f3 test designs (23,098 gate nodes) no construct is counted (0 of 144 ground-truth instances, macro-F1 0.0) and no hypothesis is committed: the best calibrated confidence in any test design is a bus at $p = 0.764$, below the 0.9 commit threshold. Without calibration the detectors count 1,166 constructs for 2 matches (macro-F1 0.038), and 2,569 counted bus hypotheses match none of the 120 ground-truth buses. The answer key itself is complete at this floor: all 50,047 gate nodes carry a label row, a cmos6 cell type and an HDL span. F3 is **DISMISSED**; it is the reason for the next experiment.

The similarity probe on f3. Gate-Breaker's AST and CFG similarity (Section II) are the measures on which a recovered text is compared with the reference RTL. With the f3 calibration L1-stop is byte-identical to L0-stop, so both measures have a delta of exactly 0.0000 in every quartile. Uncalibrated, L1 commits 1,535 hypotheses in 11 of the 15 designs and moves the means by +0.0111 (AST) and -0.0075 (CFG) with mixed signs per quartile; the commits are mostly wrong `eq_const` and bus pieces, so this is not evidence about recovery quality. A flatten control (Odin III `proc; flatten`, which also lowers the committed processes) inlines the committed hierarchy and raises both measures over the gate netlist: AST by +0.0533 over all 15 designs (0.4576 against 0.4044) and CFG by +0.0559 over the 12 designs the probe can score (0.6941 against 0.6382). Algorithms 2 and 3 of Gate-Breaker read one module, so a recovery that produces hierarchy is penalised until it is flattened: a limit of the measure and not a property of the recovery. The per-kind table and the per-quartile table are in Appendix B.

Hypothesis R2-L1 (round 2): the f3 failure is the detectors' vocabulary and not missing information. The L1 detectors read LUT covers, and the f3 netlist is a cmos6 gate netlist. The hypothesis predicts that re-mapping the same gates into LUT6 recovers close to the f2 rate. We read each f3 `netlist.blif` into Odin III, run `abc -lut 6`, and give the result to `loki`

TABLE XX
ROUND-2 L1 (RQ-L1): RECOVERY FROM THE F3 GATE NETLIST RE-MAPPED TO LUT6 AGAINST THE NATIVE F2 FLOOR, SAME 65 MICRO DESIGNS. TEST IS THE 15 HELD-OUT DESIGNS, ALL THE 65. CALIBRATION: THE MICRO-FIT F2 TABLE (GATE ROWS: THE F3 TABLE). HARD KINDS EXCLUDED FROM COUNTS AND AVERAGES; MACRO-F1 AVERAGES THE KINDS WITH A COUNTED PREDICTION.

Netlist	Calib.	Split	Runs	Counted	Matched	GT	micro-F1	macro-F1
f3 gate netlist	calibrated	test	15	0	0	116	0.000	0.000
	uncalibrated	test	15	1,166	2	116	0.003	0.038
f3 re-mapped to LUT6	calibrated	test	15	68	2	116	0.022	0.076
	calibrated	all	65	92	14	580	0.042	0.062
	uncalibrated	test	15	252	4	116	0.022	0.044
	uncalibrated	all	65	921	31	580	0.041	0.049
f2 floor (native LUT6)	calibrated	test	15	72	4	116	0.043	0.153
	calibrated	all	65	88	16	579	0.048	0.095
	uncalibrated	test	15	255	13	116	0.070	0.121
	uncalibrated	all	65	930	32	579	0.042	0.070

`re` as an f2 netlist. Odin III drops the cell names in `abc`, so each of the 8,712 LUTs is expanded back to the gates it covers: a bit-parallel simulation signature matches its output to the f3 net of the same function, and the cover is the cone of that net down to the matched inputs. 0.230 of the LUTs ABC resynthesised implement a function that is no f3 net; they take the support-contained share of their consumers' covers, and 153 are left with an empty cover. The expanded hypotheses are then scored against the unchanged f3 key, and the native f2 floor of the same 65 designs, with the same calibration, is the comparison (Table XX). On the gate netlist the calibrated flow matches 0 of 116 instances in the 15 test designs. After the LUT6 re-map it matches 2 (native f2: 4), and over all 65 designs 14 against 16 of 580, a micro-F1 ratio of 0.87 in-sample (the 65 designs include the designs the f2 calibration was fitted on) against 0.51 on the held-out test split. The decision rule asked for a ratio of at least 0.8 on both splits, so H1 is inconclusive. The in-sample recovery is that of f2, on the same kinds (adders, subtractors, muxes), but the 15-design test split holds only a handful of matches (2 against 4), so its ratio is noise in either direction. The result is consistent with the f3 failure being the detectors' vocabulary and not missing information in the gates, but does not show it; the f2 level it would recover to is itself low.

Verdicts. L1h **USEFUL**: hard adders 67 of 67 (precision 0.985) and hard multipliers 30 of 30, because the synthesis flow itself reverse engineered them. L1c **USEFUL**: ECE 0.217 calibrated against 0.650 uncalibrated. L1s **INCONCLUSIVE**: the recovery is weak (macro-F1 0.354 on f1, 0.108 on f2) but no threshold or baseline was set, so we do not call it a failure. The LUT6 re-map (round-2 hypothesis R2-L1) is **INCONCLUSIVE**: it reaches the f2 level in-sample but the held-out split has too few matches to decide.

Record. loki PR #36 (2026-10-23) and #39 (2026-10-24); register rows L1h, L1s, L1c, F3, R2-L1.

C. L2: clusters that miss the hierarchy, and a GNN that beats 1-WL (L2a, L2m, L2b)

Hypotheses. L2a: community methods find the designer's hierarchy and centroid methods do not. L2b: a GNN with node, graphlet, WL and NPN features beats the 1-WL baseline. L2m is a measurement, not a hypothesis: the modularity of the four clusterings.

TABLE XXI
L2 CLUSTERING AGAINST THE KNOWN MODULE HIERARCHY (68 DESIGNS, MEAN OVER THE RENAMED FLOORS, IDENTICAL TO BASE). MODULARITY (L2M) AND NMI(MODULE_PATH), THE HIERARCHY MEASURE (L2A), PER METHOD.

Method	Modularity	NMI
Louvain	0.4220	0.1360
Spectral	0.3620	0.1378
Affinity propagation	-0.2264	0.2708
k-means	-0.2912	0.3220

Experiment. Four clusterings of the 68 micro designs are scored against the known module hierarchy; the GNN is trained on the labelled training designs and scored on 13 test designs (3,932 test nodes), once.

Result. Table XXI scores the four clusterings against the known module hierarchy. The measure of the hierarchy is NMI against the module path of each cell, and it ranks the methods in the order opposite to the hypothesis: k-means 0.3220, affinity propagation 0.2708, spectral 0.1378, Louvain 0.1360. Modularity (L2m) is another matter: Louvain 0.4220 and spectral 0.3620 are well above zero and k-means (-0.2912) and affinity propagation (-0.2264) are negative, but modularity scores how dense the clusters are in the graph, not how well they match the designer's modules, so it is not a hierarchy measure and does not rescue L2a. 271 of 272 partition records are identical between base and renamed floors; the one difference exposed a determinism defect in the spectral path (repeated runs on the same base design give modularities -0.0079, -0.0410 and 0.0252), reported as an open issue rather than hidden.

The GNN result answers the paper's question directly. A node-only GraphSAGE loses to the 1-WL baseline on real micros (macro-F1 0.2413 against 0.2711); combining node, graphlet, WL and NPN features turns it around (0.3545 against 0.2711 on the earlier real-micro run), and the full labelled run of this experiment gives test macro-F1 0.4241 against 0.2717 (accuracy 0.8863 against 0.8492, 3,932 test nodes across 13 designs). The features carry the recovery, not the message passing alone.

Verdicts. L2a **DISMISSED**: the only hierarchy measure, NMI against the module path, ranks the centroid methods above the community methods (k-means 0.3220 against Louvain 0.1360), the reverse of the hypothesis. L2m **MEASUREMENT**: Louvain has the highest modularity (0.4220), a statement about graph density and not about the hierarchy. L2b **POSSIBLY GOOD**: macro-F1 0.4241 against 0.2717 for 1-WL, but it is one run on one split of 13 designs with no interval, and the node-only GNN loses to 1-WL (0.2413 against 0.2711); the claim is that the features carry the recovery, not that the GNN generalises.

TABLE XXII
OPERATOR-RECOVERY F1 (RQ-L7) OF THE EXISTING AND NEW L1 RUNS: THE MULTISET OF RECOVERED OPERATORS (CLASS AND OPERAND WIDTH) AGAINST THE OPERATORS OF THE L0 KEY, POOLED OVER DESIGNS. TYPE IGNORES THE WIDTH, TW REQUIRES IT.

Run set	Designs	Key ops	Got ops	P	R	type-F1	tw-F1
f1-k6, calibrated	68	468	169	0.657	0.237	0.349	0.204
f1-k6, none	68	468	964	0.233	0.481	0.314	0.166
f2, calibrated	68	468	21	0.810	0.036	0.070	0.061
f2, none	68	468	756	0.180	0.291	0.222	0.080
f3 gates, calibrated	15	108	0	–	0.000	0.000	0.000
f3 gates, none	15	108	1,161	0.009	0.102	0.017	0.008
f3 as LUT6, calibrated	65	463	19	0.895	0.037	0.071	0.062
f3 as LUT6, none	65	463	753	0.190	0.309	0.235	0.086
VerilogEval f2, calibrated	150	3,585	11	0.727	0.002	0.004	0.002
VerilogEval f2, none	150	3,585	1,151	0.196	0.063	0.095	0.042
VerilogEval f3, calibrated	150	3,585	2	1.000	0.001	0.001	0.001
VerilogEval f3, none	150	3,585	5,488	0.107	0.163	0.129	0.027

Record. loki PR #36 (2026-10-23); register rows L2a, L2m, L2b.

D. An instrument that sees recovery, and where the language model stands (R2-L7, R2-L2)

Hypotheses. R2-L7: a word-level operator F1 separates runs that AST and CFG similarity tie on. R2-L2: Loki's AST and CFG similarity is below the language model's on Q1 and Q2, does not stay flat from Q1 to Q4, and Loki's output is formally equal to its input.

Experiment. Round 2 is compute-only: no language-model call was made, and the language-model figures are the round-1 Gate-Breaker replication, read again. Every number in this subsection is a macro generated from the summary tables of `experiments/round2`. R2-L7 rescored every run set with the operator type-F1 (Section III-E); R2-L2 ran `loki re` on all 150 VerilogEval netlists and scored the output Verilog against the reference.

Result. *R2-L7: are AST and CFG blind to what Loki recovers?* Table XXII rescores every run set. The floors separate cleanly, which neither AST nor CFG can show because both exist only on f3: type-F1 0.349 on f1 (the hard detector), 0.070 on f2 and 0.000 on the f3 gates. On the f3 test designs (Table XXIII) the calibrated and uncalibrated L1 runs tie on both metrics, AST by 0.0111 and CFG by -0.0075, while operator type-F1 moves by 0.0173; the flatten control raises AST by 0.0422 and CFG by 0.0547 and leaves the recovery record, hence the operator F1, unchanged (0.0000). By the rule H7 is supported by the letter of the rule (weak evidence). The tie the rule was written for, the calibrated flow against the L0-stop netlist (both add nothing to the text), does not separate: their operator type-F1 is 0 against 0. The verdict rests on the calibrated flow against the uncalibrated one, where the calibrated side recovers no operator at all, and on the flatten clause, which holds by construction. Two cautions: the separating level is small (the uncalibrated detectors count mostly wrong constructs), and the flatten comparison holds the record fixed by construction, because the operator F1 reads the recovery record and AST and CFG read the text. The operator F1 does not replace them; it is the measure that sees recovery, where they see only form.

TABLE XXIII
AST AND CFG SIMILARITY AGAINST OPERATOR F1 ON THE 15 F3 TEST DESIGNS (THE F3-RECOVERY RUNS). CALIBRATED AND UNCALIBRATED L1 TIE ON AST AND CFG; THE FLATTEN CONTROL MOVES BOTH AND LEAVES THE RECOVERY RECORD UNCHANGED.

Condition	AST	CFG	Ops recovered	type-F1
L0-stop gate netlist	0.4044	0.6468	0	0.0000
L1 calibrated	0.4044	0.6468	0	0.0000
L1 uncalibrated	0.4155	0.6394	1,161	0.0173
L1 uncalibrated, flattened	0.4576	0.6941	1,161	0.0173

R2-L2: Loki against the language model on the same designs. Round 1 scored the language model on Q1 and Q2 of VerilogEval, and Loki's AST and CFG only on micro designs. We ran `loki re` on all 150 VerilogEval netlists at f2 and f3, calibrated (a calibration fitted on the 117 train designs of VerilogEval itself) and not, and scored the output Verilog (`design.v`) against the reference `.sv` with the round-1 AST and CFG code. 21 references that pyverilog cannot parse have no AST value for any candidate, and 6 f3 outputs exceed the CFG probe's size limit (`l2_coverage.csv`). On the designs both are scored on (Table XXIV), Loki's f3 output has AST similarity 0.763 in Q1 and 0.460 in Q2 (common designs, $n = 36$ and 36) against the model's 0.894 and 0.761, and CFG similarity 0.751 and 0.743 against 0.942 and 0.890; every gap is significant by a paired Wilcoxon test (`l2_paired.csv`). The calibration was fitted on the train designs and these means cover all 150, so we also give the held-out test split alone (`l2_testsplit.csv`): on its 16 Q1 and Q2 designs both tools score on, AST is 0.705 for Loki against 0.893 for the model and CFG 0.802 against 0.927, the same direction. (The calibrated f3 flow commits in only 7 designs, so the fit can change few outputs.) Across the four quartiles (Table XXV) the f3 AST over all scored designs ($n = 38$, 38, 32, 21; a different, larger design set than the common rows above) reads 0.754, 0.455, 0.605 and 0.505, so it falls from Q1 to Q4 without being monotone. The 21 references that do not parse fall in Q1 to Q4 as 0, 1, 5 and 15, so Q3 and Q4 are scored on the designs that parse, and the apparent fall may partly reflect which designs those are. The CFG stays flat (0.750 to 0.768). The calibrated f3 flow commits almost nothing, so its output is the L0 netlist; the uncalibrated flow, which does commit, scores lower, not higher.

What the correctness column proves is narrow. Every one of the 150 runs of each set ends `final_equiv = equivalent` (150 for the calibrated f3 set), which says that the flattened output is formally equal to the *input netlist*. It does not say the output matches the reference RTL: that netlist itself passes the official testbench in only 128 of the 150 designs (the round-1 self-check; inferred latches and asynchronous resets are the failures), so Loki's output is exactly as correct as its input, no more. The model is checked against the testbench directly, but only its correct samples enter the similarity columns. On operator recovery the model is far ahead: on the common Q1 and Q2 designs its operator type-F1

TABLE XXIV
HEAD-TO-HEAD ON THE VERILOGEVAL DESIGNS THE LANGUAGE MODEL WAS SCORED ON (Q1 AND Q2, FUNCTIONALLY CORRECT SAMPLES; AST DEPTH 3 AND CFG AS IN TABLE XII). LOKI COLUMNS: THE OUTPUT VERILOG OF THE FLOW AT THAT FLOOR AGAINST THE REFERENCE `.sv`; `L0` IS THE NETLIST BEFORE L1.

Metric	LLM (round 1)	f3 netlist (L0)	f3 L1 cal.	f3 L1 none	f2 L1 cal.	f2 L1 none
Q1 AST	0.894	0.763	0.763	0.680	0.804	0.720
Q1 CFG	0.942	0.751	0.751	0.565	0.821	0.664
Q2 AST	0.761	0.460	0.460	0.438	0.459	0.448
Q2 CFG	0.890	0.743	0.743	0.679	0.787	0.764
Q1+Q2 AST	0.827	0.611	0.611	0.559	0.632	0.584
Q1+Q2 CFG	0.916	0.747	0.747	0.622	0.804	0.714

TABLE XXV
LOKI ON ALL 150 VERILOGEVAL DESIGNS BY COMPLEXITY QUARTILE (MEAN AST AND CFG OF THE OUTPUT VERILOG AGAINST THE REFERENCE; LAST COLUMN: OPERATOR TYPE-F1 OF THE UNCALIBRATED F2 RUN). CALIBRATED F3 EQUALS THE F3 NETLIST WHEREVER L1 COMMITS NOTHING.

Quartile	n	f3 AST	f3 CFG	f3 cal. AST	f3 cal. CFG	f2 cal. AST	f2 cal. CFG	op F1
Q1	38	0.754	0.750	0.754	0.750	0.793	0.820	0.061
Q2	38	0.455	0.741	0.455	0.741	0.455	0.784	0.075
Q3	32	0.614	0.751	0.605	0.750	0.607	0.811	–
Q4	21	0.505	0.768	0.505	0.768	0.507	0.870	–

TABLE XXVI
RQ-L9, PRIMARY SET P (M2N+V2N, F2, UNCALIBRATED): COUNTED CANDIDATES, FALSE ONES, AND THE SHARE OF THE FALSE ONES PER CLASS (RULES OF `EXPERIMENT.md`).

kind	cand.	false	same ext.	frag. rel.	frag. other	straddle	partial	noise
ff_en	1,070	1,067	0.01	0.99	0.00	0.00	0.00	0.00
mux_bit	344	336	0.00	0.61	0.39	0.00	0.00	0.00
eq_const	1,105	1,096	0.00	0.18	0.82	0.00	0.00	0.00
register	155	116	0.03	0.64	0.15	0.19	0.00	0.00

TABLE XXVII
RQ-L9 (H9B), SET P: GROUND-TRUTH INSTANCES WITH AT LEAST TWO CELLS, THE SHARE WHOSE FRAGMENT UNION COVERS AT LEAST 0.8 OF THEM (q), THE SAME WITH CANDIDATES OF ANY KIND, AND THE MEAN FRAGMENT COVERAGE.

kind	GT	matched	q frag.	q any	mean cov.
ff_en	156	3	0.122	0.160	0.618
mux_bit	368	2	0.005	0.136	0.016
eq_const	995	5	0.040	0.112	0.089
register	289	39	0.194	0.228	0.497

is 0.465 (functionally correct samples elaborated with the same Odin III front as the key) against 0.081 for Loki's calibrated f2 run and 0.074 uncalibrated (the calibrated f3 run recovers no operator, 0.0, so the comparison uses f2, a different floor from the one AST and CFG use). By the rule, H2 is inconclusive: Loki is below the model on both metrics in both quartiles (yes), the equivalence holds in all runs (yes), and the similarity does not stay flat from Q1 to Q4 (no). The positioning this measures is that the model is readable and recovers the operators of small designs but is unverified, and Loki is verified but partial; neither number alone is the result, and the arm that combines them (a model given the ReverseProvenance record, L8) was not run.

Verdicts. R2-L7 **POSSIBLY GOOD**: the rule supports it, but the margins are small on few pairs (tie pair ΔF1 0.0173; the flatten control moves AST by 0.0422 and leaves the operator F1 unchanged), and the calibrated flow recovers no operator at all, so we use the operator F1 only as an added instrument beside AST and CFG and not as proof of an innovation. R2-L2 **INCONCLUSIVE**: Loki is below the model on Q1 and Q2 (AST 0.763 and 0.460 against 0.894 and 0.761) and its output is equal to its input in 150 of 150 runs, but the AST falls from Q1 to Q4, which may partly reflect which references parse, so the data position the two tools (the model readable and unverified, Loki verified and partial) without deciding between them.

Record. loki PR #44 (2026-10-24); register rows R2-L1 to R2-L7.

E. Why the detectors miss: fragments and words (H9a, H9b)

Hypotheses. H9a: most false candidates are fragments of a word of a related kind. H9b: the fragments of a word cover most of it, so grouping them recovers the word.

Experiment. Uncalibrated, the detectors propose many candidates that almost never match (Table XXVI); calibrated, they commit almost nothing. Round 3 is compute-only on the round-2 runs and the L0 answer key (no language-model call, no new detector); every figure below is a macro generated from the summary CSVs of `experiments/round3`. The round-2 counts that motivate the question (for instance `ff_en` 732 candidates and one correct on the held-out designs) were known when the thresholds were set. Analyses added after the data was seen are labelled *exploratory* and decide nothing. We added the runs the round-2 sets lacked (micro f3, calibrated and not, and micro f1-k6, uncalibrated; one micro f3 design timed out) and score every set, but the decisions use the primary set P: the micro and VerilogEval f2 runs without calibration.

Result. A false candidate is a counted candidate the one-to-one scorer does not match. We classify each (rules in the protocol): the same extent as a ground-truth instance of its family (a lost tie or a type slip), a fragment of a larger instance of its family, a fragment of an instance of another type, a straddle of several, a partial overlap, or noise. Over the 2,615 false candidates of the four kinds in P, 0.583 (1,525) are fragments of an instance of the matching or a related family. Per kind the related-fragment share is 0.988 for `ff_en`, 0.638 for `register`, 0.610 for `mux_bit` and only 0.175 for `eq_const`; 3 of four kinds reach 0.5, so H9a is supported *by the letter of the rule; not robust*. Three qualifications go with it.

First, the noise class is vacuous. The ground-truth instances cover every non-I/O, non-constant cell of the f2 datasets (9,381 of 9,381 in VerilogEval, 9,130 of 9,130 in the micro set), so rule 6 of the protocol could never yield noise, and the noise share 0.000 is structural. The arm of the H9a rule that reads "not supported when the noise share is at least 0.5" was unreachable by construction, and the sentence that no wrong

TABLE XXVIII
RQ-L9 FOR EVERY RUN SET (FOUR DECISIVE KINDS POOLED): RUNS, CANDIDATES, FALSE CANDIDATES AND THE SHARE THAT ARE RELATED FRAGMENTS, OTHER-TYPE FRAGMENTS, NOISE.

set	runs	cand.	false	frag. rel.	frag. other	noise
M2N	65	705	686	0.28	0.69	0.00
M2C	65	71	65	0.95	0.05	0.00
V2N	150	1,969	1,929	0.69	0.30	0.00
V2C	not run					
V3N	150	5,005	4,997	0.29	0.71	0.00
V3C	150	2	0	0.00	0.00	0.00
M3N	64	1,406	1,405	0.23	0.77	0.00
M3C	not run					
M1N	68	496	469	0.38	0.57	0.00

TABLE XXIX
ADDITIONAL, NON-DECIDING ANALYSES, SET P: SHARE OF FALSE CANDIDATES THAT ARE RELATED FRAGMENTS (BY CANDIDATE, BY DISTINCT CELL), THE NOISE′ SHARE (CELLS ONLY IN PLAIN-GATE GROUND-TRUTH WORDS), q OVER ALL WORDS AND OVER THE WORDS TOUCHED BY A CANDIDATE.

kind	frag. rel.	frag. rel. (cells)	noise′	q	q touched
ff_en	0.99	0.98	0.00	0.122	0.123
mux_bit	0.61	0.61	0.01	0.005	0.006
eq_const	0.18	0.18	0.52	0.040	0.042
register	0.64	0.49	0.00	0.194	0.229
pooled	0.58	0.56	0.22	0.065	0.069

candidate is noise carries no information. As an exploratory replacement we count noise′: a false candidate all of whose cells lie only in plain-gate ground-truth words (`$and`, `$or`, `$not`, `$xor`, `$xnor`, the `$logic` and `$reduce` gates). Pooled, noise′ is 0.220 (574 candidates); per kind it is 0.522 for `eq_const`, 0.006 for `mux_bit`, and 0.000 for `ff_en` and 0.000 for `register` (Table XXIX). Half of the false `eq_const` candidates are thus glue logic, not a piece of a word.

Second, the `ff_en` fragments are near-tautological. An `ff_en` candidate has two cells (a flip-flop and its enable LUT), and the ground-truth footprint of a register or `$dffe` word is the whole word, so a correct per-bit slice of a word counts as a fragment of it; 1,054 of the 1,525 related fragments are `ff_en`. For `ff_en` the detector finds correct bit slices and the miss is word-level grouping and extent. Without `ff_en` (exploratory) the pooled related-fragment share over the other three kinds is 0.304 (1,548 false candidates, 2 of three kinds at or above 0.5), and 0.271 when the largest design is also left out. For `mux_bit` and `register` the fragment reading holds at the stated shares; for `eq_const` it is a detector problem: 0.822 of its false candidates are fragments of an instance of another type (mostly single-LUT `$and` and `$or` words, registers and `fsm` instances), so the LUT computes something that looks like a constant compare and is not one. The `mux_bit` candidates are 0.390 inside an instance of another type (`$shiftx` words among them).

Third, the result rests on one design. Most `ff_en` candidates come from `ve_Prob153_gshare`, where they sit inside a mux tree, and overlapping candidates inflate their count. With that design left out (exploratory, not the rule) 1,863 false candidates remain, 0.415 of them related fragments and 0.563 fragments of another type, so the pooled share falls under the 0.5 of the rule and H9a would be inconclusive; `ff_en` stays at 0.966 and `register` drops to 0.333. Counted over the 4,004 distinct (design, cell) pairs covered by false candidates instead of over candidates (exploratory), the pooled related-fragment share is 0.558. We therefore state H9a with both numbers: supported by the letter of the rule at 0.583, not robust (0.415 without `ve_Prob153_gshare`, 0.304 without `ff_en`).

H9b asks whether grouping the fragments would recover the word: the share of ground-truth instances whose fragment union covers 0.8 of them is 0.065 pooled (`ff_en` 0.122, `register` 0.194, `eq_const` 0.040, `mux_bit` 0.005), and no kind reaches 0.5, so H9b is not supported. The denominator holds words no candidate touched; over the 1,698 words touched by at least one candidate (exploratory) the share is 0.069. The shortfall is therefore detector misses plus non-tiling, not one of them: 0.065 of all words, 0.069 of the touched ones. Even with candidates of any kind the cover stays low (`register` 0.228, `mux_bit` 0.136). The fragments exist, but they do not tile the words: a merge step over the present detector output would recover few words. The other sets (Table XXVIII) have fewer candidates when calibrated, and the gate floor f3 proposes far more false `eq_const` candidates than any f2 set.

RQ-L11: the FSM chain. Measurement only. In the f2 ground truth of VerilogEval (150 designs) 197 registers feed

TABLE XXX
RQ-L11, GROUND TRUTH: FEEDBACK REGISTERS, CONSTANT-COMPARE DECODERS ON THEM, SINKS (MUX SELECTS AND ENABLES) AND CHAINS, AND THE MUXES WITH A SELECT (ENABLED WORDS) WHOSE SELECT (ENABLE) IS A CHAIN SINK.

dataset	floor	designs	with fb.	with chain	fb. reg.	fsm flag	decoders	sinks	chains	mux S chain/all	enable chain/all
micro	f2	68	3	1	3	0	11	3	12	3/41	0/3
micro	f3	65	3	1	3	0	11	3	12	3/41	0/3
micro	f1-k6	68	3	1	3	0	11	3	12	3/41	0/3
veval	f2	150	55	12	197	16	41	30	89	15/43	15/152
veval	f3	150	55	12	197	16	41	30	89	15/43	15/152

TABLE XXXI
RQ-L11, DETECTORS: CHAIN ELEMENTS PROPOSED BY A COUNTED CANDIDATE WITH JACCARD AT LEAST 0.5 OF ANY KIND (IN BRACKETS: OF THE RIGHT SCORED TYPE), CHAINS WITH ALL THREE ELEMENTS PROPOSED, AND GROUND-TRUTH FSM INSTANCES MATCHED BY AN `FSM` CANDIDATE.

set	fb. register	decoder	sink	complete chains	fsm
M2N	0/3 (0)	1/11 (0)	1/3 (0)	0/12 (0)	0/0
M2C	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0
V2N	41/197 (26)	15/41 (9)	5/30 (0)	5/89 (0)	6/16
V2C	3/197 (0)	1/41 (0)	2/30 (0)	1/89 (0)	0/16
V3N	13/197 (3)	6/41 (2)	3/30 (0)	4/89 (0)	3/16
V3C	4/197 (0)	2/41 (0)	2/30 (0)	2/89 (0)	1/16
M3N	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0
M3C	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0
M1N	0/3 (0)	0/11 (0)	0/3 (0)	0/12 (0)	0/0

back to themselves, of which 16 the front end flags as FSM state; 41 constant-compare decoders read such a register, and the chains (register, decoder, select or enable) occur in 12 designs. Of the muxes with a select, 15 of 43 (VerilogEval) and 3 of 41 (micro) take it from such a decoder, 0.214 together, so the FSM, decode, select chain is real but covers a minority of the control muxes of these small designs. The detectors see little of it (Table XXXI): in V2N they propose 41/197 feedback registers, 15/41 decoders, 5/30 sinks, all three elements of 5/89 chains, and 6/16 FSM instances as an `fsm` candidate. The chains are triples (register, decoder, sink) and reuse their elements: the 89 triples hold only 30 distinct sinks and 41 distinct decoders. Counted on distinct elements, the detectors propose 5/30 of the sinks and 15/41 of the decoders, against 5/89 complete triples.

Implication for a word-first detector. The classification cannot say that the wrong candidates are not noise, because the ground truth covers every cell and the noise class was unreachable; the exploratory noise′ share is 0.220 pooled and 0.522 for `eq_const`. The fragment share depends on the kind and, for the pooled figure, on one design and on the `ff_en` slices: `ff_en`, `mux_bit` and (less so) `register` fail as pieces of larger constructs, while `eq_const` fails as a detector problem (the local pattern is too weak, and a word-level view, the select and enable fan-out across a word, must decide whether a compare is a decoder). Because the fragments do not tile the words, grouping the present candidates is not enough, and a detector should construct the word first and score it as a unit. Concretely, we propose to find (i) register words by shared clock, enable and reset across flip-flops; (ii) datapath words by per-bit cones that are NPN-isomorphic, seeded from the ports but not limited to them (the port words reach registers and nothing past the first layer); (iii) control nets by select and enable fan-out across a word; and (iv) FSMs by a feedback strongly connected component plus the decoders on it. Related work attacks the word problem along the same lines: WordRev [8] derives word-level structures from the gate-level netlist of a circuit and lets the user specify sequences of word-level operations whose gates it then extracts, and DANA [14] recovers high-level registers by analysing the flow of data between individual flip-flops. We ran DANA (through HAL [12]) on the round-5 sets (Appendix A-I); WordRev was not run and no comparison with it is claimed. The next round builds this detector (L8, the language model given the ReverseProvenance record, stays deferred).

Verdicts. H9a **POSSIBLY GOOD**: by the letter of the rule about half of the false candidates are fragments of a related kind (0.583 pooled), but the result fails leaving out one design (0.415) and the near-tautological `ff_en` slices (0.304), so we state it only with that fragility. H9b **DISMISSED**: a word's fragments cover it in 0.065 of the words, so the fragments exist but do not tile the words; this is the motive for the word-first detector of Appendix A-I. The FSM chain is a measurement and has no verdict.

Record. loki PR #45 (2026-10-25); register rows H9a to H11.

TABLE XXXII
STYLE-RECOVERY CEILING (CLASSES OF EQUIVALENT NETLISTS OVER CODING STYLES) ON THE TWIN SET, PER FAMILY AND FLOOR.

family	f1-k6	f2	f3
all	0.470	0.482	0.500
adder	0.667	0.667	0.667
alu	0.333	0.472	0.528
comparator	0.333	0.333	0.333
counter	0.333	0.333	0.333
dpfsm	0.500	0.500	0.500
fsm	0.500	0.500	0.500
mux	0.562	0.562	0.667
pipeline	0.333	0.333	0.333
regfile	0.500	0.500	0.500
register	0.333	0.333	0.333
shifter	0.625	0.625	0.625
subtractor	0.667	0.667	0.667

F. What is recoverable in principle (H14a)

Hypothesis. H14a: the synthesis flow destroys style information, so that the pooled style-recovery ceiling at f3 is at most 0.60 (at least 40% of style variants are indistinguishable).

Experiment. The ceiling is measured on a twin set (the same design in every coding style) at f1-k6, f2 and f3. For each twin base design, with S styles and C distinct netlist classes at the floor, the style-recovery ceiling is C/S, the accuracy of the best possible guesser of the coding style from the netlist under a uniform prior (a Weisfeiler-Lehman class, confirmed by exact isomorphism where the netlist is small).

Result. The style-recovery ceiling is 0.470 at f1-k6, 0.482 at f2 and 0.500 at f3 (Table XXXII).

Verdict. H14a **USEFUL**: the ceiling is 0.5000 at f3, so what any recovery can do is limited by what the flow destroyed, whatever the detector.

G. A synthetic test bed: the idiom corpus (H12a)

Hypothesis. H12a: a seeded corpus of synthetic idiom designs builds on every floor (at least 0.95 of the designs pass every equivalence check on every floor).

Experiment. The corpus generator has twelve idiom families, each parameterised by width, depth, encoding, reset and enable style and coding style, giving 1,200 designs split by parameter region: train values only (840), the same region with unseen text (120, iid), and at least one test-only value, such as an

TABLE XXXIII
THE ROUND-4 CORPUS: DESIGNS PER FAMILY AND SPLIT, AND DESIGNS THAT BUILD ON F1-K6, F2 AND F3 WITH EVERY EQUIVALENCE CHECK PASSING.

family	train	iid test	ood test	ok on all floors
adder	70	10	20	100
alu	70	10	20	100
comparator	70	10	20	100
counter	70	10	20	100
dpfsm	70	10	20	100
fsm	70	10	20	100
mux	70	10	20	100
pipeline	70	10	20	100
regfile	70	10	20	100
register	70	10	20	100
shifter	70	10	20	100
subtractor	70	10	20	100

unseen width or coding style (240, ood). Odin III builds each on f1-k6, f2 and f3 and the three equivalence checks are run.

Result. 1,200 of 1,200 corpus designs pass every equivalence check on all three floors (Table XXXIII); the smallest family has 100 designs. The corpus is a test bed and not a source of a lookup table: a signature dictionary mined from it wins on none of the seven eligible kinds on real designs (won=0/7 floor=f2 sets=micro,vetest), and a GNN trained on more corpus designs does not improve on real ones (slope -0.0026 per doubling, 95% interval $[-0.0097, 0.0094]$); both are DISMISSED (H12b, H12c).

Verdict. H12a **USEFUL**: the corpus builds (1200 of 1200). This is a validity check of the test bed and says nothing about transfer.

H. Repetition in big designs (H13a, H13b, H13c)

Hypotheses. H13a: big designs are mostly repeated structure (at least 0.50 of their words lie in a repeat group). H13b: big designs repeat more than small ones (by at least 0.15). H13c: the repeated instances share the dictionary signature (at least 0.70 of the repeat groups).

Experiment. A word is a group of cells that act as one multi-bit signal; a repeat group is a set of at least two words of the same type and width whose cells induce the same subgraph (a Weisfeiler-Lehman hash of radius 2 over the cell types). The share of repeated structure is the share of the words with at least two cells and width at least two that lie in a repeat group. It is measured on the large designs of the VTR suite that Odin III builds at f3 (six designs: `blob_merge`, `sha` and `stereovision0` to `stereovision3`; Table XXXIV) and, for H13b, on the small designs (the micro designs, the VerilogEval references and the training part of the corpus). For H13c the repeat groups of the big designs are checked at f2, where cells absorb their neighbours and the signatures can differ: the share of groups whose members have identical multisets of dictionary signatures.

TABLE XXXIV
BIG VTR DESIGNS AT F3: CELLS, ODIN III PEAK RSS (MB, ALL THREE FLOORS), WORDS WITH AT LEAST TWO CELLS AND WIDTH ≥ 2, SHARE OF THEM IN A REPEAT GROUP, MEMORIES, BUSES OF AT LEAST 8 BITS, NETS WITH AT LEAST 8 READERS, FSM WORDS.

design	cells	RSS	words	repeated	mem	buses	hi-fanout	fsm
blob_merge	31,778	4915.6	1,090	0.744	0	51	717	1
sha	10,629	841.7	225	0.924	0	29	99	0
stereovision0	56,208	1665.9	1,814	0.942	0	1263	182	0
stereovision1	142,932	1863.8	1,892	0.972	0	830	1430	48
stereovision2	254,419	4793.1	2,001	0.982	0	951	1724	0
stereovision3	859	75.8	145	0.834	0	9	10	1

Result. H13a: 6,653 of 7,167 words of the big designs lie in repeat groups, a share of 0.9283 (Table XXXIV gives each design: from 0.744 for `blob_merge` to 0.982 for `stereovision2`). H13c: of 911 repeat groups, 0.8606 have identical signature multisets. H13b: the big share is 0.2404 above the small share (0.9283 against 0.6879), above the 0.15 margin, but the result does not survive a change of the small baseline: against the micro designs and the VerilogEval references the difference is $+0.053$ (inconclusive by the rule), and against the VerilogEval references alone it is $+0.009$ (not supported).

Verdicts. H13a **USEFUL**: 0.9283 against the threshold 0.50, in all six designs (the smallest share is 0.744). H13c **USEFUL**: 0.8606 against the threshold 0.70. H13b **POSSIBLY GOOD**: it holds against the pooled small baseline that the rule fixed and flips with the choice of baseline, so it is stated only with that fragility and is not used as evidence on its own.

Record. loki PR #46 (2026-10-25); register rows H12a to H14d.

I. The word-first detector and DANA (H16a–H16f, RQ-H1)

Hypotheses. The word-first detector, built from the two lessons of the fragment study, makes six claims: H16a, word tiling beats the earlier detectors on held-out real designs; H16b, grouping by clock, enable and reset recovers the enabled-register words; H16c, a word-level `eq_const` beats the per-bit one; H16d, the FSM-decode-select chain is completed more often; H16e, words grown from registers reach the datapath beyond a size-matched null; H16f, repetition is evidence on big designs. RQ-H1: the detector recovers register words better than, equal to, or worse than DANA.

The fragment study says what to build: words from structure. Round 5 builds it (can words, control nets and FSMs be found first, and the rest built around them?). Two opt-in detectors, `wordfirst` and `wordgroup` (`loki/re/detectors/wordfirst.py`, `docs/user/wordfirst.md`), were added to `loki re`; the default detector list and every earlier calibration are unchanged. `wordfirst` groups flip-flops by clock, enable and reset into register words (with `ff_en` and `ff_srst` words labelled by the control net), groups 2:1 mux cells by their shared select, takes the carry chains of the `arith` detector so that the carry net joins the bits, decides `eq_const` at word level (a single-minterm cone whose

TABLE XXXV
ROUND 5: WORD TILING Q (SHARE OF GT WORDS WHOSE FRAGMENTS COVER 0.8) AT F2.

Set	R2N	R2C	W	WB	null W	null WB
MI	0.212	0.019	0.223	0.237	0.042	0.048
VT	0.022	0.004	0.017	0.023	0.017	0.203
CI	0.390	0.328	0.312	0.397	0.206	0.294
CO	0.259	0.182	0.180	0.266	0.089	0.155

support lies in one register word or the primary inputs), proposes an FSM for a register word coupled through its own D cones, and grows words along isomorphic bit slices (readers and drivers) across registers, mux words and adders into `bus` words. `wordgroup` runs last and links the per-bit hypotheses to the words that contain them. The outputs are ordinary candidates with claims, so the prover, the commit step, the scorers and the Verilog writer work unchanged. Every threshold is an agent default (no fitted parameter): groups above 32 flip-flops are split by the Q-reader and D-driver signature, `eq_const` supports are at most 16, growth needs 0.75 of the bits and two levels.

The evaluation sets are the micros (MI), the 33 held-out VerilogEval designs (VT), the corpus iid and ood test splits (CI, CO) and the 13 big designs that built at f2 (BIG). The arms are the round-2 detectors uncalibrated and calibrated (R2N, R2C), `wordfirst` alone (W) and `wordfirst` with the per-bit detectors and `wordgroup` (WB); all runs are uncalibrated.

Experiment and result. *H16a: tiling.* On VT the pooled tiling of WB is 0.023 against 0.022 for the better round-2 arm (difference 0.001, bootstrap interval 0 to 0.062, 1,388 instances over 17 designs, Wilcoxon p 1). Word grouping does not tile the held-out real words; on the micros and the synthetic splits the tiling of WB and of the round-2 arm are close (Table XXXV). The null columns replace each counted candidate by random cells of the same size. On VT the null of WB (0.203) is above WB itself and above every arm (the largest arm value is 0.023), so the null clause of the decision rule (q(WB) minus the null at least 0.10) cannot hold on VT. With the VT baseline the difference clause needs q(WB) of 0.172 and the null clause 0.303, so a supported verdict needed q(WB) of about 0.303. The status does not depend on the null: the difference is below 0.05. The null depends on the size mix of the candidates: WB has 34 counted candidates of at least ten cells on VT against 20 for R2N, the largest has 1,016 cells (R2N: 64), and the null of all of WB's counted candidates is 0.203 against 0.203 for R2N, while the `ff_en` candidates alone give 0.192 for WB and 0.004 for R2N (exploratory). The tiling measure cannot separate structure from chance when a few candidates are large.

H16b: `ff_en` words. Pooled over all four sets the recall of `ff_en` rises from 0.008 (best round-2 arm) to 0.146 for WB (precision 0.001 to 0.014); the gain in recall (0.138) and in F1 (0.025) is below the margins of the rule (0.20 and 0.10)

TABLE XXXVI
ROUND 5, FLOOR F2: PRECISION/RECALL/F1 PER KIND AND SET FOR THE ROUND-2 DETECTORS (R2N, R2C) AND THE WORD-FIRST ARMS (W, WB). POOLED OVER THE DESIGNS EACH ARM HAS; GT IS THE FIRST ARM'S COUNT.

Set	Kind	GT	R2N	R2C	W	WB
CI	adder	49	1.00/0.14/0.25	1.00/0.14/0.25	1.00/0.14/0.25	1.00/0.14/0.25
CI	comparator	167	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
CI	counter	10	0.62/0.50/0.56	1.00/0.60/0.75	0.00/0.00/0.00	0.62/0.50/0.56
CI	eq_const	161	0.02/0.02/0.02	1.00/0.02/0.04	0.04/0.02/0.02	0.02/0.02/0.02
CI	ff_en	78	0.00/0.00/0.00	0.00/0.00/0.00	0.20/0.18/0.19	0.02/0.18/0.03
CI	ff_srst	70	0.00/0.00/0.00	0.00/0.00/0.00	0.31/0.39/0.35	0.04/0.37/0.07
CI	fsm	20	0.89/0.80/0.84	1.00/1.00/1.00	0.09/0.05/0.06	0.58/0.75/0.65
CI	mux	106	0.73/0.08/0.14	1.00/0.07/0.12	0.67/0.06/0.10	0.40/0.08/0.13
CI	mux_tree	19	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
CI	register	163	0.57/0.37/0.45	0.59/0.26/0.36	0.27/0.30/0.29	0.36/0.40/0.38
CI	subtractor	32	0.50/0.09/0.16	1.00/0.09/0.17	1.00/0.09/0.17	0.50/0.09/0.16
CO	adder	90	0.22/0.02/0.04	0.22/0.02/0.04	0.30/0.03/0.06	0.22/0.02/0.04
CO	comparator	502	0.03/0.01/0.01	1.00/0.01/0.01	0.00/0.00/0.00	0.03/0.01/0.01
CO	counter	13	1.00/0.08/0.14	1.00/0.08/0.14	0.00/0.00/0.00	1.00/0.08/0.14
CO	eq_const	491	0.01/0.01/0.01	0.33/0.01/0.02	0.01/0.01/0.01	0.01/0.01/0.01
CO	ff_en	171	0.00/0.00/0.00	0.00/0.00/0.00	0.24/0.22/0.23	0.02/0.22/0.03
CO	ff_srst	87	0.00/0.00/0.00	0.00/0.00/0.00	0.20/0.51/0.29	0.02/0.48/0.04
CO	fsm	40	0.88/0.53/0.66	1.00/0.97/0.99	0.06/0.03/0.04	0.54/0.53/0.53
CO	mux	213	0.32/0.08/0.13	0.94/0.08/0.15	0.46/0.03/0.05	0.49/0.08/0.14
CO	mux_tree	32	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
CO	register	304	0.52/0.37/0.43	0.57/0.28/0.38	0.29/0.34/0.31	0.36/0.42/0.39
CO	subtractor	38	0.17/0.03/0.05	1.00/0.03/0.05	1.00/0.03/0.05	0.20/0.03/0.05
MI	adder	62	1.00/0.13/0.23	1.00/0.11/0.20	0.75/0.13/0.23	1.00/0.13/0.24
MI	comparator	116	0.01/0.01/0.01	0.00/0.00/0.00	0.00/0.00/0.00	0.01/0.01/0.01
MI	counter	2	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	eq_const	102	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	ff_en	4	0.04/0.50/0.07	0.00/0.00/0.00	0.14/0.25/0.18	0.05/0.75/0.09
MI	ff_srst	9	0.00/0.00/0.00	0.00/0.00/0.00	0.11/0.33/0.16	0.02/0.33/0.03
MI	fsm	0	–	–	0.00/0.00/0.00	–
MI	mux	49	0.14/0.02/0.04	0.14/0.02/0.04	0.00/0.00/0.00	0.12/0.02/0.04
MI	mux_tree	3	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
MI	register	94	0.28/0.12/0.16	0.00/0.00/0.00	0.18/0.27/0.22	0.22/0.16/0.19
MI	subtractor	27	0.50/0.07/0.13	1.00/0.07/0.14	1.00/0.04/0.07	0.50/0.07/0.13
VT	adder	7	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	comparator	547	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	counter	6	0.67/0.33/0.44	1.00/0.33/0.50	0.00/0.00/0.00	0.67/0.33/0.44
VT	eq_const	541	0.01/0.00/0.00	0.00/0.00/0.00	0.03/0.00/0.00	0.01/0.00/0.00
VT	ff_en	130	0.00/0.01/0.00	0.00/0.00/0.00	0.01/0.02/0.02	0.00/0.02/0.00
VT	ff_srst	6	0.00/0.00/0.00	0.00/0.00/0.00	0.03/0.83/0.06	0.01/0.83/0.02
VT	fsm	4	0.40/0.50/0.44	0.00/0.00/0.00	0.00/0.00/0.00	0.33/0.25/0.29
VT	mux	22	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	mux_tree	2	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00
VT	register	146	0.11/0.05/0.07	0.00/0.00/0.00	0.06/0.06/0.06	0.05/0.06/0.06
VT	subtractor	2	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00	0.00/0.00/0.00

although the paired test over 113 designs is significant, so the status is inconclusive. Grouping by shared control matches some enabled words that per-bit slices never matched, at a precision of 0.014 (F1 0.026); the recall stays at 0.146. Why it stops there was not measured. Three reasons are plausible and untested: the enable is often a function of several nets, the cone depth limit of the `ff_pattern` analysis cuts it, and the ground-truth footprints of such words are whole arrays that share cells.

H16c to H16f. The word-level `eq_const` changes nothing measurable (F1 0.001 for WB against 0.001 for the baseline; it is a subset of the per-bit rule). The chain test is inconclusive (WB completes 3 of 22 chains, the round-2 arm 1) and the FSM rule that was evaluated over-generates. Growth from register words reaches the datapath at 0.788 (WB) but a size-matched random null reaches 0.715, so the level alone is not evidence. Repetition on the big designs lowers F1 (0.013 to 0.010), and H16f is degenerate: no outcome but *not supported* was reachable. Their details are in the repository (`experiments/round5`) and in the research log.

Limits of the detector. It reuses the `ff_pattern` cone analysis and the `arith` chain finder, so its adders are those

of the baseline. At f3 every arm has zero recall on muxes and comparators. The rules were developed on 150 synthetic train designs and did not transfer to the real held-out set.

Against DANA. RQ-H1 asks whether an established word finder does better on the same netlists. The decision rule, the scorer and the bootstrap interval are written in the repository's HAL comparison plan. The tool is HAL v4.5.0 (commit `e7fea566b51e`) with its dataflow plugin DANA [12], [14], built without the graphical interface. Both tools read the same `netlist.blif` of a floor; a converter turns it into structural Verilog of the cells of one of two gate libraries that we wrote for HAL (a LUT library for f2, the cmos6 library for f3), with the instance name equal to the Odin III node id so that a DANA group is a set of answer-key nodes. The HAL v4.5.0 command line `--dataflow` does not run (its handler sets no gate types), so DANA was driven through the plugin's Python API with its documented default configuration; only `min_group_size` was set. The library default of `min_group_size` (8) is the headline arm; 2 is a sensitivity run. Loki's side is the round-5 `W` arm, not rerun. Ground truth are the L0 storage words with `WIDTH` ≥ 2 whose footprint, cut to flip-flops, holds at least two flip-flops, scored by one code path for both tools (Jaccard-0.5 one-to-one matching, exact match, purity, completeness).

Table XXXVIII gives the pooled result on MI and VT. On f2 the pooled F_J of DANA is 0.261 against 0.169 for Loki's `W` arm over 100 designs (41 of them with a ground-truth word); the difference $D = F_J(\text{Loki}) - F_J(\text{DANA})$ is -0.092, with a 95% cluster-bootstrap interval of [-0.317, 0.043]. The decision rule returns *inconclusive*: the point estimate favours DANA, but the interval is wide (the micro designs come in template families), the paired Wilcoxon test over the designs with a ground-truth word does not reach $p < 0.05$ (0.172 for DANA greater), and the interval is not inside the equivalence band either, so neither a lead nor a tie is shown. The comparison scorer (duplicate predictions counted once, Loki's cells and the ground-truth footprints cut to flip-flops) moves Loki's pooled f2 F_J from 0.129 to 0.169. The decision is therefore shown under both scorers (Table XXXVII; DANA's counts are the same in both rows): D is -0.092 with the comparison scorer (95% interval [-0.317, 0.043]) and -0.132 with the round-5 scorer ([-0.382, -0.044]); the rule applied to each alone gives *inconclusive* and *inconclusive*, and neither scorer is the headline. DANA's size-2 run gives a pooled F_J of 0.306; it is a sensitivity run, never the headline. Its outcome is also not a decided one, but with the round-5 scorer alone the rule gives *HAL ahead* for it (comparison scorer alone: *inconclusive*).

The two tools differ in kind. DANA proposes fewer groups (51 against 268 for Loki on f2); a larger share of them matches a word (0.745 against 0.160), but few words are recovered by either tool (recall 0.158 for DANA, 0.179 for Loki). DANA's groups cover most flip-flops of the ground-truth words (0.985 against 0.843) and are more complete (0.550 against 0.063), with a lower purity (0.853 against 0.907). Its groups are larger (median 10.0, mean 37.3 flip-flops, against 2.0 and 6.0 for Loki on f2), and 0.25 of them (13) merge words, that is, hold flip-flops of a ground-truth word and of another one, against 0.07 (18) for Loki. Loki's groups are purer but split words and leave flip-flops out. On f3 DANA's rows equal its f2 rows: it reads the flip-flop graph and control nets only, and the f2 and f3 netlists of a design have the same flip-flops and the same flip-flop connectivity; DANA's partitions are equal on 98 of 98 designs (under the original flip-flop names) and no netlist holds a `DFFSR` cell (0 in 199). Loki's f3 numbers differ from f2 (Table XXXVIII); at f3 Loki's precision is higher and its recall lower. This shift comes mostly from the scorer's flip-flop cuts (Loki's cells and the ground-truth footprints), which raise the match count only together, and not from the detector: with the round-5 scorer Loki's pooled f3 F_J is 0.065 (precision 0.149, recall 0.042), against 0.226 with the comparison scorer (`docs/results-hal.md`).

TABLE XXXVII
THE DECISION ON E_REAL AT F2 UNDER THE TWO SCORERS FOR LOKI (HAL COUNTS IDENTICAL IN BOTH ROWS).
$D = F_J(\text{LOKI}) - F_J(\text{DANA})$; p_L, p_H: ONE-SIDED WILCOXON P, LOKI GREATER AND DANA GREATER. THE LAST COLUMN APPLIES THE RULE TO THAT ROW ALONE; THE DECISION NEEDS LOKI AHEAD UNDER BOTH.

Scorer	D	95% CI	90% CI	p_L	p_H	rule alone
Comparison scorer	-0.092	[-0.317, 0.043]	[-0.266, 0.009]	0.828	0.172	inconclusive
Round-5 scorer	-0.132	[-0.382, -0.044]	[-0.337, -0.052]	0.999	0.001	inconclusive

Limits. The 13 large designs were not scored: 7 contain a RAM hard block that neither gate library can express, which is above the 20% drop limit; DANA ran on the 6 that convert (at most 46 s each) and the group counts are in `docs/results-hal.md`. No evaluation netlist contains a flip-flop with set and reset pins, so DANA's control-net evidence is the clock only on both floors; a netlist with enable and reset flip-flops is the case DANA was designed for, and we did not test it. The converter was checked against the BLIF by simulation (micro, VerilogEval and the convertible large designs); the 10545 LUT truth tables as HAL parses them show 0 mismatches with the BLIF, and the BLIF and HAL LUT counts differ on 0 of 100 designs; DANA returned the same groups on a second run (5 of 5 designs, and 5 of 5 with both runs uncached) and with permuted instance names and shuffled instance order (6 of 6 designs); these two checks cover only small VT designs (the 5 smallest designs with a ground-truth word, and 6), no MI design; HAL was not patched and DANA was not tuned. The ground truth is the Odin III answer key, so a group that matches no word is wrong only in that sense. The result is limited to register words on these sets: under the decision rule the word-first rules are not shown to be ahead of, or equal to, a published dataflow method, and neither tool recovers most of the words.

Verdicts. H16a **DISMISSED** (tiling 0.023 against 0.022 for the baseline; the null is 0.203). H16b **INCONCLUSIVE** (recall rises from 0.008 to 0.146 but F1 only by 0.025, below the margins, at a precision of 0.014). H16c and H16f **DISMISSED**; H16d and H16e **INCONCLUSIVE**. RQ-H1 **INCONCLUSIVE**: pooled F_J 0.169 for Loki against 0.261 for

TABLE XXXVIII

LOKI WORD-FIRST DETECTOR (W) AGAINST HAL'S DANA ON MI AND VT (E_REAL), WORD RECOVERY AGAINST THE L0 GROUND TRUTH; ONE SCORER FOR BOTH TOOLS. P_J, R_J, F_J: JACCARD-0.5 MATCHING; PURITY AND COMPLETENESS OVER FLIP-FLOPS. THE ROUND-5 SCORER ROW COUNTS LOKI AS ROUND 5 DID (THE SCORER USED FOR LOKI IN ROUND 5). LOKI WB HAS A RUN ON FEWER DESIGNS THAN THE OTHER ARMS (F3: 96 OF 98 DESIGNS; LISTED IN DOCS/RESULTS-HAL.MD), SO ITS ROWS POOL A DIFFERENT DESIGN SET.

	f2 (LUT6)					f3 (cmos6)				
Arm	P_J	R_J	F_J	pur.	compl.	P_J	R_J	F_J	pur.	compl.
Loki W	0.160	0.179	0.169	0.907	0.063	0.557	0.142	0.226	0.952	0.850
Loki W, round-5 scorer	0.119	0.142	0.129	-	-	0.149	0.042	0.065	-	-
Loki WB	0.141	0.133	0.137	0.983	0.129	0.320	0.071	0.116	0.956	0.874
DANA, size 8 (headline)	0.745	0.158	0.261	0.853	0.550	0.745	0.158	0.261	0.853	0.550
DANA, size 2	0.613	0.204	0.306	0.865	0.549	0.613	0.204	0.306	0.865	0.549

TABLE XXXIX

ROUND 6: UNITS PER SET AT THE STMT LEVEL AND WHAT BECAME OF THEM.

Set	Designs	Units	Kept	Lost	Born joined	Capped
ALL	221	1436	474	580	382	21
CI	120	765	179	304	282	20
MI	68	546	253	193	100	1
VT	33	125	42	83	0	0

TABLE XL

ROUND 6: FIRST LOSSES (THE TOTAL UNDER THE RANDOM-SET NULL) AND, APART FROM THEM, THE SUBSUMED ONES (THE UNIT SURVIVES ONLY AS THE ENCLOSING STATEMENT OF A FINER UNIT THAT IS STILL ALIVE) AND THE REMAINING MERGED-PLUS-REMOVED LOSSES (STMT LEVEL, MI+VT+CI).

Pass	First losses	Subsumed	Share	Merged + removed
S3_opt	320	127	0.40	193
S4_flatten	68	0	0.00	68
S5_opt_full	26	1	0.04	25
S8_abc	166	33	0.20	133
all	580	161	0.28	419

DANA, difference -0.092 with 95% interval $[-0.317, 0.043]$ and Wilcoxon $p(\text{Loki greater}) = 0.8284$; neither a lead nor a tie is shown, and neither tool recovers most of the words. We therefore make no claim that the word-first detector is an innovation that works.

Record. loki PR #47 and #48 (2026-10-25) for the word-first detector, PR #51 (2026-10-27) for the HAL comparison; register rows H16a to H16f and RQ-H1.

J. Where synthesis loses constructs: the loss audit (L15a, L15c, L15d)

Hypotheses. L15a: each construct kind is first lost at one dominant Odin III pass. L15c: the D14 labels explain the merged losses. L15d: a pass can fool a detector (its recall falls across the pass).

Experiment and result. The word-first detector finds register and enable words and does not find FSM decoders, compares and mux boundaries; we guessed that they are absorbed into LUTs at f2. The audit asks where and why: for each HDL construct, at which Odin III step is it first lost as a separable unit, joined with what, and why? The measure is a count of constructs lost per kind and pass, the share that the D14 provenance explains, and the recall of the round-5 detectors before and after a pass. It is not area, speed or power. The tools are Odin III [6] at `c82609e6`, which writes the version 2 name map (`--prov-map`: records, join labels and the construct table), and ABC [4] inside its `abc` pass; no VPR run is involved.

Protocol. Development used hand-written fixtures and the corpus train split. The f2 flow of round 5 is run pass by pass in one Python session (proc, fsm, opt, flatten, opt `-full`, wreduce, lower, abc) and a netlist plus map is written after

TABLE XLI

ROUND 6: FIRST LOSSES OF A UNIT PER HARDWARE KIND AND PASS, AS REAL/RANDOM-SET NULL/PERSISTENCE NULL (EXPECTED COUNTS, STMT LEVEL, MI+VT+CI). THE FIRST COLUMN IS THE UNITS AT RISK AT OPT. THE RANDOM-SET NULL IS PERSISTENCE-FREE AND OVERSTATES CHANCE LOSSES; THE PERSISTENCE NULL (MEAN OF 50 DRAWS) STARTS FROM THE REAL OBJECTS OF THE PREVIOUS SNAPSHOT. BOTH NULLS AND THE REAL COUNT USE THE SAME UNITS AT RISK (SEPARABLE AT THE PREVIOUS SNAPSHOT, NOT LOST EARLIER). THE LOGIC LOSS AT FLATTEN IS THE 68 INSTANCE UNITS (REMOVED), NOT DEAD LOGIC. THE LOSS MEASURE CANNOT SEE FSM RE-ENCODING (HARDWARE KIND FSM IS READ AT S1, BEFORE THE FSM PASS), SO THE FSM ROW IS EMPTY AND FSM UNITS ARE COUNTED UNDER REGISTER, MUX OR LOGIC.

Kind	n	fsm	opt	flatten	opt -full	wreduce	lower	abc
fsm	0	–	–	–	–	–	–	–
register	218	0/59/0	84/94/55	0/35/0	4/35/4	0/31/0	0/7/0	1/13/9
arith	172	0/37/0	17/51/22	0/36/0	7/36/0	0/35/0	0/0/0	31/22/20
mux	359	0/108/0	192/223/216	0/57/0	14/61/5	0/45/0	0/10/0	79/36/52
compare	33	0/6/0	18/14/16	0/3/0	0/1/0	0/2/0	0/1/0	1/1/3
logic	272	0/71/0	9/81/14	68/58/4	1/38/5	0/39/0	0/13/0	54/46/50

TABLE XLII

ROUND 6: REAL FIRST LOSSES / RANDOM-SET NULL / PERSISTENCE-NULL MEAN [2.5TH–97.5TH PERCENTILE OF THE 50 PER-DRAW TOTALS], STMT LEVEL, MI+VT+CI. THE INTERVAL COVERS THE DRAWS WITHIN THE FIXED DESIGNS, NOT VARIATION ACROSS DESIGNS. A PASS THAT REMOVES OBJECTS LOWERS THE OBJECT COUNT AND THE PERSISTENCE NULL READS EVERY REMOVAL AS A MERGE, SO AT OPT IT IS AN UPPER BOUND OF CHANCE MERGING.

Kind	opt	flatten	opt -full	abc
fsm	–	–	–	–
register	84 / 94 / 55 [48–64]	0 / 35 / 0 [0–0]	4 / 35 / 4 [1–7]	1 / 13 / 9 [6–12]
arith	17 / 51 / 22 [15–29]	0 / 36 / 0 [0–0]	7 / 36 / 0 [0–2]	31 / 22 / 20 [17–23]
mux	192 / 223 / 216 [202–226]	0 / 57 / 0 [0–0]	14 / 61 / 5 [2–10]	79 / 36 / 52 [47–58]
compare	18 / 14 / 16 [11–20]	0 / 3 / 0 [0–0]	0 / 1 / 0 [0–1]	1 / 1 / 3 [2–3]
logic	9 / 81 / 14 [9–19]	68 / 58 / 4 [4–4]	1 / 38 / 5 [3–7]	54 / 46 / 50 [45–56]

TABLE XLIII

ROUND 6: FIRST LOSSES PER REASON CLASS AND PASS (STMT LEVEL, MI+VT+CI). CLASSES WITH NO LOSS ARE KEPT.

Reason	fsm	opt	flatten	opt -full	wreduce	lower	abc
removed	0	12	68	7	0	0	10
subsumed	0	127	0	1	0	0	33
fsm-recode	0	0	0	0	0	0	0
instance-join	0	0	0	0	0	0	2
opt-merge	0	77	0	17	0	0	0
width-reduce	0	0	0	0	0	0	0
lower-share	0	0	0	0	0	0	0
lut-cone	0	0	0	0	0	0	99
unlabelled-merge	0	104	0	1	0	0	22

TABLE XLIV

ROUND 6: MERGED FIRST LOSSES, THOSE A LABELLED SAME-PASS JOIN RECORD NAMES (EXPLAINED), THE UNLABELLED REST AND HOW MANY OF THOSE ARE INHERITED BLENDS (EXPLORATORY; BY DESCENT THROUGH THE PARENT RECORDS, THEN BY THE EARLIER SUBSET CHECK). STMT LEVEL, MI+VT+CI.

Pass	Merged	Explained	Share	Unlabelled	Inherited (descent)	Inherited (subset)
S3_opt	181	77	0.425	104	104	104
S5_opt_full	18	17	0.944	1	1	1
S8_abc	123	101	0.821	22	22	22
all	322	195	0.606	127	127	127

each pass. A *unit* is the statement-level anchor of a record's construct leaves (V1.2.0 Section 3.3), reduced by dropping ancestors; a unit is *separable* at a snapshot when some logic object has exactly that unit set, and *lost* at the first later snapshot where none has. A loss is *explained* when a labelled same-pass join record names the unit; its reason class is read from the pass and that evidence. The map holds one record per object (checked on all 214,157 objects of the 1,768 maps), so reading the first record of an object loses nothing.

Two nulls. The random-set null draws, for every real object of a pass, a random unit set of the same size from the units alive at the pass. It matches the object count and the size distribution of the pass; it does not match the chains, because it forgets which units the objects of the previous snapshot already held together. A pass that changes nothing still draws fresh random sets and "loses" most units, so the random-set null overstates chance losses. The *persistence null* starts every draw from the real objects of the previous snapshot and applies random merges until the object count of the real snapshot is reached (50 draws; mean and a 95% interval of the per-draw totals; a pass that changes nothing gives zero; a pass that only removes dead objects is read as merges, so at `opt` it is an upper bound of chance merging). The rule thresholds are the same for both nulls. The sets are the 68 micros (MI), the 33 held-out VerilogEval designs (VT) and the 120 corpus iid designs (CI), each with all designs ok; the 13 big designs and the ood split were not run. The snapshot writers do not change the netlist: the final BLIF of all 68 micros (hash-style names, MI only) is byte-identical to a run without them.

Where units are lost. Of 1436 units, 580 are lost, 474 are never lost and 382 are joined from the first snapshot onwards (Table XXXIX). The losses fall on four passes only: 320 at `opt`, 68 at `flatten`, 26 at the second `opt` and 166 at `abc`. None is lost at `fsm`, `wreduce` or `lower`: `lower` splits word cells into gates, which makes units finer, not blended. The 580 is the total. It includes 161 `subsumed` losses (a statement that stays alive only as the enclosing statement of a finer unit; 0.28 of the total, 127 of the 320 at `opt` and 33 of the 166 at `abc`); without them the merged-plus-removed losses are 419 (Table XL). The 68 at `flatten` are all `INSTANCE` units (the instance cell of a dissolved module, class `removed`) and say nothing about logic: the `logic` kind is dominant at `flatten` because of them, not because dead logic is removed.

By hardware kind (Table XLI), under the random-set null, registers (dominant) and compares (dominant) are lost at `opt` (84 of 89 and 18 of 19, against random-set null shares of 0.34 and 0.49), muxes at `opt` (192 of 285, null share 0.41) and arithmetic at `abc` (31 of 55, null share 0.10). By the count rule against the random-set null, `abc` loses muxes more than chance (79 against 36) and arithmetic at chance, whereas arithmetic at `opt` (17 against 51) and muxes at the second `opt` (14 against 61) are below chance. These readings do not survive the persistence null (exploratory, Tables XLII and XLV): `opt` is also where a chance merging of the objects puts most of its losses, so the pass shares of register, compare, mux and arithmetic are `dominant-null`, the growth artefact of rule 1 (null shares 0.80, 0.84, 0.79 and 0.46). Only the `flatten` loss of `logic` (the instance cells) stays `dominant`. By the count rule, arithmetic at `opt` is

TABLE XLV

ROUND 6, RULE 1: THE PASS THAT LOSES MOST OF EACH HARDWARE KIND, ITS SHARE, AND THE OUTCOME UNDER THE RANDOM-SET NULL (PERSISTENCE-FREE) AND, BESIDE IT, UNDER THE PERSISTENCE NULL WITH THE SAME THRESHOLDS (STMT, MI+VT+CI). WHERE THEY DIFFER, NEITHER IS CONFIRMATORY.

Kind	Pass	Losses	Share/random-set null	Random-set	Share/persist.	Persist. (expl.)
arith	S8_abc	55	0.564/0.101	dominant	0.564/0.462	dominant-null
compare	S3_opt	19	0.947/0.489	dominant	0.947/0.844	dominant-null
logic	S4_flatten	132	0.515/0.168	dominant	0.515/0.055	dominant
mux	S3_opt	285	0.674/0.412	dominant	0.674/0.790	dominant-null
register	S3_opt	89	0.944/0.343	dominant	0.944/0.801	dominant-null

TABLE XLVI

ROUND 6: RECALL PER KIND OF THE THREE ARMS ON THE GATE NETLIST (S7) AND THE LUT NETLIST (S8), MI+VT+CI, ON THE PAIRED DESIGNS (THOSE WITH A RUN AT BOTH SNAPSHOTS). GT IS THE WB GROUND-TRUTH COUNT AT S7/S8; THE TWO DIFFER BECAUSE THE FOOTPRINTS ARE REBUILT FROM EACH SNAPSHOT'S MAP (THE S7 AND S8 GROUND TRUTHS ARE NOT THE SAME INSTANCES), AND THE GATE AND LUT FOOTPRINTS ARE NOT LIKE FOR LIKE.

Kind	GT S7/S8	B S7	B S8	W S7	W S8	WB S7	WB S8
ff_en	212/212	0.00	0.01	0.16	0.11	0.16	0.12
ff_srst	85/85	0.00	0.00	0.53	0.49	0.53	0.47
register	397/397	0.13	0.20	0.22	0.23	0.22	0.24
adder	110/100	0.00	0.15	0.00	0.14	0.00	0.16
subtractor	63/60	0.00	0.08	0.00	0.05	0.00	0.07
comparator	829/829	0.02	0.00	0.00	0.00	0.02	0.00
eq_const	803/803	0.00	0.01	0.00	0.00	0.00	0.01
mux	174/166	0.53	0.06	0.36	0.04	0.55	0.07
mux_tree	24/24	0.04	0.00	0.00	0.00	0.04	0.00
counter	18/18	0.00	0.39	0.00	0.00	0.00	0.39
fsm	24/24	0.12	0.75	0.04	0.04	0.17	0.67

TABLE XLVII
ROUND 6: EIGHT LOST MICRO CONSTRUCTS, 'CONSTRUCT AT FILE:LINE LOST AT PASS P BECAUSE MERGED WITH Q' (FIRST ROWS OF L15_LOST_LIST.CSV WITH ONE LABELLED PARTNER).

Design	Construct	Kind	Pass	Joined with
bm_expr_all_mod	bm_expr_all_mod.v:124.2-124.29	NONBLOCKING_ASSIGN	opt	NONBLOCKING_ASSIGN 146.2-146.122
bm_expr_all_mod	bm_expr_all_mod.v:133.2-133.39	NONBLOCKING_ASSIGN	opt	NONBLOCKING_ASSIGN 147.2-147.126
bm_lpm_all	bm_lpm_all.v:44.8-44.20	NET_ASSIGN	opt	NET_ASSIGN 51.8-51.43
bm_match1_str_arch	bm_match1_str_arch.v:49.8-49.26	NET_ASSIGN	opt	NONBLOCKING_ASSIGN 55.2-55.22
bm_match1_str_arch	bm_match1_str_arch.v:50.8-50.26	NET_ASSIGN	opt	NONBLOCKING_ASSIGN 56.2-56.22
bm_match1_str_arch	bm_match1_str_arch.v:51.8-51.26	NET_ASSIGN	opt	NONBLOCKING_ASSIGN 57.2-57.22
bm_match1_str_arch	bm_match1_str_arch.v:55.2-55.22	NONBLOCKING_ASSIGN	opt	NET_ASSIGN 49.8-49.26
bm_match1_str_arch	bm_match1_str_arch.v:56.2-56.22	NONBLOCKING_ASSIGN	opt	NET_ASSIGN 50.8-50.26

at chance (17 against a mean of 22, interval 15 to 29), not below; the "below chance" reading for it and for muxes at the second `opt` comes from the missing persistence (for muxes at the second `opt`, 14 against 5 with interval 2 to 10, the real count is above the interval). Registers at `opt` (84 against 55, interval 48 to 64) and arithmetic and muxes at `abc` (31 against 20, interval 17 to 23; 79 against 52, interval 47 to 58) are above the interval but within a factor two of the mean, so the count rule reads them as at chance. The supported statement is therefore descriptive: a compare statement is no longer a separable unit after the first `opt`, before any LUT is formed (18 of 19), but that the first `opt` is the pass that loses them is what a chance merging of the same number of objects also gives, so the data do not show that compares are lost there more than other kinds. The round-5 guess that the LUT cone is where compares go is not supported either: one compare unit is lost at `abc`. The branch level gives the same picture (548 losses against 580; `l15_kind_pass.csv`). The persistence null matches the size profile of the real snapshot only approximately: `l15_null_size_mad.csv` gives the mean absolute deviation of the sorted object sizes, over the designs where merges are drawn (0.90 at `opt`, 0.88 at `abc`, 0.32 at the second `opt`).

FSMs. The loss measure cannot see an FSM re-encoding. The hardware kind of a unit is read at the first snapshot, after `proc` and before the `fsm` pass, where no object has the type `$fsm`; FSM units are therefore counted under register, mux or logic, and the `fsm` row of Table XLI is empty. "No unit stops being separable at `fsm`" is a statement about that pass only; it does not say that FSMs are not lost, and the FSM-recode reason class has no observation.

Why. Table XLIII gives the reason classes. At `opt`, 77 losses are opt merges that a labelled join record names, 12 are removals, 127 are subsumptions and 104 are merges with no label; at `abc`, 99 are LUT-cone merges that a record names, 33 subsumptions, 10 removals, 2 instance joins and 22 unlabelled merges. Of the 322 merged losses, 195 (0.606) are explained by a same-pass join record, which the rule reads as partly explained (Table XLIV). The share differs by set and by pass: MI 0.978 (88 of 90), CI 0.462 (79 of 171) and VT 0.459 (28 of 61); `opt` 0.425 (77 of 181, mostly unexplained by the rule), second `opt` 17 of 18 and `abc` 0.821 (101 of 123, partly explained). The VT share at `opt` is 4 of 36; the VT designs are read through the slang adapter, and a gap in the labels that the adapter's records carry would give such a low share (not tested here). The pooled status is as the rule says.

All 127 unlabelled merges fit an *inherited blend* reading (exploratory, by descent through the parent records of the map): for each of them every object that holds the unit has a parent record of the pass that already blended the unit with at least one other unit, so no new join was made and none is labelled (V1.2.0 Section 3.3). The earlier subset check, which looked for any impure object, gives the same 127 of 127. The descent test shows that the unit was already in a blend in a parent; it does not show that every other unit of the new object came from the same parent, so it does not support a claim that the D14 labels name every new blend. "Style collapse" (a case, an if and a ternary of one function) leaves no record and is reported as not observable. Table XLVII shows eight lost micro constructs with their partner; `l15_lost_list.csv` holds all of them.

Detectors. The word-level snapshots are not netlists a detector reads, so recall is compared at the two netlist snapshots, the gates after `lower` (S7) and the LUTs after `abc` (S8), for the default list (B), `wordfirst` (W) and the combined list (WB). The ground truth is the round-5 derivation on the word cells of S6 with the span-subset footprint rule of L0 applied to construct keys. The gate footprints (many gates share a construct set) and the LUT footprints are not like for like, and the S7 and S8 ground truths differ in size for some kinds (Table XLVI). For the mux word the ground truth is 174 instances at S7 and 166 at S8, because the footprints are rebuilt from each snapshot's name map, so the two recalls are not fractions of the same set of instances; recall at the two snapshots is compared through rule 4 only. The table uses the paired designs (a run at both snapshots; three designs drop out: one mux design (`ve_Prob021_mux256to1v`, killed at S7 by the 6 GB slice cap) in all arms and two micro `WB` runs at S7 that timed out). Under rule 4 the mux word is the only kind that is found at S7 and lost at S8 (`fooled by abc` in all three arms; WB: recall 0.55 to 0.07 on 174 instances at S7 and 166 at S8). The related kind `mux_bit` falls from 0.201 to 0.012 (35 of 174 to 2 of 166) in arms B and WB, and rule 4 marks it `not fooled`: the rule calls a detector fooled only when recall falls by 0.20 or more, and the fall is 0.189, just under that line (arm W finds none at either snapshot). The label `not fooled` here is therefore a statement about the cut-off, not about robustness. The mux word statement stands for the mux word only: comparator, `eq_const`, counter, adder, subtractor and `mux_tree` are at 0.00 to 0.04 at S7, a floor at which a pass cannot fool a detector, so "not fooled" there means "nothing to lose", not "robust". FSM (WB: 0.17 to 0.67 on 24 instances) and counter (0.00 to 0.39 on 18) are found more often at S8 than S7, and nothing else moves by 0.20. The counts are small (mux 174, FSM 24, counter 18 instances).

What this suggests. The map already names, for every LUT, the set of statements it blends, so a detector could split a LUT region by that set instead of guessing; the mux fooling at `abc` and the compare loss at `opt` say where such a split is worth the most. This is a suggestion, not a result: no detector was changed in this round. Not done: the big designs, the ood split, a VPR flow, the instance axis across several instances of one

module (units are keyed by construct, so two instances of one module are one unit), and the word level, which the map does not observe.

Verdicts. L15d **POSSIBLY GOOD**, for one kind: the mux word is found at the gate snapshot and lost after `abc` in all three arms (WB: recall 0.55 to 0.07), but the ground truth has 174 instances at S7 and 166 at S8 and the gate and LUT footprints are not like for like, so the comparison is not like for like; `mux_bit` falls by 0.189 and is read as not fooled by a margin of 0.011; for the other kinds the recall is already at a floor, and FSM (24 instances) and counter (18) are too few to say more. L15a **POSSIBLY GOOD**: under the random-set null the register, compare, mux and arithmetic kinds are each lost at one dominant pass (`opt` 320 of 580 losses, `abc` 166), but under the persistence null that placement is what a chance merging of the same objects also gives; the paper keeps only the descriptive statement (a compare is no longer a separable unit after the first `opt`, 18 of 19). L15c **INCONCLUSIVE**: the D14 labels explain 0.606 of the merged losses, the middle class of the rule.

Record. loki PR #53 (2026-10-27); register rows L15a to L15d.

K. Language model plus Loki's hints: a pilot (LLM1, LLM2)

Hypotheses. LLM1: a model given Loki's hints recovers RTL better than without them. LLM2: the netlist helps beyond what the model knows from the interface line alone.

Every other result in this paper uses no language model. This subsection reports one small pilot that does. The measure is the functional pass rate per arm, plus the refusal rate.

Setup. The pilot is a replica of the Gate-Breaker netlist-to-RTL task [1] on VerilogEval [7]. It uses 40 problems, 10 per complexity quartile, chosen by a fixed hash rank from the quartile files, and one call per problem per arm ($K = 1$). There are three arms. Arm A gives the gate netlist and the interface line. Arm B adds the claims that Loki makes from the flat netlist alone (no provenance names, no L0 ground truth). Arm C gives the interface line only and no netlist, as a memorisation control. All arms use the same two worked examples and the same answer format. Each call is a fresh `claude-sonnet-5-5` subagent with no prior turns. The strict scorer compiles the reply with Icarus Verilog 12.0 against the reference testbench and requires exactly one line `Mismatches: 0 in M samples`; it also rejects replies that name the hidden reference module or print their own result line. A Gate-Breaker-equivalent count is reported beside it as a cross-check. A call with no reply file or an API refusal counts as a fail in the all-calls view.

Results. Table XLVIII gives the pass rate per arm in both views, and Table XLIX gives the exact two-sided McNemar tests on the paired set of 38 problems answered in all three arms. Arm B is nominally ahead of arm A in the paired view (35/38 against 33/38) and both are ahead of arm C (29/38), but *no difference reaches* $p < 0.05$: the smallest is B against C at $p = 0.0703$. This is a pilot at $n = 40$ with one draw per cell; it cannot support a claim about absolute rates or about the effect of the hints. In the all-calls view arms A and B are equal (35/40) because arm B's two missing reply files count as fails, and arm A passed both of those problems. Per-quartile cells have 8 to 10 problems each; in Q4 the all-calls rates are A 9/10, B 7/10, C 7/10, so the sign of the B-minus-A difference is not stable across quartiles (Section VI-D gives every quartile with its tests and the power limit).

TABLE XLVIII
LLM PILOT, RUN `re`: PASS RATE PER ARM (STRICT SCORER UNLESS MARKED). $n = 40$ PROBLEMS PER ARM, ONE DRAW EACH, `claude-sonnet-5-5`. ARM A: NETLIST; B: NETLIST PLUS LOKI HINTS; C: INTERFACE ONLY. GB-EQUIV.: GATE-BREAKER-EQUIVALENT SCORER.

View	A	B	C
All 40, refusal/no file = fail (strict)	35/40 (0.88)	35/40 (0.88)	30/40 (0.75)
All 40, refusal/no file = fail (GB-equiv.)	33/40 (0.82)	33/40 (0.82)	29/40 (0.72)
Answered only (strict)	35/40 (0.88)	35/38 (0.92)	30/40 (0.75)
Paired set, 38 (strict)	33/38 (0.87)	35/38 (0.92)	29/38 (0.76)
Paired set, 38 (GB-equiv.)	31/38 (0.82)	33/38 (0.87)	28/38 (0.74)

TABLE XLIX
LLM PILOT, RUN `re`: EXACT TWO-SIDED MCNEMAR TESTS, PAIRED SET $n = 38$, STRICT SCORER. "FIRST ONLY" COUNTS PROBLEMS THE FIRST ARM PASSES AND THE SECOND FAILS.

Comparison	Both	First only	Second only	Neither	p
B vs A	33	2	0	3	0.5000
A vs C	28	5	1	4	0.2188
B vs C	28	7	1	2	0.0703

Refusals. The first complete prompt that we ran, the Gate-Breaker `standard` prompt, was stopped by the protocol's stop rule at 88 of 120 calls: 8 calls came back as API safeguard refusals (no model reply) and 7 as no-file calls (the subagent declined or answered in chat without writing the reply file). Under our own prompt `re`, which explains the research, the same 120 calls gave 0 API refusals and 2 no-file calls. Table L gives the rates per arm. The two prompts differ in more than the refusal behaviour, so any comparison of pass rates between them is exploratory only. Refusal handling is part of the cost of using a general-purpose model for this task and has to be reported beside the pass rate.

Threats to validity. (i) The subagents were not tool-less. They had file and web tools loaded and the workspace instructions and a memory index in context, and the reference and test files sit on the same disk (decision 11, a blinding deviation that was accepted). The transcript audit of all 208 calls (`standard` 88, `re` 120) shows that no call used any tool other than reading its own call file and writing or editing its own reply file: no shell, search or web tool. This is evidence from behaviour, not a technical block. Two calls used more than the usual two tool calls (`standard` 073 and `re` 116); both touched only their own files and both are scored. (ii) Decision 12: the first batch used a step-by-step reasoning prompt (`cot`) that the API safeguard refused in 6 of 10 calls, even in arm C; those answers were discarded and the prompt replaced by `standard`. Decision 13: a refusal is never retried

TABLE L
LLM PILOT: API REFUSALS AND NO-FILE CALLS PER ARM. STANDARD STOPPED AT ORDER 88 OF 120 (CALLS MADE); RE RAN ALL 120.

Run	Arm	Calls	API refused	No file	Total
standard	A	30	2	4	6/30 (0.20)
standard	B	29	3	1	4/29 (0.14)
standard	C	29	3	2	5/29 (0.17)
re	A	40	0	0	0/40 (0.00)
re	B	40	0	2	2/40 (0.05)
re	C	40	0	0	0/40 (0.00)

or reworded, and pass rates are reported with refusal as a fail and on the paired set. Decision 14: the re system prompt is ours, not Gate-Breaker's, so exact Gate-Breaker fidelity is dropped; the worked examples, user content and answer format are unchanged (120 of 120 user messages byte-equal). The re prompt mentions Loki hints in every arm, including arms A and C that carry none; this cue is the same in all arms. (iii) The prompts therefore differ from Gate-Breaker's. Sampling is one draw with no controlled temperature, and the 40-problem subset has a prompt-size cap that under-represents big designs, so these rates are not comparable with the published ones. The hints are uncalibrated at f3, and arm C controls for memorisation of VerilogEval; it is not a measure of recovery.

Verdicts. LLM1 **INCONCLUSIVE**: arm A 35/40 and arm B 35/40 (paired set of 38: A 33, B 35; exact McNemar $p = 0.5000$). LLM2 **INCONCLUSIVE**: arm C 30/40 (paired A against C $p = 0.2188$, B against C $p = 0.0703$). One draw per cell on 40 problems cannot support a claim about the hints in either direction.

Record. loki PR #49, #50 and #52 (2026-10-27); register rows LLM1, LLM2.

L. Per-quartile tables: Loki alone and Loki with a model (Q1, Q2)

Hypothesis and measurement. Q1: the hints help in some complexity quartiles (hypothesis; verdict **INCONCLUSIVE**, the cells hold 8 to 10 problems and no quartile has enough discordant pairs for $p < 0.05$). Q2: what Loki alone recovers per quartile (a measurement with no verdict).

Two questions ask the same thing per Gate-Breaker complexity quartile: is the model with Loki's hints better than the model without them, and what does Loki alone recover? All numbers here come from files already committed; no new run or model call was made for this subsection. Every table below uses one partition, the Gate-Breaker quartile files (41, 36, 36 and 31 problems), which are also the partition of the full language-model run. The 138 built designs that appear in a quartile file are binned by them; 12 built designs appear in no file and are left out. The paper's absolute complexity bins give a different partition, shown once in Table LVI: of the 138 designs, 35 sit in a different quartile there (103 agree), and 9 of the 40 pilot designs would change quartile.

Loki with a model, per quartile. Table LI splits the pilot of Appendix A-K by quartile, and Table LII gives the exact two-sided McNemar test in each quartile. Conditioning on the discordant pairs of each quartile, each discordant pair is equally likely to favour either arm under the null, so the sum of the first-only counts over quartiles is binomial over all discordant pairs: the stratified exact test equals the overall exact McNemar test, which is the All row of the table. Arm B is ahead of arm A by one problem in Q2 and in Q3, level in Q1, and behind by two in Q4 (all calls: 7/10 against 9/10). Both problems for which arm B has no reply file are Q4 problems, so in the paired Q4 view A and B are level at 7/8. Over all quartiles the paired difference is 2 problems for B and 0 for A ($p = 0.50$), and in the all-calls view 2 against 2 ($p = 1.0$). Arm C, which sees no netlist, is behind arm B in the point estimate in every quartile except Q4 (level 7/10), but the overall test of B against C is $p = 0.070$ in the paired view and $p = 0.18$ in the all-calls view, and no quartile reaches $p < 0.05$.

TABLE LI
LLM PILOT (RUN RE) PASS RATE PER GATE-BREAKER COMPLEXITY QUARTILE (THE QUARTILE FILES), STRICT SCORER. ALL CALLS: REFUSAL OR MISSING REPLY FILE COUNTS AS A FAIL. PAIRED: PROBLEMS ANSWERED IN ALL THREE ARMS. ARM A NETLIST, B NETLIST PLUS LOKI HINTS, C INTERFACE ONLY; ONE DRAW PER CELL. AST AND CFG SIMILARITY WERE NOT SCORED IN THE PILOT.

Quartile	View	n	A	B	C
Q1	all calls	10	10/10	10/10	8/10
Q1	paired	10	10/10	10/10	8/10
Q2	all calls	10	8/10	9/10	8/10
Q2	paired	10	8/10	9/10	8/10
Q3	all calls	10	8/10	9/10	7/10
Q3	paired	10	8/10	9/10	7/10
Q4	all calls	10	9/10	7/10	7/10
Q4	paired	8	7/8	7/8	6/8
All	all calls	40	35/40	35/40	30/40
All	paired	38	33/38	35/38	29/38

What the tests can show. A two-sided exact McNemar test needs at least 6 discordant pairs, all in one direction, for $p < 0.05$; with 8 discordant pairs it needs 8 to 0, and with 10 it needs 9 to 1. The largest number of discordant pairs in any single quartile comparison here is 3. Overall, B against C in the all-calls view has 9 discordant pairs split 7 to 2, where 8 to 1 would have been needed. These data therefore do not show that Loki's hints help in any quartile, and do not show that they do not. No claim beyond these tests is made. A larger run (all designs of the quartile files, 3 draws per arm) is planned and is what can test an effect.

Loki alone, per quartile. Table LIII shows every Loki result that is defined on the VerilogEval designs, in the four quartiles. The L0 ceiling is flat: every node of every built design has a label row, a cell_type and an HDL span, and the module coverage is 1.000 in every quartile. Signal coverage has mean (median) 1.000 (1.000), 0.986 (1.000), 1.008 (1.000) and 0.896 (0.866) in Q1 to Q4. The 138 designs of the quartile files include the 6 references that are not built (3 in Q3, 3 in Q4: two that Odin III's SystemVerilog reader skips and four

TABLE LII
LLM PILOT (RUN RE): EXACT TWO-SIDED MCNEMAR TEST PER QUARTILE, STRICT SCORER. CELLS ARE FIRST-ONLY/SECOND-ONLY DISCORDANT PAIRS (ALL-CALLS VIEW IN THE FIRST GROUP, PAIRED VIEW IN THE SECOND); p IN BRACKETS. THE ALL ROW POOLS THE DISCORDANT PAIRS OF Q1–Q4; IT IS ALSO THE STRATIFIED EXACT TEST, BECAUSE THE SUM OF INDEPENDENT BINOMIAL(d_i, 1/2) COUNTS IS BINOMIAL($\sum d_i$, 1/2). A TWO-SIDED EXACT TEST NEEDS AT LEAST 6 DISCORDANT PAIRS, ALL IN ONE DIRECTION, FOR $p < 0.05$.

View	Stratum	B vs A	B vs C	A vs C
all calls	Q1	0/0 (1.000)	2/0 (0.500)	2/0 (0.500)
all calls	Q2	1/0 (1.000)	2/1 (1.000)	1/1 (1.000)
all calls	Q3	1/0 (1.000)	2/0 (0.500)	1/0 (1.000)
all calls	Q4	0/2 (0.500)	1/1 (1.000)	2/0 (0.500)
all calls	All (strat.)	2/2 (1.000)	7/2 (0.180)	6/1 (0.125)
paired	Q1	0/0 (1.000)	2/0 (0.500)	2/0 (0.500)
paired	Q2	1/0 (1.000)	2/1 (1.000)	1/1 (1.000)
paired	Q3	1/0 (1.000)	2/0 (0.500)	1/0 (1.000)
paired	Q4	0/0 (1.000)	1/0 (1.000)	1/0 (1.000)
paired	All (strat.)	2/0 (0.500)	7/1 (0.070)	5/1 (0.219)

that fail Loki's own equivalence gate), so all three floors are built for 41, 36, 33 and 28 designs. The AST similarity of the re-emitted Verilog is lowest in Q2: calibrated `f3` gives 0.691, 0.508, 0.602 and 0.533 for Q1 to Q4 on $n = 41$, 34, 30 and 15 designs, and the CFG similarity is 0.740, 0.750, 0.763 and 0.757 on $n = 41$, 36, 33 and 28. Designs whose reference pyverilog cannot parse have no AST value (0, 2, 3 and 13 in Q1 to Q4), so the Q3 and Q4 AST means rest on fewer, plainer designs and are not a clean effect of complexity. L1 changes little: calibrated and L0-stop `f3` AST agree to the third decimal in Q1, Q2 and Q4 (0.602 against 0.612 in Q3), because the calibrated flow commits 14 hypotheses in 7 designs (round 2). The recovery experiment on the micro designs (Appendix A-B) is ranked by node count, not by Gate-Breaker quartile, and the loss audit of Appendix A-J reports pooled counts per unit kind; neither is defined per quartile, so neither appears in the tables.

Word recovery against HAL, per quartile. Table LIV splits the round 5b comparison of Appendix A-I on the 28 test-split VerilogEval designs that are in a quartile file. Q1 and Q2 have no ground-truth word and every tool predicts none. In Q3 (5 designs) the three tools match 5, 5 and 6 of 7 ground-truth words; in Q4 (6 designs) Loki W matches 5 of 6 with 7 predicted words, Loki WB 4 of 6 with 11 and HAL 4 of 6 with 5. The counts are small, and they depend on the partition: with the absolute bins (Table LVI) Q4 holds 10 designs and 139 ground-truth words.

Side by side. Table LV puts the Loki-only AST and CFG, the round-1 Gate-Breaker harness replication and the pilot's pass rates in one table, with Loki and the round-1 model both on the common set (designs scored by both). The metrics differ: AST and CFG measure the similarity of recovered text, and the pilot measures functional correctness. On the common set Loki alone is below the round-1 model on AST and CFG in Q1 (AST 0.699 against 0.907, CFG 0.741 against 0.913) and Q2 (AST 0.512 against 0.733, CFG 0.754 against 0.917); the common set has one design in each of Q3 and Q4, so no mean is compared there (Section VI-A). The pilot has no AST or CFG score, so the pilot and these columns are not compared on a common metric.

M. Language model plus Loki's hints: the full run, as run (LLM3, LLM4)

Hypotheses. LLM3: a model given Loki's hints recovers RTL better than the same model given the netlist alone. LLM4: the netlist helps beyond what the model knows from the interface line alone.

The pilot could not separate the arms, so the same question was asked on every eligible problem of the four quartile files. The plan (arms, scorer, primary outcome, tests, stop rules) was written before the runs and committed first.

Setup. 137 problems (144 in the quartile files, 7 excluded: four prompts over the 40000-byte cap, one reference that fails its own testbench, two whose hints cannot be built), arms A (netlist), B (netlist and Loki's hints) and C (interface only), three independent draws per problem and arm: 1233 calls to fresh `claude-sonnet-5-5` subagents, with the pilot's prompts, scorer and answer format. The primary outcome is that a problem passes in at least 2 of its 3 draws. Per quartile, the exact two-sided McNemar test; across quartiles, the exact stratified form (discordant problems pooled); Holm adjustment over the ten tests. A claim needs an adjusted $p < 0.05$.

Results. All 1233 calls were answered (0 API refusals, 0 no-file calls, 0 timeouts). Table XIII gives the pass rates and Table XIV the tests. Arms A and B are level (mean pass@1 0.888 against 0.878; A minus B $+0.010$, 95% interval $[-0.017, +0.036]$), with 3 problems only A passes and 4 only B passes. Arm C, with no netlist, still passes 0.560, so over half of the passes need only the interface line. The netlist adds the rest: B minus C is $+0.319$ ($[+0.241, +0.394]$), 52 problems only B passes against 5 only C passes (Holm $p < 0.0001$). By quartile the B-against-C difference passes the rule in Q1 and Q2; Q3 and Q4 have the same sign but Holm $p = 0.1504$ and 0.2109.

(Table XIII is in the main text.)

(Table XIV is in the main text.)

Threats to validity. (i) Dispatch: a Sonnet dispatcher subagent spawned the call subagents, so the controller session did not. (ii) Blinding: the call subagents were not tool-less. A probe lists Agent, Artifact, Bash, Edit, Read, Skill, ToolSearch, Write and the Docs tools, with the workspace instructions, a memory index and a terse-style hook text in context; the reference files are on the same disk. The transcript audit of all 1233 calls found 0 calls touching any path other than their own call file and reply file; 37 used more than two tool calls (paged reads, double writes, and in 3 calls a shell search of their own call file). All are scored as made. This is evidence from behaviour, not a technical block. (iii) The 7 excluded problems and the prompt-size cap under-represent the largest designs. (iv) Arm A is near the ceiling in Q1 and Q2, which limits how much any hint could show. (v) The system prompt mentions Loki hints in every arm, and sampling is not controlled.

TABLE LIII

LOKI WITHOUT A LANGUAGE MODEL ON THE VERILOGEVAL REFERENCES, PER GATE-BREAKER QUARTILE (THE QUARTILE FILES; 12 BUILT DESIGNS THAT ARE IN NO QUARTILE FILE ARE LEFT OUT). REFERENCES: PROBLEMS IN THE QUARTILE FILE (41, 36, 36, 31). BUILT: DESIGNS WITH THE FLOOR BUILT; THE 6 REFERENCES THAT ARE NOT BUILT (3 IN Q3, 3 IN Q4) ARE 2 THAT ODIN III'S SYSTEMVERILOG READER SKIPS (AN EXPLICIT CAST) AND 4 THAT FAIL LOKI'S OWN EQUIVALENCE GATE, SEE THE VEVAL-L0 RECORD. COVERAGE: PER-DESIGN COVERAGE ON `f3` (LABEL ROWS, `cell_type`, HDL SPAN, MODULE SCOPES, DECLARED SIGNALS), MEAN (MEDIAN). SIGNAL COVERAGE EXCEEDS 1 FOR `ve_prob080_timer` (5 RECOVERED AGAINST 4 DECLARED; FLATTENED ARRAY ELEMENTS ADD NAMES). AST AND CFG: MEAN SIMILARITY OF THE RE-EMITTED VERILOG TO THE REFERENCE (ROUND 2); n IS THE DESIGNS WITH A VALUE (REFERENCES THAT PYVERILOG CANNOT PARSE HAVE NO AST VALUE), SO THE CFG MEAN CAN REST ON FEWER DESIGNS.

Measure	Q1	Q2	Q3	Q4
References in the quartile file	41	36	36	31
Built, `f1-k6`	41	36	33	28
Built, `f2`	41	36	33	28
Built, `f3`	41	36	33	28
Label coverage, `f3`	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
`cell_type` coverage, `f3`	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
HDL-span coverage, `f3`	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Module coverage, `f3`	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Signal coverage, `f3`	1.000 (1.000)	0.986 (1.000)	1.008 (1.000)	0.896 (0.866)
n (designs with an AST value)	41	34	30	15
n (designs with a CFG value)	41	36	33	28
L0-stop `f3`, AST	0.691	0.508	0.612	0.533
L0-stop `f3`, CFG	0.740	0.750	0.765	0.757
L1 calibrated `f3`, AST	0.691	0.508	0.602	0.533
L1 calibrated `f3`, CFG	0.740	0.750	0.763	0.757
L1 uncalibrated `f3`, AST	0.633	0.463	0.589	0.511
L1 uncalibrated `f3`, CFG	0.598	0.645	0.739	0.749
L0-stop `f2`, AST	0.734	0.527	0.609	0.541
L0-stop `f2`, CFG	0.828	0.840	0.837	0.856
L1 calibrated `f2`, AST	0.713	0.525	0.600	0.541
L1 calibrated `f2`, CFG	0.774	0.829	0.823	0.856
L1 uncalibrated `f2`, AST	0.652	0.497	0.591	0.510
L1 uncalibrated `f2`, CFG	0.649	0.787	0.797	0.811

TABLE LIV

ROUND 5B ON THE VERILOGEVAL TEST-SPLIT DESIGNS AT `f2` THAT ARE IN A GATE-BREAKER QUARTILE FILE (28 OF THE 33), PER QUARTILE: WORD MATCHES AT JACCARD 0.5 OVER PREDICTED WORDS AND OVER GROUND-TRUTH WORDS (MATCHED/PREDICTED/GROUND TRUTH). LOKI W: WORD-FIRST DETECTOR; WB: WITH THE BLOCK RULE; HAL: DANA. QUARTILES HOLD 5 TO 10 DESIGNS, AND DESIGNS WITHOUT A GROUND TRUTH WORD CONTRIBUTE ZEROS, SO READ THESE AS COUNTS, NOT RATES.

Quartile	Designs	Loki W	Loki WB	HAL
Q1	10	0/0/0	0/0/0	0/0/0
Q2	7	0/0/0	0/0/0	0/0/0
Q3	5	5/6/7	5/7/7	6/6/7
Q4	6	5/7/6	4/11/6	4/5/6

Verdicts. LLM3 **DISMISSED**: A minus B $+0.010$ $[-0.017, +0.036]$, stratified discordant 3 against 4, Holm $p = 1.0000$; the interval rules out a gain larger than about 0.04. LLM4 **USEFUL**: B minus C $+0.319$, stratified 52 against 5, Holm $p < 0.0001$; per quartile Q1 and Q2 pass the rule and Q3 and Q4 are **INCONCLUSIVE**. Arm C passing 0.56 bounds what the netlist adds: B minus C $= +0.319$.

N. Language model plus Loki's hints, version 2, as run (LLM5, LLM6)

Hypotheses. LLM5: a model given Loki's linked hints (version 2) recovers RTL better than the netlist alone. LLM6: the model uses the hints, so hints with wrong content hurt.

The full run found no gain from the version-1 hints. Two defects were fixed before this run: the claims were weak (no counter kind, empty state-machine claims, duplicates), and the model was never told what the hints are. Version 2 writes one plain-English line per claim with a probability, links it into the netlist as a comment on the line of each cell the claim covers, and the prompt explains the hints and asks the model to name the claims it used. The plan was written and committed before the calls.

Setup. The 64 problems of Q3 (34) and Q4 (30), where arm A of the full run was not at the ceiling. Four arms: A2 (netlist only), B1' (the version-1 hints with the new prompt), B2 (version-2 hints) and W (version-2 claims moved to wires they do not describe; 5 problems whose wrong claims stayed too close to the true ones are left out of the W tests). Three draws per problem and arm: 768 calls to fresh `claude-sonnet-5-5` subagents, the full-run scorer and tests (at least 2 of 3 draws; exact stratified McNemar; Holm over nine tests). The stop was set at 97 percent of the weekly meter (Peter, before the calls); the meter never passed 89.

Results. 762 calls produced a reply file; 6 did not (the model answered in chat), with 0 API refusals and 0 timeouts. Outcomes: pass 645, runtime 109, compile 8. Table XV gives the pass rates. In Q4 the model passes more often with the

TABLE LV

PER GATE-BREAKER QUARTILE, THE LOKI-ONLY FLOW AND THE LANGUAGE-MODEL ARMS SIDE BY SIDE. LOKI AND ROUND-1 LLM ARE BOTH TAKEN ON THE COMMON SET, THE DESIGNS SCORED BY THE ROUND-1 GATE-BREAKER HARNESS REPLICATION (TABLE XII) AND BY LOKI, WITH THE SAME AST AND CFG METRIC; LOKI IS THE CALIBRATED `f3` L1 OUTPUT (ROUND 2). n IS THE COMMON DESIGNS WITH AN AST VALUE; n_{CFG} IS THE DESIGNS WITH A CFG VALUE FOR LOKI / THE ROUND-1 MODEL. A DASH IS A QUARTILE WITH NO COMMON DESIGN. PILOT ARMS: FUNCTIONAL PASS RATE, ALL CALLS (STRICT SCORER), n PROBLEMS, A DIFFERENT METRIC THAT IS NOT COMPARABLE WITH AST OR CFG. GATE-BREAKER'S PUBLISHED NUMBERS APPEAR IN THE PAPER ONLY AS DELTAS (SECTION VI-A), SO NO ABSOLUTE PUBLISHED COLUMN IS SHOWN.

	Common set, Loki				Round-1 LLM		Pilot pass			
Quartile	n_{AST}	n_{CFG}	AST	CFG	AST	CFG	n	A	B	C
Q1	40 / 40	40 / 40	0.699	0.741	0.907	0.913	10	10/10	10/10	8/10
Q2	30 / 31	31 / 31	0.512	0.754	0.733	0.917	10	8/10	9/10	8/10
Q3	1 / 1	1 / 1	0.360	0.744	0.727	0.990	10	8/10	9/10	7/10
Q4	1 / 1	1 / 1	0.350	0.763	0.606	0.901	10	9/10	7/10	7/10

TABLE LVI

SECONDARY VIEW: THE SAME LOKI-ONLY MEASURES PER QUARTILE OF THE PAPER'S ABSOLUTE COMPLEXITY BINS (TABLE XII), NOT OF THE GATE-BREAKER QUARTILE FILES. OF THE 150 BUILT DESIGNS, 12 ARE IN NO QUARTILE FILE AND 35 OF THE OTHER 138 SIT IN A DIFFERENT QUARTILE IN THE TWO PARTITIONS (CROSS-TABULATION IN `PARTITION_CROSSTAB.CSV`). CELLS AS IN TABLE LIII; HAL CELLS ARE LOKI W MATCHED/PREDICTED/GROUND TRUTH. THE Q4 SIGNAL-COVERAGE MEAN INCLUDES `VE_PROB153_GSHARE`: 140 RECOVERED SIGNALS AGAINST 15 DECLARED (REFERENCE METHOD: REGEX), A RATIO OF 9.33. THE REGEX EXTRACTOR DOES NOT UNDERCOUNT THE SOURCE DECLARATIONS (15 NAMES: 11 PORTS, 3 INTERNAL NETS, THE ARRAY `PHT`, COUNTED ONCE); THE RECOVERED COUNT IS CONSISTENT WITH THE 128-ENTRY ARRAY `PHT` (`2**N`, $n = 7$) FLATTENED INTO ELEMENTS PLUS 12 OTHER NAMES, BUT THE RECOVERED SIGNAL LIST ITSELF WAS NOT INSPECTED.

Measure	Q1	Q2	Q3	Q4
Built designs, `f3`	38	39	37	36
Signal coverage, `f3`	0.987 (1.000)	1.000 (1.000)	0.974 (1.000)	1.161 (0.908)
AST n, L1 calibrated `f3`	38	38	32	21
AST, L1 calibrated `f3`	0.754	0.455	0.605	0.505
CFG n, L1 calibrated `f3`	38	38	36	32
CFG, L1 calibrated `f3`	0.750	0.741	0.750	0.768
HAL designs	11	7	5	10
Loki W, matched/pred/GT	0/0/0	0/0/0	5/6/7	8/142/139

version-2 hints (pass@1 0.933 for B2 against 0.744 for A2, 0.800 for B1' and 0.789 for W); in Q3 all arms sit between 0.85 and 0.87. Over all 64 problems B2 minus A2 is $+0.099$ with 95% interval $[+0.052, +0.151]$ and B2 minus B1' is $+0.073$ $[+0.031, +0.115]$. Pass@3 is 0.906 for A2, B1' and B2 (W 0.875): the hints raise the consistency of a single draw, not the best-of-three ceiling. The primary test does not reach the rule: 4 problems only B2 passes against 0 only A2 passes, Holm $p = 1.0000$. A2 already passes 54 of 64 problems by that measure, so few problems can differ.

Use of the claims. The model names a claim in 107 of 192 B2 calls. FSM claims were used in 57 B2 calls and all 57 passed; counter claims in 24 and all passed. In W, FSM claims were used in 33 calls with 27 passes, against 24 calls where an FSM claim was offered and not used with 19 passes (0.82 against 0.79, no difference); a used wrong state-machine hint passed less often than a used right one (27 of 33 against 57 of 57), but on different problems, so this is not a paired test. W minus A2 is $+0.026$ ($[-0.021, +0.073]$), with 2 problems only W passes and 2 only A2 passes.

(Table XV is in the main text.)

Threats to validity. (i) Underpowered: the primary test counts discordant problems, A2 is near the ceiling, and 4 against 0 cannot pass after Holm; the pass@1 interval and the claim-use table are not the pre-stated rule. (ii) Blinding: as in the full run, the call subagents were not tool-less and saw the workspace instructions, a memory index and a terse-style hook text. The transcript audit of all 768 calls found 0 calls touching any path other than their own call file and reply file and 0 using other tools; 26 used more than two tool calls (paged reads, double writes, and 6 calls that edited their own reply file with Edit instead of Write). All are scored as made. (iii) Confound: A2 has neither the hint explanation nor the claim-use line, so B2 against A2 differs by both; B2 against B1' holds the prompt fixed and isolates the hint content. (iv) The probability of a sequential claim is a per-kind constant checked on the netlist, not a measure of designer intent, and it was not used in any analysis. The checks are in-sample. (v) Wrong claims (W) stay partly close to the true ones in small designs. (vi) Memorisation is not controlled in this run. Sampling is not controlled.

Verdicts. LLM5 **INCONCLUSIVE**: B2 against A2 stratified discordant 4 against 0, Holm $p = 1.0000$, although mean pass@1 is $+0.099$ $[+0.052, +0.151]$. The interval and the claim-use table suggest that version-2 hints help on Q4 and that the model reads and uses state-machine and counter claims; this is a pointer for a better-powered test, not a verdict. LLM6 **DISMISSED**: W against A2 discordant 2 against 2, Holm $p = 1.0000$; the hypothesis that wrong hints lower the pass rate is not supported by the pre-stated test.

O. State-machine encoding (FSM2a, FSM2b, FSM3)

Question. Question (hub L Q3.3): does recoding a state machine one-hot change the footprint Loki sees, and are the state-machine claims selective enough to use?

Experiment and result. On the 100 generated state machines of round 4, built three ways (the fsm pass recoding one-hot, recoding binary, or leaving the HDL encoding) on the gate and the LUT floor, the version-2 claims of `loki re --intent`, which read how the state bits turn each other on, recover 98 of 100 one-hot machines and 99 of 100 binary ones, against 0.13 and 0.35 for the stage-one detector. The recall difference one-hot minus binary is -0.010 with 95% interval $[-0.025, 0.000]$, inside ± 0.10 (hypotheses FSM2a, FSM2b). The encoding call is right for 97 of 98 one-hot machines, but a binary machine whose state count is not a power of two is called other. The cost is precision: 446 of 577 counted claims sit on memory words of the 20 register-file negatives, so the pooled precision is 0.17 against 0.70 to 0.92 (per design 0.82, and 0.75 without the register-file claims); FSM2a is therefore DISMISSED on its precision half, and FSM2b holds for recall only until the claims are selective.

A rule on the next-state function makes the claims more selective, but not enough. A memory word never decides where it goes next: for every control value its next state is its old state or one value whatever the state is (hold or load), and a counter with a load port can reach every code in one step. Rejecting these two patterns (chosen on the even-numbered half of the designs, scored on the odd-numbered half, fixed before the change) removes all 256 register-file claims per cell on the test half and loses no state machine (recall stays 1.00), and the pooled claim precision rises from 0.15 to 0.68 (per design 0.81 to 0.88). The bar of 0.80 is not reached: 21 claims on pipeline negatives and 3 on counters remain, and no tuning design produced one, so the rule has not seen them (hypothesis FSM3, DISMISSED).

Verdict and PR. FSM2a DISMISSED (the precision half is falsified); FSM2b POSSIBLY GOOD (recall holds); FSM3 DISMISSED (precision 0.676 against the bar of 0.80, recall 1.000). loki PRs #64 and #65 (2026-10-30).

P. The L3 census (P3C)

Question. Question: which seed patterns occur in which designs, and how does the netlist footprint of each change with the tool and the target?

Experiment and result. A first census (hypothesis register row P3C, a measurement) counts the seed patterns in 519 designs (the 156 VerilogEval references, 360 round-4 designs, three VTR designs), labelled from the register-transfer side, and synthesises each with Odin III (gates, LUT6, LUT6 with hard blocks) and Yosys (gates, LUT6). A state machine occurs in 78 designs, a counter in 44, a shift register in 11, a timer (counter with terminal compare) in 12, and a composite is rare: three designs have a state machine that drives a shift register and one has a state machine with counters and a shift register. The footprint changes with the target and the tool. On the hard-block target no counter claim survives (0 of 43; a hard-adder claim appears in all 43), against 37 of 43 on gates and LUTs. At the same gate target the terminal compare of a timer is a separate claim in 7 of 9 Yosys netlists and 2 of 11 Odin III netlists, and the Yosys gate netlists are about four times larger. Both tools recode state machines to one-hot, so the encoding call does not differ. The motivating fancy timer yields no state-machine, counter or shift-register claim in any Odin III cell, and the Gate-Breaker Yosys script turns it into constant outputs, which is the gap an L3 composite is meant to explain. No two claims of a composite were joined by a register-input path.

Verdict and PR. P3C is a measurement and has no verdict. loki PR #66 (2026-10-30).

Q. The L3 v0 detector (L3A, L3C)

Question. Question: do rule detectors on the level-one claims find single patterns and the parts and wires of composites?

Experiment and result. A first detector (v0; hypothesis register rows L3A and L3C, measurements with a stated rule) reads the level-one claims and the netlist through a JSON catalogue with one signature per target; on the hard-block target a counter is a hard-adder claim whose sum bits loop through flip-flops back to the adder with a one-bit or constant operand. On 252 held-out single designs it works for state machines (precision 0.85 to 0.95, recall 0.95), counters (no false alarm; recall 0.86 to 0.90 on gates and LUTs, 21 of 21 on the hard-block target) and glue logic; it does not work for shift registers (precision 0.14 to 0.17), timers (recall at most 0.17) or register files, and the LFSR has one held-out design. Most misses are a missing level-one claim, not a wrong composition. We then built 182 composite benchmarks by wiring 27 VerilogEval references (named by their problem numbers, for example Q035_146) with a known wiring; on the 82 held-out composites the parts are found 41 to 50% of the time and the wires between parts 12 to 18% (38 to 61% when both ends were found), below the bar of 0.70, so the combined benchmarks do not yet show that the composite idea works; they show which single claims must be repaired first. The Odin III and Yosys cells behave alike.

Verdict and PR. L3A-fsm to L3A-register-glue and L3C-composite are measurements with a stated rule and have no verdict. loki PR #67 (2026-10-31).

R. Graphlet signatures and a library of every width (GL1 to GL7)

Question. Question: can a partition's graphlet signature say that it is a given pattern when exact matching cannot?

Experiment and result. The plan is in `experiments/graphlets/EXPERIMENT.md`), we built a library of 700 generated circuits (counters, shift registers, LFSRs, registers, timers, register files, state machines and glue, every width from 1 to 32; register files at four depths) in the five cells, all synthesised (3500 of

3500), and took their undirected (73 orbits) and directed (129 orbits) graphlet signatures with the Odin III `graphlets` pass; signatures of widths above 32 are made on demand and cached (tested at 33, 48 and 64 bits, 165 of 165 builds). The signature of a whole single design is a weak identifier: on 243 held-out single-pattern designs it does not reach precision and recall 0.80 for any pattern in any cell, and the rule detector of v0 stays better for counters, state machines and glue; a signature-or-rule hybrid recovers 39 of 60 register files (the rule alone, 6) with extra false alarms. Directed graphlets were more precise in one of 25 tests (state machines on Yosys gates) after Holm correction. Library circuits with a width left out are classified correctly 0.77 (undirected) and 0.80 (directed) of the time when only the flip-flop count is used and the nearest size stands in, and 0.95 to 1.00 when all other widths are searched. Cutting the design into state groups (the registers the claims found, grown to the union of their input cones) and matching each cut with the library by its flip-flop count does better but not well enough: on single designs no pattern reaches 0.80 (one hybrid cell, counters on the hard-block target, does, where the rule already works); on the 82 held-out composites the signature alone finds 0.33 to 0.61 of the parts (rule 0.41 to 0.50) and, with the rule added, 0.52 to 0.74, with more false alarms (precision 0.25 to 0.36, rule 0.35 to 0.54). Directed graphlets were more precise for state machines on the two LUT cells and nowhere else (2 of 25). The Bayesian posterior stays below 0.3 for almost every partition, so it does not support a confident accept. Stopping each cone at the neighbouring registers, so that a partition holds one part and its own glue, raises the part precision of the signature alone on the composites from 0.23–0.35 to 0.35–0.87 and lowers its recall from 0.33–0.61 to 0.21–0.42; with the rule added, the parts are found 0.46–0.65 of the time (rule alone 0.41–0.50) at precision 0.32–0.57 (rule 0.35–0.54). Links between matched partitions recover the wires between parts at most 0.33 of the time, with precision 0.17–0.46 (rule: recall 0.12–0.18, precision 0.36–0.45), so they are not yet a usable wire detector. Partitions from the second-level clustering and a learned classifier are not yet tested.

Verdict and PR. GL1, GL2, GL4, GL5, GL6 and GL7 INCONCLUSIVE; GL3 a measurement. loki PRs #68, #69 and #70 (2026-10-31).

S. Fuzzy graphlet signatures and a learned classifier (GL8, GL9)

Question. Question: does a classifier that learns what the signatures of real structures look like find structures that nearest-entry matching misses?

Experiment and result. The plan is in `experiments/graphlets-fuzzy/EXPERIMENT.md`), we described each pattern by the orbits that carry it in the library, then compared signatures more loosely: a distance between per-orbit count distributions, a cosine of log totals, and a logistic regression and a random forest on normalised orbit totals, count quantiles and the flip-flop count, on undirected, directed and direction-tolerant directed signatures (each orbit mixed with the orbits that differ from it by one reversed edge, at four strengths, and a collapse to undirected), all fitted on TUNE only. The classifiers are clearly better than the nearest-entry match: their probabilities are close to the observed rates (a probability of 0.6 to 0.7 is right 0.89 of the time, where the earlier posterior never exceeded 0.3), the signature alone reaches precision and recall 0.80 in four cells at one tolerance level (state machines on three cells, a counter on one; exploratory among 28 methods), and with the rule added the composite parts are found 0.54 to 0.80 of the time (rule 0.41 to 0.50). The rule alone still beats every signature method on single designs, a per-pattern choice fitted on TUNE kept the rule for every pattern, and wires remain weak (recall up to 0.47 at precision 0.28). Exact directed beat undirected on TUNE F1 (0.759 against 0.715); loosening the direction did not help on TUNE.

Verdict and PR. GL8 and GL9 INCONCLUSIVE. loki PR #71 (2026-10-31).

T. Q-L4: why the signature detectors miss

Question. Question: why are high-level things not recovered?

Experiment and result. The plan and rule are in `experiments/ql4-misses/EXPERIMENT.md`, fixed before the run), we took every construct of the answer key that five signature detectors of the earlier rounds did not find (1739 constructs: 695 single designs and 1044 composite parts, five cells) and asked two questions of each miss: did the matcher compare the right block, and does the right block, cut out by the Odin III provenance, look like its pattern in directed signature distance? Most misses of the cone detectors compared the wrong block (fewer than 40 percent compared the right one), so the partition is the largest gap; by the rule fixed before the run, 80 percent of the misses of the bounded-cone detector and 79 percent of the random forest are near misses (wrong partition, no library entry of that width, or the threshold), which supports the near-miss hypothesis. That rule counts a wrong partition as near without measuring the true block; measured, the true block is close to its pattern for only 39 percent and 26 percent of those misses (67 to 75 percent for the three other detectors), and the share of structure that the signature cannot see is 20 percent by the rule and up to about 70 percent if ”far from the pattern” counts. Parts that Odin III optimised away entirely (about 9 percent of the misses) and most of the structure misses lose their source units first in the `opt` pass.

Verdict and PR. QL4a POSSIBLY GOOD (80 percent near misses by the fixed rule, which counts a wrong partition as near); QL4b INCONCLUSIVE; QL4c a measurement. loki PR #72 (2026-11-01).

U. Q-L6: growing blocks from anchors

Question. Hypothesis: blocks grown from anchors by a Bayes factor on directed graphlet signatures are the right block more often than bounded cones.

Experiment and result. The plan, method and verdict rules are in `experiments/ql6-grow/EXPERIMENT.md`, fixed before the run). Anchors are the places where a known pattern is easy to see (a group of flip-flops that feed only each other, a shift chain, a triangular counter, a decoder-selected memory, a register, an adder chain). Every anchor grows at once, one neighbouring cell per step, as long as the Odin III directed graphlet signature of the bigger block raises the odds of a known construct against glue, and at every step the block keeps a probability over six construct kinds. The posterior is learned from answer-key blocks of held-out designs only. On the 1255 constructs of the earlier rounds that have a provenance key, the grown block is the right one for 61 (0.049), against 466 (0.371) for the bounded cones, so the hypothesis that growing beats cutting is dismissed as specified: the blocks are correct as far as they go (they stay inside the construct) but stop early, a median of 11 cells against 35. Reading the same growth paths without the stop (post hoc) shows that some prefix is the right block for 0.556 of the constructs, against 0.385 for the cones in the same units, so the growth order is good and the Bayes factor stop, resting on a likelihood that picks the right kind for about half of the held-out pieces, is what fails. The posterior does sharpen with size (mean entropy 1.55 bits at 4 to 7 cells, 0.96 bits at 32 cells and more) but the argmax is not more often right (verdict inconclusive), and pieces of 4 to 7 cells give a top-3 guess that is 0.10 better than guessing the three most frequent kinds (lower bound 0.011, possibly good). The anchor census bounds the approach: an anchor of the matching class survives for 0.35 (shift) to 0.75 (register file) of the constructs.

Verdict and PR. QL6a DISMISSED (0.049 against 0.371); QL6b INCONCLUSIVE; QL6c POSSIBLY GOOD (lower bound 0.011); QL6e a measurement. loki PR #73 (2026-11-01).

V. Q-L7: where a grown block stops

Question. Hypothesis: a stop rule that does not depend on the weak per-step likelihood recovers the right block more often than the cones.

Experiment and result. The stop rule was the failing part of Q-L6, so we grew every block to the end of its path (frontier exhausted, or 300 cells) and chose the cut afterwards (the plan, split, four candidate rules and verdict rules are in `experiments/ql7-stop/EXPERIMENT.md`, fixed before the test run). The rules were the posterior odds of a known construct against glue with a size term, the first prefix at a local minimum of the conductance of the block (wires crossing its border over its wires), a classifier on trajectory features trained on held-out designs, and the larger of the last two, chosen on a validation split of designs that trained nothing else. On the 1141 test constructs that the cones and the earlier grower also scored, the combined rule is right for 0.528, against 0.053 for the earlier stop, 0.385 for the cones and 0.556 for the answer-key ceiling, so it closes 94 percent of the gap to the ceiling (0.927 to 0.961) and beats the cones (geomean over design and cell units 1.146, 1.105 to 1.190; gate netlists 1.348, four-input LUT 1.122, but k6 hard-map 0.968 and composite parts 1.058 are inconclusive). The gain is however only that blocks are no longer stopped early: growing every block to the end of its path is right for the same 0.528, and none of the three rules beats it (they are worse by 1 to 9 percent, and the combination equals it). On single-design units the end of the path is already the ceiling (0.597), and on composite parts it is 0.441 against a ceiling of 0.504, so the stop has little room. What is left to recover lies in the growth order (449 of 1141 constructs have no right prefix on any block) and in the anchors (58 constructs are not touched by any block), and the kind guess stays weak (precision 0.27, recall 0.19).

Verdict and PR. QL7a and QL7b USEFUL; QL7c a measurement; QL7e INCONCLUSIVE (no rule beats never stopping). loki PR #74 (2026-11-02).

W. Q-L8c: a stronger kind likelihood for grown blocks

Question. Hypothesis: a likelihood on richer directed graphlet features raises kind precision and recall well above Q-L7's (fixed as +0.10 macro F1).

Experiment and result. The grown blocks are now often right, but the guess of what a block is stays weak (precision 0.27, recall 0.19). We gave the likelihood more to look at: orbit instances per cell and not only orbit shares, how the block touches the rest of the netlist (conductance, frontier fan-in and fan-out, primary inputs read), the share of flip-flops, hard and adder cells, and the feedback structure of the block (strongly connected components, cycle rank, longest combinational path) (plan, features, three models and verdict rules in `experiments/ql8c-kind/EXPERIMENT.md`, fixed before the test run; the Odin III graphlets pass supplies every orbit count). Three likelihoods were fitted on the answer-key blocks of the even half of the tuning designs (naive Bayes on binned features, a full-covariance Gaussian on 24 principal components with a size line, a diagonal mixture per kind), their temperature on the grown blocks of 97 further designs, and the model was chosen on 82 more; a fourth, the Gaussian fitted with wrong grown blocks as the glue class, was added after a first look. The Gaussian won. On the test designs it recovers 232 of 1163 constructs with their kind against 213 for the earlier likelihood (per-unit geomean 1.011, 1.001 to 1.021, W74 / L55 / T1221), and the macro F1 over the five kinds rises from 0.211 to 0.262 (gain 0.051, 0.019 to 0.081), which is real but about half of the 0.10 we had set as "well above"; the hypothesis is not supported. Counters, registers and register files gain, finite-state machines and shift registers do not. The posterior is better calibrated on all grown blocks (ECE 0.139 against 0.197, difference -0.057, -0.085 to -0.019) but not on the right blocks alone. Given the answer-key block, the kind is guessed with macro F1 0.487 against 0.419 and recall below 0.41, so most of the remaining error is the kind guess itself and not the block, and no model here has learned what a wrong block looks like.

Verdict and PR. QL8c-a POSSIBLY GOOD; QL8c-b DISMISSED (gain +0.051); QL8c-c and QL8c-d measurements. loki PR #75 (2026-11-02).

X. Q-L8a: a growth order that respects structure

Question. Hypothesis: a growth order that chooses the next cell by wiring recovers the right block more often than the cones and than Q-L7's order.

Experiment and result. 449 of 1141 constructs have no right prefix on any block. We tried three orders that choose the next cell by wiring and keep the weak likelihood only for ties (attachment strength, a feedback and sibling-net frontier, input cones first; plan, split and verdict rules in `experiments/ql8a-order/EXPERIMENT.md`, fixed before the test run). The order was chosen on 82 designs outside the test set. The attachment order recovers the right block for 0.526 of the constructs against 0.528 for the earlier order grown to the end and 0.385 for the cone detector; it gives a right prefix to 12 of the 449 (2.7 percent) and loses 14 of the 602 that were right, and k6 and hard-map netlists (0.965 against the cones) and composite parts (1.044) stay inconclusive. The sibling-net frontier helps single designs (0.625 against 0.597) and gate netlists (0.624 against 0.587) and hurts composite parts (0.393 against 0.441). The hypothesis is not supported. What remains is mostly constructs cut across several blocks (251 of the failures), then constructs never fully reached (103) and constructs swallowed by a block that grows past them (95): a problem of joining blocks, not of the order inside one.

Verdict and PR. QL8a-a, QL8a-b and QL8a-c INCONCLUSIVE; QL8a-d a measurement. loki PR #76 (2026-11-02).

Y. Q-L8b: anchors for the constructs no block touches

Question. Hypothesis: extra anchor classes make a share of the 58 constructs that no block touches touched and right, at no cost elsewhere.

Experiment and result. Fifty-eight of the 1141 test constructs were never touched by any block. Described first (plan and census in `experiments/ql8b-anchors/EXPERIMENT.md`, fixed before the test run): 37 are the constant cells of the netlist that the provenance gives to a part, two are six-cell combinational state machines, and 19 are flip-flop groups that read several other flip-flops, 14 of them the 64-flip-flop shifter of one problem. We added three anchor classes (flip-flops sharing the same global control, open feedback loops of flip-flops, and the constant cells as one anchor that does not grow), kept all three on 82 validation designs, and ran the same growth once on the test designs. All 58 are now touched and 47 are right (19 of the 21 structural ones, 28 of the 37 constants), so the hypothesis that anchors can cover them is supported. The price is not zero: 89 constructs that were right are lost and 115 are gained, the right-block share rises from 0.528 to 0.550, and the per-unit comparison with the old anchors is a tie (geomean 1.001, 0.980 to 1.024) with a loss on single designs (0.960) and gate netlists (0.948) and a gain on composite parts (1.129) and k6 and hard-map netlists (1.042). Against the cone detector composite parts are now better (1.184, 1.101 to 1.273; it was inconclusive). The loss comes from new blocks that take cells of a design that is itself the construct, and from the constant block taking tie-off cells that sit inside other constructs; both are a joining problem, not a missing anchor.

Verdict and PR. QL8b-a POSSIBLY GOOD (47 of 58 right against the bar of 10); QL8b-b INCONCLUSIVE; QL8b-c a measurement. loki PR #77 (2026-11-02).

Z. Q-L9: joining and splitting grown blocks

Question. Hypothesis: a post-growth join and split step raises the right-block share above 0.550 without losing single designs.

Experiment and result. Most missed constructs are cut across several blocks (251 constructs). We added a step after growth (the plan is in `experiments/ql9-join/EXPERIMENT.md`, rules, grids and verdict rules fixed before any run): a join step that merges two touching blocks when a rule says they are one construct, and a split step that cuts a block along the Fiedler vector of its cell graph when two parts explain it better than one. Four join rules were tried: a Bayes factor of one construct against two with the #73 likelihood, the same with the #75 Gaussian, the cut conductance between the two blocks, and shared global control nets; each had its threshold set on 383 training designs and compared on 320 validation designs. The conductance rule won (join when at least a fifth of the link ends of the smaller block lie on links to the other). No split threshold had a net gain on either set, so the headline has no split. On the test designs the right-block share rises from 0.550 to 0.715 (cone detector 0.385); per unit the geomean is 1.149 (1.123 to 1.177) against the blocks before the step, W194 / L9 / T654, with single designs 1.151 (they were the cost of the extra anchors in Q-L8b) and gate, 4-LUT, k6 and composite sets all better. Of the 251 constructs split across blocks, 157 are now right. The swallowed ones are not helped (the 42 of 95 right after the step were already right before it), and 29 constructs that were right are lost (all in composite parts, a joined block swallows a neighbour). The Bayes-factor joins are weaker (0.680 and 0.677) and shared control nets weaker still (0.617). The macro F1 of the kind guess on the blocks rises from 0.237 to 0.392 (the #75 Gaussian posterior on the blocks before and after the step).

Verdict and PR. QL9-a and QL9-b USEFUL; QL9-c POSSIBLY GOOD; QL9-d DISMISSED; QL9-e a measurement. loki PR #78 (2026-11-02).

AA. Q-L10: Loki's blocks decoded one at a time by a language model

Question. Hypothesis (plan in `experiments/ql10-blocks/EXPERIMENT.md`, rules fixed before any model call): when Loki cuts the

TABLE LVII

F3 L1 PER KIND ON THE 15 TEST DESIGNS. THE CALIBRATED COLUMNS ARE THE PROTOCOL'S SETTING; THE `--calib none` COLUMNS ARE A DIAGNOSTIC OF WHAT THE DETECTORS PROPOSE BEFORE CALIBRATION PRUNES IT. A KIND WITH NO COUNTED PREDICTION HAS NO F1 AND IS LEFT OUT OF THE MACRO AVERAGE.

		calibrated			`--calib none`			
Kind	GT	counted	matched	F1	counted	matched	precision	F1
adder	17	0	0	–	1	1	1.000	0.111
adder_cin	17	0	0	–	0	0	–	–
comparator	20	0	0	–	14	0	0.000	0.000
eq_const	16	0	0	–	1,141	0	0.000	0.000
ff_srst	2	0	0	–	7	0	0.000	0.000
hard_add	17	0	0	–	0	0	–	–
hard_mul	11	0	0	–	0	0	–	–
mux	7	0	0	–	0	0	–	–
mux_bit	7	0	0	–	0	0	–	–
register	22	0	0	–	3	1	0.333	0.080
subtractor	8	0	0	–	0	0	–	–
macro-F1	144	0	0	0.000	1,166	2		0.038
buses	120	2,569	0	0.000	2,835	0	0.000	0.000
datapath role	20,541	20,541	20,541	1.000	20,541	20,541	1.000	1.000
control role	0	0	0	–	0	0	–	–

TABLE LVIII

F3 AST (GATE-BREAKER ALGORITHM 2, DEPTH 3, PYVERILOG) AND CFG (ALGORITHM 3, ODIN3 PROBE) SIMILARITY AGAINST THE MICRO REFERENCE RTL, L0-STOP (GATE NETLIST) AGAINST L1-STOP (`DESIGN.V`); QUARTILES BY F3 NODE COUNT. WITH THE F3 CALIBRATION L1-STOP EQUALS L0-STOP, SO EVERY DELTA IS 0.0000; THE TABLE THEREFORE SHOWS THE L0 VALUES AND THE DIAGNOSTIC (`--CALIB NONE`) DELTAS.

Quartile	AST L0	AST L1	ΔAST	CFG L0	CFG L1	ΔCFG
0 (4 designs)	0.4428	0.4428	0.0000	0.7909	0.7909	0.0000
1 (4)	0.3765	0.3765	0.0000	0.4224	0.4224	0.0000
2 (4)	0.3969	0.3969	0.0000	0.7013	0.7013	0.0000
3 (3 AST / 1 CFG)	0.4003	0.4003	0.0000	0.7507	0.7507	0.0000
mean (15 / 13)	0.4044	0.4044	0.0000	0.6468	0.6468	0.0000
diagnostic `--calib none` (L1 commits hierarchy)						
0 (4)	0.4428	0.4277	−0.0151	0.7909	0.7776	−0.0133
1 (4)	0.3765	0.3756	−0.0009	0.4224	0.4099	−0.0124
2 (4)	0.3969	0.4542	+0.0573	0.7013	0.7014	+0.0001
3 (3 AST / 1 CFG)	0.4003	0.4007	+0.0004	0.7507	0.7561	+0.0055
mean	0.4044	0.4155	+0.0111	0.6468	0.6394	−0.0075

netlist into its joined blocks and a model decodes one block per prompt, (a) the assembled design is as correct as the whole-netlist prompt of arm A, (b) designs too large for one prompt still decode, (c) the kind guess as a hint does not help.

Experiment and result. The controller ran the model calls (pilot of 75 calls, then all multi-block designs with three draws; the 358 single-block calls were deliberately not run, a dated amendment of the plan; 1,414 replies, none empty, no API error). Twenty-two calls used a shell or an editor on their own reply file, which the call template forbids; none read another file, and the analysis is repeated without the 27 rows they touch (same verdicts). Of 137 problems, 15 have no gate (scored with no call), 27 are multi-block (D 72 of 81 draws pass, A 73 of 81; difference -0.012, -0.074 to $+0.037$), 94 single-block problems were not run, and one pilot design (Prob088) has one draw. Over-cap designs: 8 of 12 draws pass (mux256to1 3 of 3, gates100 3 of 3, rotate100 2 of 3, shift18 0 of 3); processor modules: 1 of 5 equivalent (tinyriscv_div), 4 not equivalent, and replacing the blocks that failed the block check by their own gates makes all four equivalent. Block check: 954 of 969 replies ok. Hint: D+h minus D -0.042 (-0.125 to 0.000) on 8 designs. AST and CFG similarity, multi-block, D flattened against A flattened: -0.037 (23 designs) and -0.042 (27 designs). Estimated tokens per multi-block design over three draws: D about 14 500, A about 10 300.

Verdict and PR. QL10-a INCONCLUSIVE; QL10-b USEFUL; QL10-c INCONCLUSIVE; QL10-d (AST and CFG) and QL10-e (answer key) measurements. loki PR #82 (2026-11-02).

APPENDIX B
THE F3 EXPERIMENT IN DETAIL

Hypothesis F3 (Appendix A-B): L1 recovers structure from Gate-Breaker's own gate input. Table LVII gives the recovery per kind on the 15 f3 test designs, and Table LVIII gives the AST and CFG similarity per quartile of f3 node count.

APPENDIX C
RESEARCH LOG

This appendix lists every hypothesis, one row each, including the DISMISSED and INCONCLUSIVE ones (Tables LIX to LXIII); the rows are generated from `docs/paper/hypotheses.md`. A MEASUREMENT row has no hypothesis and no verdict. UNTESTED rows are the Future work.

Sources. Every row's numbers are in a committed file, named by the Round column: R1 in `experiments/rename-l2/EXPERIMENT.md`, `experiments/f3-recovery/EXPERIMENT.md` and `experiments/veval-l0/EXPERIMENT.md`; R2 to R6 in `experiments/round2` to `experiments/round6` (each has an `EXPERIMENT.md` and the CSV files); HAL in `experiments/hal` and `docs/results-hal.md`; LLM in `experiments/llm-pilot/EXPERIMENT.md`; Q in `experiments/quartiles`. The exceptions are the success-rate and Success@k columns of the Gate-Breaker replication (rows GB1 and GB2), which come from run directories of the controller machine that are not committed; the AST and CFG columns of GB2 are in `experiments/round2/l2_headtohead.csv`.

Order of the work. Round 0 (2026-10-21 to 22) built L0 (provenance-labelled, anonymised datasets and the renamed-instance check), L1 (Bayesian hypotheses with calibration and proofs) and L2 (graph analysis). Round 1 (2026-10-23) gave the first results: hard blocks recovered, soft structures weak, nothing recovered on the gate netlist, the GNN ahead of 1-WL. Round 2 (2026-10-24) asked five follow-up questions (the LUT6 re-map, the operator F1, the head-to-head with the language model, the evidence families and the must/may bracket and structural prefilter) and found that only the operator F1 is supported, weakly. Its puzzle, why wrong candidates are not real constructs, came from the author's question (2026-10-24): "for Loki - the question is from a reverse engineering perspective why is it hard to reconstruct to HDL - going from if structures (which we know are muxes) and then down it

seems like many of the patterns are constant - so hypothesizing that things are a bus (and by a bus I mean a signal that could be a traditional bus, but also a wide wordsize signal that has regular oprtations to it) [...] Control flow is usually in the form of muxin and FSMs have pattern of feedback (state equals next state) and then control based on state. How do we find these patterns better?" and "Also, the outputs (words) and the inputs as words also provide hints of where the words are". Round 3 (2026-10-24) measured fragments and I/O propagation; round 4 (2026-10-25) built a synthetic corpus, a signature dictionary, a look at big designs and the identifiability ceiling; round 5 (2026-10-25) built the word-first detector; round 5b (2026-10-26) compared it with DANA; the LLM pilot and the per-quartile tables (2026-10-27 to 28) put a model beside Loki; round 6 (2026-10-27) is the loss audit. The full language-model run (2026-10-29) and the hints-version-2 run (2026-10-30) followed, then the state-machine encoding study (2026-10-30), the L3 census and detector (2026-10-30 to 31), the graphlet studies (2026-10-31) and the Q-L series, Q-L4 to Q-L9 (2026-11-01 and 2026-11-02).

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TABLE LIX
RESEARCH LOG, HYPOTHESES 1 TO 12 OF 107: EVERY HYPOTHESIS, ONE ROW EACH, WITH THE VERDICT.

ID	Round	Hypothesis	Result	Verdict
L0a	R1	The pipeline uses no names: renaming every identifier changes nothing.	all pairs byte-identical for L1 (204 + 204 + 68 sets); L2 clusters identical on 271 of 272 partitions (one spectral nondeterminism, also present on base)	USEFUL
L0b	R1	The provenance answer key reconstructs the reference at every floor with no recovery step.	54,306 gate nodes all labelled; module coverage 1.000; signal coverage geomean 0.986; 6 of 156 not built	USEFUL
L1h	R1	Structures the synthesis flow already reverse-engineered (hard adders and multipliers) are recovered almost perfectly.	hard adders 67/67 (P 0.985), hard multipliers 30/30	USEFUL
L1s	R1	The Bayesian flow recovers the structures the flow erased.	f1 0.354 (271 counted, 107 matched, 688 GT); f2 0.108 (91/19/686); uncalibrated f1 0.202, f2 0.075	INCONCLUSIVE
L1c	R1	Calibration makes the predicted probabilities honest.	calibrated 0.217 vs uncalibrated 0.650	USEFUL
F3	R1	L1 recovers structure from Gate-Breaker's own gate input.	0 of 144 counted, 0 committed; buses proposed 2,569, none match; similarity probe: with the f3 calibration L1-stop equals L0-stop (every delta 0.0000); uncalibrated, 1,535 hypotheses in 11 of 15 designs move AST by +0.0111 and CFG by -0.0075; the flatten control raises AST by +0.0533 and CFG by +0.0559 (experiments/f3-recovery/EXPERIMENT.md)	DISMISSED
L2a	R1	Community methods find the designer's hierarchy; centroid methods do not.	NMI against module_path: k-means 0.3220, affinity propagation 0.2708, spectral 0.1378, Louvain 0.1360 (the reverse of the hypothesis); modularity is not a hierarchy measure (row L2m)	DISMISSED
L2m	R1	(measurement, no hypothesis) the modularity of the four clusterings of the netlist graph	Louvain 0.4220, spectral 0.3620, affinity propagation -0.2264, k-means -0.2912	MEASUREMENT
L2b	R1	A GNN with node, graphlet, WL and NPN features beats the 1-WL baseline.	0.4241 vs 0.2717; accuracy 0.8863 vs 0.8492; node-only GNN loses (0.2413 vs 0.2711)	POSSIBLY GOOD
R2-L1	R2	The f3 failure is detector vocabulary, not missing information.	0.87 in-sample (14 vs 16 constructs of 580), 0.51 held-out (2 vs 4)	INCONCLUSIVE
R2-L2	R2	Loki's AST/CFG is below the LLM's on Q1/Q2 and does not fall Q1 to Q4, with output equal to input.	Loki AST 0.763/0.460 vs LLM 0.894/0.761 (Q1/Q2); AST falls Q1 to Q4 (0.754 ... 0.505); equivalence 150/150	INCONCLUSIVE
R2-L4	R2	One or two evidence families carry the calibrated performance.	full 0.0038 vs priors-only 0.0051 (difference -0.0013; the gap is below the 0.02 floor)	INCONCLUSIVE

TABLE LX
RESEARCH LOG, HYPOTHESES 13 TO 24 OF 107: EVERY HYPOTHESIS, ONE ROW EACH, WITH THE VERDICT.

ID	Round	Hypothesis	Result	Verdict
R2-L5a	R2	Bracket width falls with per-design F1.	rho 0.087, p 0.417 (exploratory calibrated: -0.346, p 0.0008)	DISMISSED
R2-L5b	R2	A cheap structural rule prunes $\geq$ 30% of prover calls with no lost proof.	best zero-loss rule prunes 0.067	DISMISSED
R2-L7	R2	Word-level operator F1 separates runs that AST and CFG tie on.	tie pair dF1 0.0173; flatten control moves AST by 0.0422, operator F1 unchanged	POSSIBLY GOOD
H9a	R3	Most false candidates are fragments of a word of a related kind.	pooled 0.583 (1,525 of 2,615); without the largest design 0.415, without `ff_en` 0.304 (exploratory); noise class unreachable by construction	POSSIBLY GOOD
H9b	R3	A word's fragments cover most of it, so grouping recovers it.	0.065 (touched words only 0.069)	DISMISSED
H10a	R3	Datapath words are reached from the I/O words within two hops.	0.095 (per-design mean 0.326 exploratory)	DISMISSED
H10b	R3	Control words are not reached.	0.605 (needs $\leq$ D - 0.20)	DISMISSED
L11	R3	(measurement, no hypothesis) 0.214 of muxes with a select are chain sinks; complete chains 5 of 89 on VerilogEval		MEASUREMENT
H12a	R4	The corpus builds on all floors (P_ok $\geq$ 0.95).	P_ok 1.0000 (1200/1200), min family 100	USEFUL
H12b	R4	The dictionary beats the round-2 detectors on real designs.	won on 0 of 7 kinds (V0, V1, V2)	DISMISSED
H12b2	R4	The dictionary generalises over unseen regions.	R 0.3288 (iid 0.1866, ood 0.0614)	INCONCLUSIVE
H12c	R4	GNN transfer to real designs improves with corpus size.	-0.0026, CI [-0.0097, 0.0094], 29 designs	DISMISSED

TABLE LXI
RESEARCH LOG, HYPOTHESES 25 TO 36 OF 107: EVERY HYPOTHESIS, ONE ROW EACH, WITH THE VERDICT.

ID	Round	Hypothesis	Result	Verdict
H13a	R4	Big designs are mostly repeated structure.	0.9283 (6,653 of 7,167)	USEFUL
H13b	R4	Big designs repeat more than small.	0.2404 (0.9283 vs 0.6879); flips with the small baseline (experiments/round4/EXPERIMENT.md)	POSSIBLY GOOD
H13c	R4	Repeated instances share the dictionary signature.	0.8606	USEFUL
H14a	R4	The flow destroys style information.	0.5000 (f2 0.4823, f1 0.4697)	USEFUL
H14b	R4	The coarser floor destroys more.	0.0177	INCONCLUSIVE
H14c	R4	The MDL score ranks references above recovered RTL.	l0-f2 0.860 ... f3-none 0.940 (rule not met)	INCONCLUSIVE
H14d	R4	The MDL score tracks round-2 similarity.	rho -0.5110, partial 0.3029	INCONCLUSIVE
H16a	R5	Word tiling beats the detectors on held-out real designs.	q_WB 0.023 vs baseline 0.022, null 0.203	DISMISSED
H16b	R5	Grouping recovers the enabled words (R +0.20, F1 +0.10).	recall 0.008 to 0.146, F1 0.002 to 0.026, precision 0.014	INCONCLUSIVE
H16c	R5	Word-level eq_const beats per-bit.	dF1 -0.000	DISMISSED
H16d	R5	The chain is complete more often.	WB 3 vs R2N 1	INCONCLUSIVE
H16e	R5	Words grown from registers reach the datapath beyond a null.	D 0.829, null 0.715, gap 0.114	INCONCLUSIVE

TABLE LXII
RESEARCH LOG, HYPOTHESES 37 TO 48 OF 107: EVERY HYPOTHESIS, ONE ROW EACH, WITH THE VERDICT.

ID	Round	Hypothesis	Result	Verdict
H16f	R5	Repetition evidence raises F1 on big designs.	-0.0033; degenerate (margin unreachable, F1 bound 0.0165)	DISMISSED
RQ-H1	HAL	Loki's word-first detector recovers register words better than, equal to or worse than DANA.	Loki 0.169 vs HAL 0.261, D = -0.092, 95% CI [-0.317, 0.043]	INCONCLUSIVE
GB1	R1	(measurement, no hypothesis) what the GateBreaker harness scores on the simple micro benchmarks with the model available on the host	success rate 0.933 (Q1) and 0.829 (Q2), Success@10 1.000 in both, AST 0.576 and 0.444, CFG 1.000 and 0.736; 169 of 174 candidates correct	MEASUREMENT
GB2	R1	(measurement, no hypothesis) the baseline the model-based rows compare against	Q1 (36): SR 0.903, S@1 0.944, S@5 1.000, AST 0.894, CFG 0.942; Q2 (37): SR 0.778, S@1 0.811, S@5 1.000, AST 0.761, CFG 0.890	MEASUREMENT
LLM1	LLM	A model given Loki's hints recovers RTL better than without.	A 35/40, B 35/40; paired 38: A 33, B 35 (B vs A discordant 2/0)	INCONCLUSIVE
LLM2	LLM	The netlist helps beyond what the model knows from the interface line.	C 30/40; paired A 33 vs C 29	INCONCLUSIVE
LLM3	Q	A model given Loki's hints recovers RTL better than without.	A 0.888, B 0.878; MAJ A 123/137, B 124/137; A-B +0.010 [-0.017, +0.036]; discordant 3 vs 4	DISMISSED
LLM4	Q	The netlist helps beyond what the model knows from the interface line.	C 0.560; B-C +0.319 [+0.241, +0.394]; MAJ B 124/137, C 77/137; discordant 52 vs 5	USEFUL
LLM5	Q	A model given Loki's linked hints v2 recovers RTL better than the netlist alone.	B2 vs A2: pass@1 0.901 vs 0.802, +0.099 [+0.052, +0.151]; Q4 0.933 vs 0.744; MAJ B2 58/64, A2 54/64; discordant 4 vs 0	INCONCLUSIVE
LLM6	Q	A model that uses the hints is hurt by hints with wrong content (W below A2).	W vs A2: pass@1 0.828 vs 0.802 (+0.026 [-0.021, +0.073]); discordant 2 vs 2; B2 names a claim in 107/192 calls; W FSM claims used 33, pass 27	DISMISSED
Q1	Q	The hints help in some quartiles.	B ahead by one in Q2 and Q3, level in Q1, behind by two in Q4	INCONCLUSIVE
Q2	Q	(measurement) what Loki alone recovers per quartile	see quartile tables	MEASUREMENT

TABLE LXIII
RESEARCH LOG, HYPOTHESES 49 TO 60 OF 107: EVERY HYPOTHESIS, ONE ROW EACH, WITH THE VERDICT.

ID	Round	Hypothesis	Result	Verdict
L15a	R6	Each construct kind is first lost at one dominant Odin III pass.	register, compare, mux, arithmetic `dominant` under the random-set null (opt 320, abc 166, flatten 68, second opt 26); under the persistence null (exploratory) `dominant-null` for all four (null shares 0.80, 0.84, 0.79, 0.46); only flatten of logic stays dominant	POSSIBLY GOOD
L15c	R6	The D14 labels explain the merged losses.	0.606 pooled "partly explained"; MI 0.978, CI 0.462, VT 0.459; opt 0.425, abc 0.821	INCONCLUSIVE
L15d	R6	A pass fools a detector.	mux word fooled by abc in all arms (WB 0.55 to 0.07; the ground truth is 174 instances at S7 and 166 at S8, so the two recalls are not on the same set); mux_bit: B and WB 0.201 to 0.012 (35 of 174 to 2 of 166), delta -0.189, which rule 4 reads as "not fooled" because it needs a delta of -0.20 or less; others at a floor	POSSIBLY GOOD
L15b	R6	(descriptive) which constructs are joined with which, why		MEASUREMENT
FSM2a	Q	The v2 FSM claims recover more FSMs than the stage 1 detector in every arm and floor without losing precision.	recall 0.98/0.99 against 0.13 to 0.51 (+0.48 to +0.85, Holm $p < 0.0001$); pooled precision 0.17 against 0.70 to 0.92 (446 of 577 counted claims on the 20 regfile negatives)	DISMISSED
FSM2b	Q	Reading how state bits turn each other on lets Loki recover one-hot FSMs about as well as binary ones.	-0.010 [-0.025, 0.000] pooled; encoding call right for 97 of 98 one-hot FSMs	POSSIBLY GOOD
FSM3	Q	An FSM's next state depends on its own state bits through logic other than a hold/load mux; a rule that separates state machines from memory words keeps recall ≥ 0.95 and raises claim precision to ≥ 0.80. (controller's proposal, to be confirmed by Peter)	recall 1.000 (50 of 50, unchanged from v2); pooled claim precision 0.676 against 0.149 for v2 (regfile false claims 256 to 0; pipeline 21 and counter 3 remain); design precision 0.877	DISMISSED
P3C	Q	(measurement, no hypothesis) which composite patterns occur in the corpora and how their L1 claim footprint changes between Odin III and Yosys and between gate, LUT and hard-block targets	fsm 78 designs, counter 44, shift 11, timer 12, fsm_shift 3, fsm_counter 1, glue 199; counter claim 37/43 (Odin III gate, LUT), 38/41 (Yosys), 0/43 on hardmap (hard_add 43/43); timer compare claim 7/9 Yosys gate against 2/11 Odin III gate; FSM claim one-hot on both tools; 0 of 4 composites linked by a D path	MEASUREMENT
L3A-fsm	Q	Does the L3 FSM detector find FSMs in single designs?	fsm TP 41 FP 7 FN 2 (O-gate) precision 0.85 recall 0.95; WORKING in O-lut, O-hard, Y-gate, Y-lut (precision 0.89 to 0.95, recall 0.95); O-gate INCONCLUSIVE (precision lower bound 0.77)	MEASUREMENT
L3A-counter	Q	Does the L3 counter detector (hard-adder loop on the hard-block target) find counters?	counter FP 0 in all cells; recall 0.86 (O-gate, O-lut), 0.90 (Yosys), 1.0 (O-hard, 21/21); WORKING on O-hard, INCONCLUSIVE elsewhere (recall lower bound 0.71 to 0.78)	MEASUREMENT
L3A-shift	Q	Do the L3 shift-register and LFSR detectors find them?	shift precision 0.14 to 0.17 (FP 12 to 17) recall 0.4 to 0.5: NOT WORKING in 5 of 5; lfsr 1 TEST design, no claim: INCONCLUSIVE	MEASUREMENT
L3A-timer	Q	Does the L3 timer detector (counter plus terminal compare reaching the counter) find timers?	timer recall 0 to 0.17 (6 to 7 TEST timers), L1 compare claim absent in about half: NOT WORKING in 5 of 5	MEASUREMENT

TABLE LXIV
RESEARCH LOG, HYPOTHESES 61 TO 72 OF 107: EVERY HYPOTHESIS, ONE ROW EACH, WITH THE VERDICT.

ID	Round	Hypothesis	Result	Verdict
L3A-regfile	Q	Does the L3 register-file detector find register files?	regfile recall 0 to 0.31, precision 0 to 0.8: NOT WORKING in 5 of 5	MEASUREMENT
L3A-register-glue	Q	Do the L3 plain-register and glue detectors work?	glue precision 0.98 to 1.0 recall 1.0: WORKING in 5 of 5; register recall 0.78 to 0.85: NOT WORKING in 3 Odin III cells, INCONCLUSIVE in 2 Yosys cells	MEASUREMENT
L3C-composite	Q	Does the L3 composite detector find parts and wires in 82 TEST composites (Q-named benchmarks)?	part recall 0.41 to 0.50, wiring recall 0.12 to 0.18 (0.38 to 0.61 given both ends found), exact match 0 to 2 of 82: NOT WORKING in 5 of 5 (rule: wiring recall $\geq$ 0.70); 62 to 69 composites miss a part, 5 to 14 miss only a wire	MEASUREMENT
GL1	Q	Graphlet signatures find the pattern kinds that the level-one claim chain misses.	whole-design signature alone: NOT WORKING in all patterns and cells (for example counter on gates, O-gate: precision 0.14 to 0.27, recall 0.43 to 0.86, the rule detector 1.00 and 1.00); signature OR rule recovers register files (TP 39 of 60 against 6) at the cost of false alarms; cones and composite partitions not yet tested	INCONCLUSIVE
GL2	Q	Directed graphlets beat undirected ones in precision at equal or better recall.	directed more precise in 1 of 25 (fsm on Yosys gates: precision 0.78 against 0.50, recall 0.49 against 0.22, Holm p 0.0125); not in the other 24	INCONCLUSIVE
GL3	Q	(measurement, no hypothesis) what a missing width costs: library without widths 5, 11, 19, 27 against those instances	undirected 339 of 440 (0.77), directed 352 of 440 (0.80) when the flip-flop count has no entry and the nearest size is used; 0.95 to 1.00 per pattern when all other widths are searched (undirected register 0.78)	MEASUREMENT
GL4	Q	Partition signatures find the patterns the level-one claim chain misses.	singles: signature alone 0 of 40 rows WORKING; composites: part recall 0.33 to 0.61 (rule 0.41 to 0.50), 0.52 to 0.74 with the rule added, precision 0.23 to 0.36 (rule 0.35 to 0.54); neither reaches 0.70 on both	INCONCLUSIVE
GL5	Q	Directed graphlets beat undirected ones in precision at equal or better recall on partitions.	directed more precise in 2 of 25 (state machines on O-lut: precision 0.33 to 0.73; on Y-lut: 0.13 to 0.64; Holm p $<$ 0.0001)	INCONCLUSIVE
GL6	Q	A partition that holds one part plus its glue is matched more precisely, without a large loss of recall.	composites, count known: part precision 0.23 to 0.35 -¿ 0.35 to 0.87, recall 0.33 to 0.61 -¿ 0.21 to 0.42; with the rule added recall 0.46 to 0.65 (rule 0.41 to 0.50), precision 0.32 to 0.57 (rule 0.35 to 0.54); singles: signature alone still 0 of 40 rows WORKING, hybrid 4	INCONCLUSIVE
GL7	Q	Links between matched partitions recover the wires of the composites.	rule recall 0.12 to 0.18; signature, all widths: recall 0.14 to 0.33 (precision 0.17 to 0.46); count known: recall 0.04 to 0.13	INCONCLUSIVE
GL8	Q	A classifier that learns what the signatures of real structures look like detects structures that nearest-entry signature matching misses.	#70 signature alone 0 WORKING cells, hybrid 4; rule 14	INCONCLUSIVE
GL9	Q	A directed signature with a tolerance for reversed single edges detects structures better than the exact directed or the undirected signature.	exact directed TUNE F1 0.759 (rf), undirected 0.715	INCONCLUSIVE

TABLE LXV
RESEARCH LOG, HYPOTHESES 73 TO 84 OF 107: EVERY HYPOTHESIS, ONE ROW EACH, WITH THE VERDICT.

ID	Round	Hypothesis	Result	Verdict
QL4a	Q	Most failed detections are near misses: the gap is the partition, a missing width entry or the threshold, not missing structure.	D70 0.801 (0.771 to 0.831) of 864 classified misses; D71c 0.792 (0.753 to 0.830) of 506; partition misses 486 and 286	POSSIBLY GOOD
QL4b	Q	When the answer-key block is cut out, it is close to its pattern in directed signature distance for most misses.	true block near: D68 0.667, D69 0.679, D70 0.389, D71h 0.749, D71c 0.255; right block for 0.34 to 0.41 of the misses of the cone detectors (D68: 1.0 by construction)	INCONCLUSIVE
QL4c	Q	A minority of misses are structure the signature cannot see (complex decoding); the rest a better detector could recover.	D70 172 of 864 (0.20), D71c 105 of 506 (0.21); first-loss pass opt for absent constructs (37) and 69 of 82 structure misses	MEASUREMENT
QL6a	Q	Blocks grown one cell at a time from anchors (fsm / counter / shift / memory / register / hard) by a Bayes factor on directed graphlet signatures are the right block more often than the D70 cones.	grower 61 right (0.049) against D70 466 (0.371); W13 / L364 / T561 of 938 units; geomean 0.758 (0.735 to 0.781); 0.044 of the D70 misses	DISMISSED
QL6b	Q	The posterior over construct kinds sharpens (entropy falls) as a block grows, and the bigger grouping says what it is.	median rho -0.143 (-0.467 to 0.012) over 379 blocks; mean entropy 1.55 bits at 4-7 cells, 0.96 at 32+; accuracy difference -0.031 (-0.123 to 0.084); ECE 0.178 (final blocks)	INCONCLUSIVE
QL6c	Q	Even pieces of 4 to 7 cells give a top-3 guess of construct kinds better than the prior-only top-3.	0.652 against 0.550, lift 0.102 (0.011 to 0.201); lift 0.00 at 1-3 cells	POSSIBLY GOOD
QL6e	Q	Growth from anchors reaches the right block at some size for a large share of the constructs; the stop rule is what fails.	highest-odds cut 0.080 (WORSE than D70 0.385); oracle prefix 0.556 (geomean against D70 1.166, 1.125 to 1.209)	MEASUREMENT
QL7a	Q	A stop rule that does not depend on the weak per-step likelihood recovers the right block more often than the D70 cones.	d 0.528 against D70 0.385; geomean 1.146 (1.105 to 1.190), W296 / L130 / T431 of 857 units; gate 1.348 BETTER, k4 1.122 BETTER, k6 / hardmap 0.968 INCONCLUSIVE; composite parts 1.058 (0.991 to 1.125) INCONCLUSIVE	USEFUL
QL7b	Q	The new stop closes most of the gap between the #73 stop (0.053 on the same constructs) and the answer-key ceiling (0.556).	0.944 (0.927 to 0.961)	USEFUL
QL7c	Q	Which of the four candidate rules (size-penalised posterior a, lowest conductance b, learned stop c, combination d) recovers the right block.	a 0.422, b 0.462, c 0.514, d 0.528, E73 0.053; geomean against #73 1.39 to 1.52, all BETTER; precision of the kind claim 0.19 to 0.27, recall 0.16 to 0.19	MEASUREMENT
QL7e	Q	The headline rule recovers more constructs than simply growing every block to the end of its path.	d 1.000 (0.997 to 1.004), W5 / L5 / T847; composite parts 1.008 (0.997 to 1.022); a 0.912, b 0.943, c 0.990, each WORSE	INCONCLUSIVE
QL8c-a	Q	A likelihood built on orbit density, boundary, feedback and typed-cell features recovers more constructs together with their kind than the #74 likelihood, on the same grown blocks.	geomean 1.011 (1.001 to 1.021), W74 / L55 / T1221 of 1350 units; k4 1.015 BETTER, gate 1.013 and k6 / hardmap 0.999 INCONCLUSIVE; single 1.011 BETTER, composite parts 1.013 INCONCLUSIVE; recall 0.199 against 0.183	POSSIBLY GOOD

TABLE LXVI
RESEARCH LOG, HYPOTHESES 85 TO 96 OF 107: EVERY HYPOTHESIS, ONE ROW EACH, WITH THE VERDICT.

ID	Round	Hypothesis	Result	Verdict
QL8c-b	Q	A likelihood on richer directed graphlet features raises kind precision and recall well above #74's at the same blocks (fixed as +0.10 macro F1).	0.262 against 0.211, gain +0.051 (0.019 to 0.081); precision 0.287 against 0.269, recall 0.199 against 0.183; F1 per kind counter 0.212 against 0.173, register 0.231 against 0.176, regfile 0.393 against 0.196, fsm 0.242 against 0.265, shift 0.230 against 0.244	DISMISSED
QL8c-c	Q	(measurement, no hypothesis) the top-label calibration error of the chosen model against the #74 likelihood with its temperature refitted on the same blocks.	all grown blocks 0.139 against 0.197, difference -0.057 (-0.085 to -0.019); right blocks 0.202 against 0.228 (-0.026, -0.057 to 0.011); true blocks 0.170 against 0.206; top-3 contains the kind 0.834 against 0.810 (right blocks)	MEASUREMENT
QL8c-d	Q	(measurement, no hypothesis) what the kind guess reaches when the block is already right.	macro F1 0.487 against 0.419, gain +0.068 (0.032 to 0.101); precision 0.690 against 0.678, recall 0.407 against 0.365; counter recall 0.504 against 0.372; top-3 0.823 against 0.781 (+0.042, 0.022 to 0.060)	MEASUREMENT
QL8a-a	Q	The headline growth order OA (grow where the block is most strongly tied) recovers the right block more often than the D70 cones.	OA 0.526 against D70 0.385; geomean 1.145 (1.102 to 1.188), W293 / L127 / T437 of 857 units; gate 1.341 and k4 1.125 BETTER, k6 / hardmap 0.965 (0.914 to 1.016) and composite parts 1.044 (0.979 to 1.111) INCONCLUSIVE; single 1.182 BETTER	INCONCLUSIVE
QL8a-b	Q	The headline order OA recovers the right block more often than the #74 order (grow to the end).	geomean 0.999 (0.994 to 1.004), W12 / L13 / T832; gate 0.990 (0.980 to 0.998) WORSE; k4 1.006, k6 / hardmap 0.999, composite parts 0.996 INCONCLUSIVE; OB gate 1.034 BETTER but composite parts 0.939 WORSE; per-target mix 1.013 (1.006 to 1.021) BETTER (extra comparison)	INCONCLUSIVE
QL8a-c	Q	A growth order that respects structure gives a right prefix to a share of the 449 constructs with none in #74 and lifts k6 / hardmap and composite parts above D70.	12 of 449 recovered at the path end by OA (2.7 percent; 30 by OB, 31 by the per-target mix), 14 of 602 lost; k6 / hardmap and composite parts not BETTER than D70	INCONCLUSIVE
QL8a-d	Q	(measurement, no hypothesis) why the constructs with no right prefix have none.	no right prefix on any block, #74 / OA / OB: NOSEED 58 / 58 / 58, BLOATED 95 / 76 / 75, SPLIT 251 / 250 / 269, UNDERREACH 103 / 103 / 80; oracle share 0.556 / 0.573 / 0.578	MEASUREMENT
QL8b-a	Q	Extra anchor classes (flip-flop banks sharing control, open feedback loops, constants) make a share of the 58 constructs that no block touches touched and right.	58 of 58 touched (0 before), 47 right: structural 19 of 21 (14 are one HDL problem, the 64-flip-flop shifter), constant cells given to a part 28 of 37	POSSIBLY GOOD
QL8b-b	Q	The extra anchors cost no right construct elsewhere: the right-block share does not fall against the #74 anchors.	OLD 0.528, NEW 0.550 (D70 0.385); per unit against OLD geomean 1.001 (0.980 to 1.024), W78 / L66 / T713; gate 0.948 WORSE, single 0.960 WORSE (W0 / L37), composite parts 1.129 BETTER, k6 / hardmap 1.042 BETTER; 89 constructs right before are lost, 115 gained (61 of the lost hold constant cells that the const block takes; post hoc bank and loop alone: 0.527, 86 lost)	INCONCLUSIVE
QL8b-c	Q	(measurement, no hypothesis) the new anchors against the D70 cones, and the precision of the anchor classes.	against D70 geomean 1.148 (1.103 to 1.197) BETTER, composite parts 1.184 (1.101 to 1.273) BETTER (was 1.058 INCONCLUSIVE in #74), k6 / hardmap 1.007 INCONCLUSIVE; anchor precision (touch / right at the end): bank 107 blocks, 107 / 34; loop 167, 167 / 52; const 1186, 807 / 30	MEASUREMENT
QL9-a	Q	A post-growth join / split step raises the right-block share above #77's 0.550 without losing single designs.	share 0.550 to 0.715 (D70 0.385); per unit against the #77 blocks geomean 1.149 (1.123 to 1.177), W194 / L9 / T654; single designs 1.151 BETTER (W129 / L0), gate 1.061, k4 1.134, k6 / hardmap 1.295, composite parts 1.142 all BETTER; against D70 1.318 (1.276 to 1.364); 217 constructs gained, 29 lost (all in composite parts, 17 shift and 11 fsm); the join rule is the cut conductance between blocks at 0.2, the Bayes-factor joins give 0.680 and 0.677	USEFUL
QL9-b	Q	The step recovers a large share of the 251 constructs split across blocks (#76 SPLIT class).	157 of 251 right after the step (0 before; Bayes-factor joins 138 and 124 of 251, shared control nets 56)	USEFUL

TABLE LXVII
RESEARCH LOG, HYPOTHESES 97 TO 107 OF 107: EVERY HYPOTHESIS, ONE ROW EACH, WITH THE VERDICT.

ID	Round	Hypothesis	Result	Verdict
QL9-c	Q	The step recovers a large share of the 95 constructs swallowed by a block that grows past them (#76 BLOATED class).	42 of 95 right after the step, but 44 were already right before it (anchors of #77); the join gains none of the swallowed ones and loses 2; the 29 constructs the step loses are swallowed by a joined block	POSSIBLY GOOD
QL9-d	Q	A split step (cut a block along the Fiedler vector where two constructs explain it better than one) helps.	every threshold loses constructs on TRAIN (net -120 at 0 to -10 at 8 for the #73 evidence; strictest 16: 4 splits, net 0) and on VALIDATION; so no split in the headline; on TEST the split alone gains 0 and loses 1 (geomean 0.999) and adds nothing to the join	DISMISSED
QL9-e	Q	(measurement, no hypothesis) the kind guess on the blocks after joining, and the three other join rules.	macro F1 0.237 to 0.392 (fsm 0.282 to 0.476, counter 0.132 to 0.509, shift 0.287 to 0.317, register 0.188 to 0.266, regfile 0.296 to 0.393); right-block share by join rule: cut conductance 0.715, Bayes factor #73 0.680, Bayes factor M2 0.677, shared control nets 0.617	MEASUREMENT
QL10-a	Q	On designs that fit Gate-Breaker's prompt, a model that decodes Loki's joined blocks one at a time (block netlist and boundary nets, no hints) and a script that assembles them give RTL at least as correct as the whole-netlist prompt (arm A).	multi-block D 72 of 81 draws pass, A 73 of 81; pass@1 difference -0.012 (-0.074 to +0.037), draws W3 / L4 / T74, designs W2 / L3 / T22; pass@3 26 of 27 both; all 42 run designs +0.008 (-0.032 to +0.056) (15 are gate-free, no model); same with the 27 template-deviation rows removed	INCONCLUSIVE
QL10-b	Q	On designs too large for one prompt (arm A cannot run), decoding Loki's blocks one at a time still produces a correct design, or at least correct blocks.	over-cap 8 of 12 draws pass (mux256to1 3/3, gates100 3/3, rotate100 2/3, shift18 0/3); processor modules 1 of 5 equivalent; block check ok for 954 of 969 block replies (0.985); the 10 processor assemblies with a failed block become equivalent when that block is replaced by its own gates	USEFUL
QL10-c	Q	Adding Loki's one-word kind guess for a block to its prompt does not improve correctness (the hints-do-not-help result carried to blocks).	pass@1 D 0.750, D+h 0.708, difference -0.042 (-0.125 to 0.000), draws W1 / L2 / T21; hinted blocks check ok 95 of 102 in each arm; 5 processor modules (1 draw): D 1 pass, D+h 0	INCONCLUSIVE
QL10-d	Q	(measurement, no hypothesis) Gate-Breaker's second measure, AST and CFG similarity of the assembled design to the reference RTL, D against A.	multi-block D-flat minus A-flat (same writer, headline): AST -0.037 (-0.063 to -0.014; 23 designs, W6 / L17), CFG -0.042 (-0.067 to -0.021; 27 designs, W6 / L21); against A as written -0.132 and -0.065; as assembled (top module only) -0.396 and -0.552; D+h minus D 0.000; over-cap D-flat AST 0.748, CFG 0.874	MEASUREMENT
QL10-e	Q	(measurement, no hypothesis) the share of the Odin III provenance answer key (flip-flop sets) held by one right block, and decoded correctly, per arm.	multi-block: right block 12 of 36, decoded ok 10 of 36, the same for D+h; over-cap 6 of 6 and 6 of 6; the structure-only 39 of 47 over all 42 designs with a key (no model) includes 34 single-block designs that were not run	MEASUREMENT
L3	-	A GNN trained on one suite transfers to the other suite.	not run; needs: Train on the micro designs, test on the VerilogEval designs and the reverse, with an interval over designs.	UNTESTED
L6	-	The graphlet counts differ from a motif null model.	not run; needs: Compare the ORCA graphlet counts of each netlist with degree-preserving random rewirings of the same graph.	UNTESTED
L8	-	A language model given Loki's ReverseProvenance record recovers RTL better than the model alone.	not run; needs: The Gate-Breaker task with a record arm and a no-record arm on the full quartile files, three draws per arm (deferred by Peter; the pilot of the LLM rows is the small first step).	UNTESTED